# Chapter 4

# Topology and Sheaves

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## 4.1 Topological Spaces

References:

- General Topology [Kel71] (Chap. 1)
- General Topology [Wil70]

Topology is useful in algebraic geometry, but often the natural topologies are usually much coarser so the theory looks rather different.

**Definition 4.1.1 (Topological Space)**
*A topological space $(X, \mathcal{T}_X)$ consists of a set $X$ and family of open sets $\mathcal{T}_X \subseteq \mathcal{P}(X)$ satisfying the following properties*

- $X, \emptyset \in \mathcal{T}_X$
- $U_i \in \mathcal{T}_X \implies \bigcup_{i \in I} U_i \in \mathcal{T}_X$
- $U, V \in \mathcal{T}_X \implies U \cap V \in \mathcal{T}_X$

*A subset $Z \subset X$ is said to be closed iff $X \setminus Z$ is open. We may equivalently define the topology in terms of closed sets.*

**Proposition 4.1.2**
*Let $X$ be a topological space. Both the open sets and closed sets form a distributive lattice under inclusion.*

**Definition 4.1.3 (Subspace topology)**
*Let $Y \subset X$, then we may define the **subspace topology** on $Y$ by*

$$\mathcal{T}_Y := \{U \cap Y \mid U \in \mathcal{T}_X\}$$

*when $Y$ is open then this is given by*

$$\mathcal{T}_Y = \{U \subseteq Y \mid U \in \mathcal{T}_X\}$$

**Definition 4.1.4 (Base)**
*We say $\mathcal{B} \subseteq \mathcal{P}(X)$ is a base (of open sets) on $X$ if*

- *For every $x \in X$ there is a $U \in \mathcal{B}$ such that $x \in U$*
- *Suppose $U, V \in \mathcal{B}$ and $x \in U \cap V$ then there exists $W \in \mathcal{B}$ such that $x \in W \subseteq U \cap V$*

**Proposition 4.1.5 (Topology generated by a base)**
*Let $\mathcal{B}$ be a base, then the following is a topology on $X$*

$$\mathcal{T}_{\mathcal{B}} := \left\{ \bigcup_{U_i \in I} U_i \;\middle|\; I \subseteq \mathcal{B} \right\}$$

*i.e. the set of arbitrary unions of sets in $\mathcal{B}$.*

**Proposition 4.1.6 (Base generating topology)**
*A base $\mathcal{B}$ satisfies $\mathcal{T}_{\mathcal{B}} = \mathcal{T}_X$ if and only if*

*a) $\mathcal{B} \subset \mathcal{T}_X$*
*b) For every $x \in U$ and $U \in \mathcal{T}_X$ there exists $V \in \mathcal{B}$ such that $x \in V \subseteq U$.*

*In this case we say $\mathcal{B}$ is a base for (the topology on) $X$.*

**Proposition 4.1.7 (Subbase for a topology)**
*Let $X$ be a set and $\mathcal{B}$ be an arbitrary family of subsets. Then there exists a topology $\mathcal{T}_{\mathcal{B}}$ such that*

*a) $\mathcal{B} \subset \mathcal{T}_{\mathcal{B}}$*
*b) For every topology $\mathcal{T}'$ containing $\mathcal{B}$ we have $\mathcal{T}_{\mathcal{B}} \subset \mathcal{T}'$.*

*We say that $\mathcal{B}$ is a **subbase** for the topology $\mathcal{T}_{\mathcal{B}}$. Explicitly $\mathcal{T}_{\mathcal{B}}$ consists of arbitrary unions of finite intersections of elements of $\mathcal{B}$.*

**Proposition 4.1.8 (Base for subspace topology)**
*Let $(X, \mathcal{T})$ be a topological space, and $Y \subset X$ has the subspace topology $\mathcal{T}_Y$ (4.1.3). If $\mathcal{B}$ is a base for $(X, \mathcal{T})$ then*

$$\mathcal{B}_Y := \{U \cap Y \mid U \in \mathcal{B}\}$$

*is a base for $(Y, \mathcal{T}_Y)$.*

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**Definition 4.1.9 (Adherent point)**
*For $Y \subset X$ we say $x$ is an **adherent point** of $Y$ if every open neighbourhood of $x$ intersects $Y$.*

*Similarly we say $x$ is a **limit point** (or **accumulation point**) for $Y$ if every open neighbourhood of $x$ intersects $Y$ at a point other than $x$.*

**Proposition 4.1.10 (Closed point)**
*Let $x \in X$ then TFAE*

- *$\{x\}$ is closed*
- *For every $y \neq x$ there is $U \ni y$ such that $x \notin U$.*

**Definition 4.1.11**
*For a topological space $X$ let $X_0$ denote the subset of closed points.*

**Definition 4.1.12 (Open Embedding)**
*Let $f : X \to Y$ be a map of topological spaces. We say it is an **open embedding** if one of the following equivalent conditions holds*

*a) $f(X)$ is open and $f$ is a homeomorphism onto its image*
*b) $f$ is continuous, open and injective.*

### 4.1.1 Axioms of Countability

**Definition 4.1.13 (Neighbourhood)**
*Let $X$ be a topological space and $x \in X$. We say a set $V \subset X$ is a **neighbourhood** of $x$ if $x \in U \subset V$ for some open set $U$. We write $\mathcal{U}_x$ for the neighbourhoods of $x$, and $\mathcal{O}_x$ for the open neighbourhoods of $x$.*

*We say that a collection of neighbourhoods $\mathcal{B}_x$ is a **local base** for $x$ if every neighbourhood $U \in \mathcal{U}_x$ contains some neighbourhood $V \in \mathcal{B}_x$. For example $\mathcal{O}_x$ is a local base.*

**Definition 4.1.14 (First-Countable)**
*Let $X$ be a topological space. $X$ is **first countable** if every point $x \in X$ has a countable local base.*

**Definition 4.1.15 (Second-Countable)**
*Let $X$ be a topological space. $X$ is **second countable** if there exists a countable basis. Evidently this is stronger than first countability.*

**Example 4.1.16**
*For example a metric space is always first countable as we may consider balls of radius $\frac{1}{n}$.*

*Further $\mathbb{R}$ is second countable as we may consider open intervals with rational endpoints.*

### 4.1.2 Closure

**Definition 4.1.17 (Adherent Point)**
*Let $X$ be a topological space and $Y \subset X$. We say that $x \in X$ is an **adherent point** of $Y$, if $U \in \mathcal{U}_x \implies U \cap Y \neq \emptyset$.*

*We say that $x$ is a **sequential limit point** of $Y$ if there exists a sequence $x_n \in Y$ such that $x_n \to x$.*

**Proposition 4.1.18 (Adherent Point)**
*Let $X$ be a topological space. Then every sequential limit point is an adherent point. If $X$ is first countable the converse holds.*

*Proof.* The first implication is straightforward. For the latter, consider a countable local base at $x$ given by $U_n$. We may assume wlog that it is decreasing. Then by assumption we may choose $x_n \in U_n \cap Y$. For any $V \in \mathcal{B}_x$ we have $U_N \subset V$ for some $N$, and therefore $n \geq N \implies x_n \in U_n \implies x_n \in V$. Therefore $x_n \to x$. ∎

**Proposition 4.1.19 (Topological Closure)**
*Let $Y \subset X$ then the following equality holds*

$$\bigcap_{\substack{Z \supseteq Y \\ Z \text{ closed}}} Z = \{x \in X \mid x \text{ adherent point of } Y\}$$

*We denote this by $\overline{Y}$ (or $\mathrm{cl}_X(Y)$ to emphasise the ambient space $X$) and refer to it as the **closure** of $Y$ in $X$. Furthermore the following properties hold*

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*a)* $Y \subseteq \overline{Y}$ *and* $\overline{Y}$ *is closed*
*b)* $Y = \overline{Y}$ *if and only if* $Y$ *is closed*
*c)* $\left(Y \cap U \neq \emptyset \iff \overline{Y} \cap U \neq \emptyset\right)$ *for any* $U$ *open*

*When* $X$ *is first countable then* $\overline{Y}$ *is the set of sequential limit points of* $Y$*.*

*Proof.* Suppose $Z$ is a closed set containing $Y$ and $x$ is an adherent point of $Y$. Then $x \notin Z \implies x \in X \setminus Z \implies (X \setminus Z) \cap Y \neq \emptyset$ a contradiction. Conversely assume $x \notin \overline{Y}$ then there exists $Z \supseteq Y$ closed such that $x \notin Z \implies x \in X \setminus Z$. This means $x$ is not an adherent point.

a) An arbitrary intersection of closed sets is closed
b) This follows because $\overline{Y}$ is the smallest closed superset.
c) One implication is clear because $Y \subseteq \overline{Y}$. Conversely if $x \in \overline{Y} \cap U$ then $x$ must be a limit point of $Y$ hence $U \cap Y \neq \emptyset$ as required.

■

**Corollary 4.1.20**
*Let* $X$ *be a first countable topological space. Then* $\mathrm{cl}_X(Y)$ *is the set of sequential limit points of* $Y$*.*

**Proposition 4.1.21 (Closure in Subspace Topology)**
*Let* $X$ *be a topological space and* $Y$ *a subset. For* $Z \subset Y$ *we have*

$$\mathrm{cl}_Y(Z) = \mathrm{cl}_X(Z) \cap Y$$

*Proof.* By (4.1.19) $\mathrm{cl}_X(Z)$ is a closed subset of $X$ and therefore $\mathrm{cl}_X(Z) \cap Y$ is a closed subset of $Y$ in the subspace topology. Therefore $\mathrm{cl}_Y(Z) \subset \mathrm{cl}_X(Z) \cap Y$ by minimality. Similarly $\mathrm{cl}_Y(Z) = W \cap Y$ for $W$ closed in $X$. By minimality $\mathrm{cl}_X(Z) \subset W \implies \mathrm{cl}_X(Z) \cap Y \subset W \cap Y = \mathrm{cl}_Y(Z)$. ■

**Proposition 4.1.22**
*Let* $X$ *be a topological space with* $X = \bigcup_{i \in I} U_i$ *an open cover. Let* $Y \subset X$ *be a subset then* $Y$ *is closed in* $X$ *if and only if* $Y \cap U_i$ *is closed in* $U_i$ *for all* $i \in I$*.*

*Proof.* One direction is obvious by definition of the subspace topology. Conversely $U_i \setminus (Y \cap U_i)$ is open in $U_i$ and therefore in $X$. We see that

$$X \setminus Y = \bigcup_{i \in I} U_i \setminus (Y \cap U_i)$$

is open and therefore $Y$ is closed. ■

**Proposition 4.1.23**
*Let* $Y_1, ..., Y_n$ *be subsets of a topological space* $X$*. Then*

$$\overline{Y_1 \cup ... \cup Y_n} = \overline{Y_1} \cup ... \cup \overline{Y_n}$$

*Proof.* By induction it is sufficient to demonstrate the case $n = 2$. $\overline{Y_1 \cup Y_2}$ is a closed set containing both $Y_1$ and $Y_2$. Therefore by definition

$$\overline{Y_1} \cup \overline{Y_2} \subset \overline{Y_1 \cup Y_2}$$

On the other hand $\overline{Y_1} \cup \overline{Y_2}$ is closed and contains $Y_1 \cup Y_2$. Therefore the reverse inclusion follows immediately. ■

**Proposition 4.1.24 (Dense subset)**
*Let* $Y \subset X$ *then the following are equivalent*

*a)* $\overline{Y} = X$
*b)* $Y \cap U \neq \emptyset$ *for any* $U \subset X$ *open*

*In this case we say* $Y$ *is* ***dense****.*

*Proof.* *a)* $\implies$ *b)* Follows from (4.1.19).c)

*b)* $\implies$ *a)*. Suppose $Y \subseteq \overline{Y} \subsetneq X$ then $X \setminus \overline{Y}$ is an open set not intersecting $Y$ a contradiction. ■

### 4.1.3 Continuous Maps

**Proposition 4.1.25 (Continuous at a point)**
*Let* $f : X \to Y$ *a map of topological spaces and* $x \in X$*. Then the following are equivalent*

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*a)* $V \in \mathcal{U}_{f(x)} \implies f^{-1}(V) \in \mathcal{U}_x$
*b)* $V \in \mathcal{B}_{f(x)} \implies f^{-1}(V) \in \mathcal{U}_x$
*c)* $V \in \mathcal{U}_{f(x)} \implies \exists U \in \mathcal{U}_x$ s.t. $f(U) \subseteq V$
*d)* $V \in \mathcal{B}_{f(x)} \implies \exists U \in \mathcal{B}_x$ s.t. $f(U) \subseteq V$

*where $\mathcal{B}_{f(x)}$ is any local base at $f(x)$ and $\mathcal{B}_x$ is any local base at $x$. We say that $f$ is* ***continuous*** *at $x$.*

*Proof.* *a)* $\implies$ *b)* is obvious. Suppose *b)* holds and $V \in \mathcal{U}_{f(x)}$ then by definition there exists $V' \in \mathcal{B}_{f(x)}$ such that $V' \subset V$. By hypothesis $f^{-1}(V') \in \mathcal{U}_x$ from which it follows that $f^{-1}(V) \in \mathcal{U}_x$.

*a)* $\implies$ *c)* $f(f^{-1}(V)) \subseteq V$

*c)* $\implies$ *d)* By definition there exists $U' \in \mathcal{B}_x$ such that $U' \subseteq U \implies f(U') \subseteq f(U) \subseteq V$

*d)* $\implies$ *b)* Observe that $f(U) \subseteq V \implies U \subseteq f^{-1}(V)$ whence $f^{-1}(V) \in \mathcal{U}_x$ as required. ∎

**Proposition 4.1.26**
*Let $f : X \to Y$ and $g : Y \to Z$ be maps of topological spaces such that $f$ is continuous at $x$ and $g$ is continuous at $f(x)$. Then $g \circ f$ is continuous at $x$.*

**Proposition 4.1.27 (Characterisation of Continuous Maps)**
*Let $f : X \to Y$ be a map of topological spaces. Then the following are equivalent*

*a)* *$f$ is continuous at all points $x \in X$*
*b)* *$V \subset Y$ open $\implies f^{-1}(V) \subset X$ is open*
*c)* *$Z \subset Y$ closed $\implies f^{-1}(Z) \subset X$ is closed*

*Let $\mathcal{B}_1, \mathcal{B}_2$ be bases of open sets for $X$ and $Y$ respectively. Then this is equivalent to the following condition*

$$\forall V \in \mathcal{B}_2, \quad \text{and } x \in f^{-1}(V), \quad \exists U \in \mathcal{B}_1 \text{ s.t. } x \in U \subset f^{-1}(V)$$

**Proposition 4.1.28**
*Let $f : X \to Y$ and $g : Y \to Z$ be maps of topological spaces such that $f$ is continuous and $g$ is continuous. Then $g \circ f$ is continuous.*

**Proposition 4.1.29**
*Let $f : X \to Y$ be a map of topological spaces and $X = \bigcup_{i \in I} U_i$ an open cover. Then $f$ is continuous iff $f|_{U_i}$ is continuous for all $i \in I$.*

*Proof.* Observe that $f^{-1}(V) = \bigcup_{i \in I} f^{-1}(V) \cap U_i = \bigcup_{i \in I} \left(f|_{U_i}\right)^{-1}(V)$. ∎

**Proposition 4.1.30**
*Let $f : X \to Y$ be a map of topological spaces. Then $f$ is continuous iff*

$$f\left(\overline{Z}\right) \subset \overline{f(Z)}$$

*for all subsets $Z \subset X$.*

*Proof.* Suppose $f$ is continuous and $f(Z) \subset W$ is closed. Then by (4.1.27) $Z \subset f^{-1}(W)$ is closed. By definition $\overline{Z} \subset f^{-1}(W) \implies f\left(\overline{Z}\right) \subset f(f^{-1}(W)) \subset W$. As $W$ was an arbitrary closed set containing $f(Z)$ we find by definition $f\left(\overline{Z}\right) \subset \overline{f(Z)}$.

Conversely suppose $W \subset Y$ is closed and $Z \coloneqq f^{-1}(W)$. Then $f\left(\overline{Z}\right) \subset \overline{f(Z)} = \overline{W \cap \operatorname{Im}(f)} \subset W \implies f^{-1}\left(f\left(\overline{Z}\right)\right) \subset Z \implies \overline{Z} \subset Z \implies \overline{Z} = Z$ and therefore $Z$ is closed. Therefore $f$ is continuous by (4.1.27). ∎

### 4.1.4 Quasi-Homeomorphism

**Proposition 4.1.31 (Quasi-Homeomorphism)**
*Let $f : X \to Y$ be a map. The following are equivalent*

*a)* *The map $W \to f^{-1}(W)$ is a bijection between open sets*
*b)* *The map $Z \to f^{-1}(Z)$ is a bijection between closed sets*

*In this case we say $f$ is a* ***quasi-homeomorphism****. Such a map induces a bijection between irreducible sets, respectively irreducible components.*

*Proof.* This follows by taking complements and observing $X \setminus f^{-1}(A) = f^{-1}(Y \setminus A)$. ∎

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**Proposition 4.1.32 (Induced Quotient)**
*Let $\pi : X \to Y$ be a surjective map where $X$ is a topological space. The family*

$$\mathcal{T}_Y := \{U \subset Y \mid \pi^{-1}(U) \in \mathcal{T}_X\}$$

*is a well-defined topology on $Y$. Suppose that $\pi^{-1}(\pi(V)) \subset V$ for all $V$ open (resp. closed) subsets of $X$. Then $\pi$ is both open, closed and a quasi-homeomorphism.*

*Proof.* As arbitrary intersections and unions commute with inverse image we see that $\mathcal{T}_Y$ is a well-defined topology.

As $\pi$ is surjective we have $\pi(\pi^{-1}(U)) = U$ for all open subsets $U \subset Y$. Generically $V \subset \pi^{-1}(\pi(V))$ and by assumption the reverse inclusion holds. Therefore we see $\pi$ induces a quasi-homeomorphism. This also shows $\pi$ is open and therefore closed. ∎

## 4.1.5 Kolmogorov Spaces

**Definition 4.1.33**
*Let $X$ be a topological space. We say that $X$ is **Kolmogorov** or $T_0$ if for all distinct pairs $x, y \in X$ there exists an open set $U$ such that either $x \in U$ and $y \notin U$, or $x \notin U$ and $y \in U$.*

**Proposition 4.1.34 (Indistinguishable points)**
*Let $X$ be a topological space and $x, y \in X$. The following are equivalent*

*a) $\overline{\{x\}} = \overline{\{y\}}$*
*b) $x \in \overline{\{y\}}$ and $y \in \overline{\{x\}}$*
*c) Every closed $Z \subset X$ contains both or neither of $x, y$*
*d) Every open $U \subset X$ contains both or neither of $x, y$*

*We say $x$, $y$ are **topologically indistinguishable**, otherwise we say $x$ and $y$ are **distinguishable**.*

*$X$ is Kolmogorov iff every pair of points are distinguishable.*

*Being indistinguishable determines an equivalence relation on $X$, and we denote by $X_0 := X/\sim$ the **Kolmogorov Quotient**.*

*Proof.* $a) \implies b)$ Clear

$b) \implies a)$ By definition of closure $x \in \overline{\{y\}} \iff \overline{\{x\}} \subset \overline{\{y\}}$.

$c) \iff d)$ Follows by taking complements

$c) \implies b)$ Suppose $x \in Z$ for $Z$ closed then by hypothesis $y \in Z$. As $Z$ was arbitrary we see that $y \in \overline{\{x\}}$, and symmetrically $x \in \overline{\{y\}}$.

$b) \implies c)$ By assumption $x \in Z \implies y \in Z$ for $Z$ closed and symmetrically $y \in Z \implies x \in Z$. ∎

**Definition 4.1.35 (Kolmogorov Quotient)**
*Let $X$ be a topological space. The set $X_0$ of equivalence classes of topologically indistinguishable points is the **Kolmogorov Quotient***

**Proposition 4.1.36 (Kolmogorov Quotient Properties)**
*Let $X$ be a topological space and $X_0$ be the Kolmogorov Quotient with quotient map $\pi : X \to X_0$. Relative to the induced topology*

$$\mathcal{T}_{X_0} := \{W \subset X_0 \mid \pi^{-1}(W) \in \mathcal{T}_X\}$$

*$\pi$ is a quasi-homeomorphism and an open and closed map. Further $X_0$ is Kolmogorov.*

*Proof.* By (4.1.32) it is sufficient to show that $\pi^{-1}(\pi(U)) \subset U$ for all open subsets $U \subset X$. Suppose $x \in \pi^{-1}(\pi(U))$ then $\pi(x) = \pi(y)$ for some $y \in U$. Then by definition of $\pi$, $x$ and $y$ are indistinguishable and so $x \in U$.

Suppose $\pi(x) \neq \pi(y)$ then by definition there exists an open set $U \subset X$ such that $x \in U$ and $y \notin U$. As $\pi$ is open then $\pi(U)$ is open. By construction $\pi(x) \in \pi(U)$ and $\pi(y) \notin \pi(U)$. This shows that $X_0$ is Kolmogorov. ∎

**Proposition 4.1.37 (Kolmogorov Quotient Functoriality)**
*Let $f : X \to Y$ be a continuous map, then there is a unique (continuous) map $f_0 : X_0 \to Y_0$ such that the following diagram commutes*

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$$\begin{array}{ccc} X & \xrightarrow{f} & Y \\ \downarrow{\scriptstyle \pi} & & \downarrow{\scriptstyle \pi} \\ X_0 & \xrightarrow{f_0} & Y_0 \end{array}$$

*Proof.* Given $x, x' \in X$ and suppose $f(x) \nsim f(x')$. Then there exists an open set $U$ such that $f(x') \in U$ and $f(x) \notin U$. Therefore $x' \in f^{-1}(U)$ and $x \notin f^{-1}(U)$ and $x \nsim x'$. Equivalently $x \sim x' \implies f(x) \sim f(x')$. Therefore the map

$$f_0([x]) = [f(x)]$$

is well-defined.

For $W_0 \subset Y_0$ we have $\pi^{-1}(f_0^{-1}(W_0)) = f^{-1}(\pi^{-1}(W_0))$ is open. As $\pi$ is surjective and open $f_0^{-1}(W) = \pi(\pi^{-1}(f_0^{-1}(W_0)))$ is open, and therefore $f_0$ is continuous.

Any such $f_0$ satisfies $\{f_0(x_0)\} = \pi(f(\pi^{-1}(\{x_0\})))$ and so is unique. ∎

**Proposition 4.1.38 (Bijections with Homeomorphic Kolmogorov Quotients are Homeomorphisms)**
*Let $f : X \to Y$ be a continuous bijection of topological spaces such that the induced map $f_0 : X_0 \to Y_0$ (4.1.37) is a homeomorphism. Then $f$ is a homeomorphism.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

Write $\pi_X : X \to X_0$, $\pi_Y : Y \to Y_0$ for the quotient maps, so $\pi_Y \circ f = f_0 \circ \pi_X$. For $x, x' \in X$ we may show

$$x \sim x' \iff f(x) \sim f(x')$$

For $x_0 \in X_0$ with $x_0 = \pi_X(x)$ we have

$$f(\pi_X^{-1}(x_0)) = \{f(x') \mid x' \sim x\} = \{f(x') \mid f(x') \sim f(x)\} = \pi_Y^{-1}(f_0(x_0))$$

where the last equality requires surjectivity of $f$.

It follows that $f$ is open: for $W \subset X$ open, indistinguishable points lie in exactly the same open sets, so $x \in W$ and $x \sim x'$ forces $x' \in W$; that is, $W$ is a disjoint union of the equivalence classes it meets, which are precisely the fibres of $\pi_X$ (4.1.35), $W = \bigsqcup_{x_0 \in \pi_X(W)} \pi_X^{-1}(x_0)$, so

$$f(W) = \bigcup_{x_0 \in \pi_X(W)} f(\pi_X^{-1}(x_0)) = \bigcup_{x_0 \in \pi_X(W)} \pi_Y^{-1}(f_0(x_0)) = \pi_Y^{-1}\left( \bigcup_{x_0 \in \pi_X(W)} \{f_0(x_0)\} \right) = \pi_Y^{-1}(f_0(\pi_X(W)))$$

which is open because both $f_0$ and $\pi_X$ are open and $\pi_Y$ is continuous.

A bijective, continuous, open map is a homeomorphism. ∎

**Proposition 4.1.39**
*Let $X$ be a topological space with open cover $X = \bigcup_{i \in I} U_i$. If every $U_i$ is Kolmogorov with respect to the subspace topology then so is $X$.*

*Proof.* Consider $x, y \in X$. We may assume wlog that $x, y \in U_i$ for some $i$ (otherwise we would be done). Then by assumption there exists $V \subset X$ such that $x \in U_i \cap V$ and $y \notin U_i \cap V$, whence $x \in V$ and $y \notin V$ as required. ∎

### 4.1.6 Symmetric Spaces

**Definition 4.1.40**
*Let $X$ be a topological space. We say that $x$ is **quasi-closed** if $y \in \overline{\{x\}} \implies x \in \overline{\{y\}}$.*

**Proposition 4.1.41 (Symmetric Topology)**
*Let $X$ be a topological space. Then the following are equivalent*

*a) For all $x \in X$ we have*

$$\overline{\{x\}} = \{y \in X \mid y \text{ topologically indistinguishable from } x\},$$

*b) For all $x, y \in X$ $\quad x \in \overline{\{y\}} \implies y \in \overline{\{x\}}$ (i.e. every point is **quasi-closed**),*
*c) Every open neighbourhood $U$ of $x$ contains $\overline{\{x\}}$,*
*d) $X$ is partitioned by sets of the form $\overline{\{x\}}$, indexed by $X_0$.*

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*In this case we say that $X$ is **symmetric** or $R_0$. Further $x \in \overline{\{y\}}$ if and only if $x$ and $y$ become equal in $X_0$.*

**Proposition 4.1.42**
*Let $X$ be a symmetric topological space and $x, y \in X$. Then the following are equivalent*

*a) $x$ and $y$ are topologically indistinguishable,*
*b) $x \in \overline{\{y\}}$,*
*c) $y \in \overline{\{x\}}$,*
*d) $\overline{\{x\}} = \overline{\{y\}}$,*
*e) $x$ and $y$ become equal in the Kolmogorov Quotient $X_0$*

**Proposition 4.1.43 (Open Subspace is Symmetric)**
*Let $X$ be a symmetric space and $U \subset X$ an open subset. Then $U$ is symmetric and for every $x \in U$ we have*

$$\mathrm{cl}_X(\{x\}) = \mathrm{cl}_U(\{x\})$$

*Proof.* By (4.1.21) we have $\mathrm{cl}_U(\{x\}) = \mathrm{cl}_X(\{x\}) \cap U$, so these are equal by (4.1.41).c). We conclude that the sets $\overline{\{x\}}$ partition $U$ so that $U$ is also symmetric. ∎

**Proposition 4.1.44 (Symmetric Open Cover $\implies$ Symmetric)**
*Let $X$ be a topological space with open cover $\bigcup_{i=1}^n U_i$ such that $U_i$ are symmetric in the subspace topology. Then $X$ is symmetric.*

*Proof.* Suppose $x \in \mathrm{cl}_X(\{y\})$ then by (4.1.43) we have $x \in \mathrm{cl}_{U_i}(\{y\})$, whence $y \in \mathrm{cl}_{U_i}(\{x\})$ by (4.1.41) and $y \in \mathrm{cl}_X(\{x\})$. Therefore by (4.1.41) we have $X$ is symmetric. ∎

**Proposition 4.1.45**
*Let $X$ be a topological space. Then the following are equivalent*

*a) $X$ is both Symmetric and Kolmogorov*
*b) Every point $x \in X$ is closed.*

**Corollary 4.1.46**
*Let $X$ be a symmetric topological space. Then every point of $X_0$ is closed.*

### 4.1.7 Hausdorff Spaces

**Definition 4.1.47**
*Let $X$ be a topological space. We say that $X$ is **Hausdorff** if for all distinct pairs $x, y \in X$ there exists open neighbourhoods $U, V$ of $x$ and $y$ respectively such that $U \cap V = \emptyset$.*

*Observe that **Hausdorff** $\implies$ **Kolmogorov**, **Symmetric**, but not conversely (e.g. co-finite topology).*

**Proposition 4.1.48**
*Let $X$ be a Hausdorff topological space. Then $X$ is both Kolmogorov and Symmetric. Further singletons are closed.*

### 4.1.8 Convergent Sequences

**Proposition 4.1.49 (Convergent Sequence)**
*Let $X$ be a topological space, $(x_n)$ a sequence in $X$ and $x \in X$. Let $\mathcal{B}_x$ be a local base for $x$. Then the following are equivalent*

*a) For every $V \in \mathcal{U}_x$ there exists $N$ such that*

$$n \geq N \implies x_n \in V$$

*b) For every $V \in \mathcal{B}_x$ there exists $N$ such that*

$$n \geq N \implies x_n \in V$$

*c) The function*

$$\begin{aligned} \mathbb{N} \cup \{\infty\} &\to X \\ n &\to x_n \\ \infty &\to x \end{aligned}$$

*is continuous at $\infty$ with respect to the topology on $\mathbb{N} \cup \{\infty\}$ given by open sets of the form $\{N, N+1, ...,\} \cup \{\infty\}$ and arbitrary subsets of $\mathbb{N}$.*

*In this case we say that $x_n \to x$.*

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**Proposition 4.1.50 (Continuous $\implies$ Sequentially Continuous)**
*Let $f : X \to Y$ be a continuous map and $x_n \to x$ a sequence in $X$. Then $f(x_n) \to f(x)$.*

*Proof.* If we regard the sequence as a continuous map $\mathbb{N} \cup \infty$ (4.1.49) then the result follows from (4.1.26). ∎

**Proposition 4.1.51 (Limits are Unique in Kolmogorov Spaces)**
*Let $X$ be a Kolmogorov topological space. Then limits of sequences are unique.*

**Proposition 4.1.52 (Sequentially Continuous $\implies$ Continuous)**
*Let $X, Y$ be topological spaces, $f : X \to Y$ a map and $x \in X$ a point with a* countable *local base $\mathcal{B}_x$. Suppose that for all sequences $x_n \to x$ we have $f(x_n) \to f(x)$. Then $f$ is continuous at $x$.*

*In particular if $X$ is first countable then $f : X \to Y$ is continuous iff it is sequentially continuous.*

*Proof.* Consider a point $x \in X$ and $y \coloneqq f(x)$. By hypothesis $\mathcal{B}_x = \{U_n\}$ is countable. By replacing with the base $U_n' \coloneqq \bigcap_{k=1}^{n} U_k$ we may suppose that

$$U_1 \supseteq U_2 ... \supseteq U_n \supseteq ...$$

is decreasing. Suppose $f$ is not continuous at $x$ then by (4.1.25).d) there exists $V \in \mathcal{U}_{f(x)}$ such that $f(U_n) \setminus V \neq \emptyset$ for all $n$. Therefore we may choose $x_n \in U_n$ such that $f(x_n) \notin V$. Consider $U \in \mathcal{U}_x$ then by definition there is some $N$ such that $U_N \subseteq U$. Furthermore $n \geq N \implies U_n \subseteq U_N \subseteq U \implies x_n \in U$. Therefore $x_n \to x$. By construction $f(x) \in V$ but $f(x_n) \notin V$ for all $n$ and in particular $f(x_n) \nrightarrow f(x)$. ∎

## 4.1.9 Connected Topological Spaces

**Proposition 4.1.53 (Connected Topological Space)**
*Let $X$ be a topological space. Then the following are equivalent*

*a) $X$ cannot be written as the disjoint union $U \sqcup V$ of two non-empty open subsets $U, V \subset X$*
*b) There are no proper non-empty subsets which are both open and closed*

*In this case we say that $X$ is **connected**.*

**Proposition 4.1.54 (Connected Subset)**
*Let $X$ be a topological space and $Z \subset X$. Then the following are equivalent*

*a) $Z$ is connected in the subspace topology*
*b) $Z$ cannot be written as a disjoint $(Z \cap U) \sqcup (Z \cap V)$ with both $Z \cap U, Z \cap V \neq \emptyset$*

**Proposition 4.1.55**
*Let $X$ be a topological space and $\{Z_i\}_{i \in I}$ a family of connected subsets such that $\bigcap_{i \in I} Z_i \neq \emptyset$. Then $\bigcup_{i \in I} Z_i$ is connected.*

**Definition 4.1.56 (Connected Component)**
*Let $X$ be a topological space. We say $Z \subset X$ is a **connected component** if it is both connected and not properly contained in a connected subset.*

**Proposition 4.1.57**
*Let $X$ be a topological space. Then every connected subset $Z$ is contained in a unique connected component.*

*Proof.* Define

$$\overline{Z} \coloneqq \bigcup \{Z' \supset Z \mid Z' \subset X \text{ connected}\}$$

then by (4.1.55) this is connected. If $Z'' \supset \overline{Z} \supset Z$ is connected then by construction $Z'' \subset \overline{Z}$, so by definition $\overline{Z}$ is a connected component. Let $\overline{Z}'$ be another connected component containing $Z$, then by the same argument $\overline{Z}' \subset \overline{Z}$ and by definition $\overline{Z}' = \overline{Z}$. ∎

**Proposition 4.1.58**
*Let $X$ be a topological space, then the connected components partition $X$.*

*We denote this collection by $\pi_0(X)$.*

*Proof.* Evidently $\{x\}$ is connected so that the connected components cover $X$ by (4.1.57).

Suppose we have two connected components $Z_1, Z_2$ are not disjoint. Then by (4.1.55) $Z_1 \cup Z_2$ is connected. By maximality we have $Z_1 \cup Z_2 = Z_1 = Z_2$ as required. ∎

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### 4.1.10 Irreducible Topological Spaces

References:

- [Bou98b] (Chapter II §4.2)
- [Stacks] (Tag 004U)

**Proposition 4.1.59 (Irreducible space)**
*Let $X$ be a topological space. Then the following are equivalent*

*a) $X = Z_1 \cup Z_2$ closed implies either $Z_1 = X$ or $Z_2 = X$*
*b) $U, V \neq \emptyset \implies U \cap V \neq \emptyset$ for open sets $U, V$*
*c) $U \neq \emptyset \implies \overline{U} = X$ i.e. every non-empty open set is dense*

*and we say $X$ is irreducible.*

*Proof.* *a)* $\implies$ *b)* Suppose $U, V$ are open sets such that $U \cap V = \emptyset$. Then $X = (X \setminus U) \cup (X \setminus V)$. By hypothesis $X = (X \setminus U)$ or $X = (X \setminus V)$ whence either $U$ or $V$ is empty.

*b)* $\implies$ *a)* Similar.

*c)* $\iff$ *b)* Follows directly from (4.1.24) ■

**Proposition 4.1.60 (Irreducible Subset)**
*Let $Y \subset X$ be a subset of a topological space. Then the following conditions on $Y$ are equivalent*

*a) $Y$ is irreducible in the subspace topology.*
*b) $Y \subseteq Z_1 \cup Z_2 \implies Y \subseteq Z_1$ or $Y \subseteq Z_2$ where $Z_1, Z_2$ are closed subsets of $X$*
*c) $U \cap Y \neq \emptyset, V \cap Y \neq \emptyset \implies (U \cap V) \cap Y \neq \emptyset$ for $U, V$ open*

*and we say $Y$ is an **irreducible subset**.*

*Proof.* *a)* $\implies$ *b)*. Suppose that $Y$ is irreducible in the subspace topology and $Y \subseteq Z_1 \cup Z_2$. This implies $Y = (Z_1 \cap Y) \cup (Z_2 \cap Y)$ is a decomposition of closed sets. So either $Z_1 \cap Y = Y$ or $Z_2 \cap Y = Y \implies Y \subseteq Z_1$ or $Y \subseteq Z_2$ as required.

*b)* $\implies$ *a)*. Suppose that $Y = (Z_1 \cap Y) \cup (Z_2 \cap Y)$. Then $Y \subseteq Z_1 \cup Z_2$, and for example $Y \subseteq Z_1$, which implies $Z_1 \cap Y = Y$. ■

**Proposition 4.1.61**
*Let $Y \subset X$ be a **closed** subset then the following are equivalent*

*a) $Y$ is an irreducible subset*
*b) $Y = Z_1 \cup Z_2 \implies Y = Z_1$ or $Y = Z_2$ where $Z_1, Z_2$ are closed subsets of $X$*
*c) $Y \subseteq Z_1 \cup Z_2 \implies Y \subseteq Z_1$ or $Y \subseteq Z_2$ where $Z_1, Z_2$ are closed subsets of $X$*

*In other words in the lattice of closed subsets, the irreducible subsets are precisely the join-prime subsets.*

*Proof.* *a)* $\implies$ *b)*. Clearly $Z_1, Z_2$ are also closed subsets of $Y$, so the result follows by definition.

*b)* $\iff$ *c)*. This is (2.4.2).

*c)* $\implies$ *a)*. This was already proven in (4.1.60). ■

**Remark 4.1.62**
*Singletons $\{x\}$ are always irreducible.*

**Definition 4.1.63 (Irreducible Component)**
*We say that $Y$ is an irreducible component if it is a maximal irreducible subset.*

**Proposition 4.1.64**
*Let $f : X \to Y$ be a continuous map and $Z \subset X$ irreducible. Then $f(Z)$ is irreducible in $Y$.*

*Proof.* Suppose $f(Z) \subset Y_1 \cup Y_2$ for $Y_1, Y_2$ closed. Then $Z \subset f^{-1}(f(Z)) \subset f^{-1}(Y_1) \cup f^{-1}(Y_2)$. Then by (4.1.60).b) wlog $Z \subset f^{-1}(Y_1)$ therefore $f(Z) \subset f(f^{-1}(Y_1)) \subset Y_1$. Therefore by the same criterion $f(Z)$ is irreducible. ■

**Proposition 4.1.65**
*Let $Y \subset X$. Then $Y$ is irreducible iff $\overline{Y}$ is.*

*Proof.* This follows from (4.1.60).b) and (4.1.19) that $Y \subset Z \iff \overline{Y} \subset Z$. ■

**Proposition 4.1.66 (Decomposition into Irreducible Components)**
*A topological space $X$ may be decomposed into irreducible components. More precisely*

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*a) Every irreducible component is closed*
*b) Every irreducible closed subset is contained in an irreducible component*
*c) $X$ is the union of irreducible components*

*Proof.* We prove each in turn

a) By (4.1.65) $\overline{Y}$ is irreducible and closed, so by maximality $Y = \overline{Y}$ is closed.
b) We may show that the lattice of closed subsets is chain complete so we may use to show that the lattice of irreducible closed subsets is also chain complete (2.4.4). Therefore we may apply (2.1.56) to find an irreducible component.
c) As $\{x\}$ is irreducible every element is contained in an irreducible component by $b$).

■

**Corollary 4.1.67**
*For $x \in X$ the closure $\overline{\{x\}}$ is an irreducible closed subset.*

**Corollary 4.1.68**
*$X$ is irreducible if and only if it has a single irreducible component.*

**Definition 4.1.69 (Generic Point)**
*Let $Z$ be an irreducible closed subset of $X$, then we say $\eta \in X$ is a generic point of $Z$ if $Z = \overline{\{\eta\}}$.*

**Proposition 4.1.70**
*Let $X$ be an irreducible space then every open subset $U$ is irreducible and connected.*

*Proof.* By (4.1.59) $U$ is dense in $X$. Therefore by (4.1.65) $U$ is irreducible.

Suppose that $U$ is not connected then by (4.1.54) $U = (U \cap V) \sqcup (U \cap W)$ with $V, W$ open subsets of $X$ and $U \cap V, U \cap W$ non-empty. Then as $U$ is irreducible we have $U \cap V \cap W \neq \emptyset$, a contradiction. ■

**Proposition 4.1.71**
*Let $X = \bigcup_{i=1}^{n} X_i$ where $X_i$ are irreducible closed subsets, and mutually incomparable. Then the $X_i$ are the irreducible components of $X$.*

*Proof.* For any irreducible closed subset $E$ we have by (4.1.61) $E \subset X_i$ for some $i$. In particular every irreducible component is contained in some $X_i$. On the other hand every $X_j$ is contained in an irreducible component by (4.1.66). By incomparability we deduce that each $X_i$ is an irreducible component, and that the set is complete. ■

**Proposition 4.1.72**
*Let $Y \subset X$ such that $Y = \bigcup_{i=1}^{n} Y_i$ where $Y_i$ are the irreducible components of $Y$. Then the set of irreducible components of $\overline{Y}$ is $\left\{\overline{Y_i}\right\}_{i=1\ldots n}$, and these are distinct.*

*Proof.* By (4.1.65) $\overline{Y_i}$ are irreducible subsets which, by (4.1.23) cover $\overline{Y}$. By (4.1.66) $Y_i$ is closed in $Y$ and so by (4.1.21) $\overline{Y_i} \cap Y = Y_i$. This shows that the $\overline{Y_i}$ are distinct and also incomparable. Then (4.1.71) shows that these must be the irreducible components. ■

**Proposition 4.1.73**
*Let $U \subset X$ be an open subset. There is an order isomorphism*

$$\begin{aligned} \{Y \subset U \mid Y \text{ irreducible, closed and non-empty}\} &\longleftrightarrow \{Z \subset X \mid Z \text{ irreducible and closed and } Z \cap U \neq \emptyset\} \\ Y &\to \operatorname{cl}_X(Y) \\ Z \cap U &\leftarrow Z \end{aligned}$$

*which restricts to irreducible components.*

*Proof.* By (4.1.65) the first map is well-defined. Conversely $Z \cap U$ is closed in $U$ and open in $Z$, so irreducible by (4.1.70).

By (4.1.59) $Z \cap U$ is dense in $Z$, that is $Z = \operatorname{cl}_Z(Z \cap U) \overset{(4.1.21)}{=} \operatorname{cl}_X(Z \cap U) \cap Z \implies Z \subset \operatorname{cl}_X(Z \cap U)$, and so by minimality they are equal.

By (4.1.21) $\operatorname{cl}_X(Y) \cap U = \operatorname{cl}_U(Y) = Y$. Therefore the maps are mutually inverse and order preserving. ■

**Proposition 4.1.74**
*Suppose $X$ is a topological space with open cover $\bigcup_{i \in I} U_i$ such that*

*a) $U_i$ is irreducible for all $i \in I$ and*

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*b)* $U_i \cap U_j$ *is non-empty for all* $i, j \in I$.

*Then* $X$ *is irreducible.*

*Proof.* Suppose that $X = Z_1 \cup Z_2$. Pick some $i \in I$ then by (4.1.60) wlog $U_i \cap Z_1 \neq \emptyset$. Similarly for all $j \in J$ by assumption $U_i \cap U_j$ is a non-empty open subset of $U_j$ and therefore dense (4.1.59). Further $U_i \cap U_j$ is contained in the closed subset $Z_1 \cap U_j$. We conclude that $U_j \subset Z_1 \cap U_j$. Therefore $X \subset Z_1$. By (4.1.60) we further conclude that $X$ is irreducible. ∎

## 4.1.11 Noetherian Topological Spaces

**Definition 4.1.75 (Noetherian)**
*A topological space* $X$ *is* ***Noetherian*** *if the lattice of closed subsets is Artinian (i.e. satisfies the descending chain condition).*

**Proposition 4.1.76 (Decomposition into Irreducibles)**
*Let* $X$ *be a Noetherian topological space. Then every closed subset* $Y$ *may be expressed uniquely as a finite, incomparable union of irreducible closed subsets. These are precisely the irreducible components of* $Y$.

*In particular* $X$ *has only finitely many irreducible components.*

*Proof.* The lattice of closed subsets is distributive and Artinian by definition. Therefore the result follows from (4.1.61) and (2.4.7). ∎

**Proposition 4.1.77**
*Let* $X$ *be a Noetherian topological space and* $Y \subset X$. *Then* $Y$ *is Noetherian.*

**Proposition 4.1.78**
*Let* $X = \bigcup_{i=1}^{N} X_i$ *be a topological space such that* $X_i$ *is Noetherian under the subspace topology. Then* $X$ *is Noetherian.*

*Proof.* Let $\{Z_n\}_{n\in\mathbb{N}}$ be a descending chain of closed subsets of $X$. By definition $\{Z_n \cap X_i\}_{n\in\mathbb{N}}$ terminates for $n \geq m_i$. Then for $n \geq \max(m_1, ..., m_N)$ the sequence $\{Z_n\}_{n\in\mathbb{N}}$ also terminates. ∎

**Proposition 4.1.79**
*Let* $X$ *be a Noetherian topological space and* $U \subset X$ *an open subset. Then there is a bijection*

$$\begin{aligned} \{Y \subset U \mid Y \text{ closed and non-empty}\} &\longleftrightarrow \{Z \subset X \mid Z \text{ closed and every irreducible component meets } U\} \\ Y &\to \ \mathrm{cl}_X(Y) \\ Z \cap U &\leftarrow \ Z \end{aligned}$$

*which is order preserving and restricts to closed irreducible subsets and irreducible components.*

*Proof.* By (4.1.77) and (4.1.76) $Y = Y_1 \cup ... \cup Y_n$ where $Y_i$ are the irreducible components of $Y$. Then by (4.1.72) the irreducible components of $\overline{Y}$ are $\overline{Y_i}$, which evidently intersect $U$. Furthermore by definition $Z \cap U$ is closed in $U$ and the maps are well-defined.

By (4.1.73) the maps restrict to closed irreducible subsets (resp. components) and are bijections, in particular for $Z$ irreducible closed we have $\overline{Z \cap U} = Z$.

By (4.1.21) $\overline{Y} \cap U = Y$. For the case $Z$ is closed then as before $Z = Z_1 \cup ... \cup Z_n$ is a decomposition into irreducible components. By assumption $Z_i \cap U \neq \emptyset$ for all $i = 1...n$. Therefore by the irreducible case (4.1.73) and (4.1.23)

$$\overline{Z \cap U} = \overline{Z_1 \cap U} \cup ... \cup \overline{Z_n \cap U} = Z_1 \cup ... \cup Z_n = Z.$$

Therefore the maps are mutually inverse as required. ∎

## 4.1.12 Krull Dimension

References:

- Éléments de géométrie algébrique IV [Gro64] (Chap. 0 §**14.4.1**)
- Some remarks on biequidimensionality of topological spaces and Noetherian schemes [Hei17]

For a Noetherian topological space $X$ the closed subsets form an Artinian, distributive lattice. Furthermore the irreducible subsets are precisely the join-prime elements by (4.1.61). Therefore we may use the notions of Krull Lattice developed in Foundations.tex's Krull Dimension section (2.5.2).

**Definition 4.1.80 (Chain of irreducibles)**
*Let* $X$ *be a Noetherian topological space. A* ***chain*** *of irreducible closed subsets*

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$$Z_0 \subsetneq Z_1 \subsetneq ... \subsetneq Z_n$$

*is said to have* ***length*** $n$*. A chain is* ***saturated*** *if there is no proper refinement, that is if* $Y$ *is irreducible then*

$$Z_i \subseteq Y \subseteq Z_{i+1} \implies Y = Z_i \text{ or } Y = Z_{i+1}.$$

*If in addition* $Z_n$ *(resp.* $Z_0$*) is maximal (resp. minimal) then the chain is* ***maximal****.*

**Definition 4.1.81 (Krull Lattice of Closed Subsets)**
*Let* $X$ *be a topological space.*

- *The* ***Krull dimension*** $\dim(X)$ *of* $X$ *is the maximal length of all chains of irreducible closed subsets. Note this may be* $\infty$*.*
- *The* ***height*** *or* ***codimension*** *of an irreducible closed subset* $Y \subseteq X$*, denoted* $\operatorname{codim}(Y, X)$*, is the maximal length of chains of irreducible subsets containing* $Y$*.*
- *When* $Y$ *is not irreducible we write*

$$\operatorname{codim}(Y, X) = \inf_\alpha \operatorname{codim}(Y_\alpha, X)$$

*where* $Y_\alpha$ *varies among the irreducible components of* $Y$*.*

*If* $\dim(X) < \infty$ *then we say* $X$ *is* ***finite-dimensional****. In this case it's clear the closed subsets of a topological space form a Krull Lattice where the irreducible closed subsets are precisely the join-prime elements of the lattice.*

*Note any saturated chain for* $Y \subset X$ *must start at* $Y$ *and terminate at an irreducible component of* $X$*. In particular if* $X$ *is irreducible then a saturated chain must terminate at* $X$*.*

**Proposition 4.1.82 (Extending Chains)**
*Let* $X$ *be a finite-dimensional topological space. Then*

*a) Every chain is contained in a saturated chain with the same endpoints*
*b) Every chain is contained in a maximal chain*

**Proposition 4.1.83 (Simple properties of (co-)dimension)**
*Let* $X$ *be a topological space. Then the following properties hold*

*a)* $\dim(X) = \sup_\alpha \dim(X_\alpha)$ *where* $X_\alpha$ *are the irreducible components of* $X$
*b)* $\operatorname{codim}(Y, X) = \sup_\alpha \operatorname{codim}(Y, X_\alpha)$ *where* $X_\alpha$ *are the irreducible components of* $X$ *containing* $Y$
*c)* $\dim(Y) + \operatorname{codim}(Y, X) \leq \dim(X)$ *for every closed subset* $Y \subset X$
*d)* $\operatorname{codim}(Y, Z) + \operatorname{codim}(Z, T) \leq \operatorname{codim}(Y, T)$ *for* $Y \subset Z \subset T$ *irreducible closed subsets*
*e)* $Y \subsetneq Z$ *is a saturated chain of irreducible closed subsets if and only if* $\operatorname{codim}(Y, Z) = 1$*.*
*f) Let* $Y$ *be a closed subset of* $X$*. Then* $\operatorname{codim}(Y, X) = 0 \iff$ *every irreducible component of* $Y$ *is an irreducible component of* $X$*. In particular if* $X$ *is irreducible then* $\operatorname{codim}(Y, X) = 0 \iff Y = X \iff \dim(X) = \dim(Y)$*.*

*In particular if* $X$ *is finite-dimensional then all codimensions are also finite.*

**Proposition 4.1.84**
*Let* $X$ *be an irreducible topological space and* $Y \subset X$ *a closed subset. Then* $\dim(Y) \leq \dim(X)$ *with equality iff* $Y = X$*.*

*Proof.* The inequality follows from (4.1.83).b). If $\dim(Y) = \dim(X)$ then by c) $\operatorname{codim}(Y, X) = 0$ and so by f) we have $Y = X$. ∎

**Definition 4.1.85 (Properties)**
*Let* $X$ *be a topological space of finite dimension. Then we say* $X$ *is*

- ***Equidimensional*** *if all irreducible components of* $X$ *have the same dimension*
- ***Equicodimensional*** *if* $\operatorname{codim}(Y, X)$ *is constant as* $Y$ *varies over minimal irreducible subsets of* $X$
- ***Biequidimensional*** *if all maximal chains of irreducible subsets have the same length.*
- ***Quasi-Biequidimensional*** *if every irreducible component is biequidimensional*
- ***Catenary*** *if any two saturated chains with the same endpoints, say* $Y$ *and* $Z$*, have the same length, namely* $\operatorname{codim}(Y, Z)$

*Observe that* ***quasi-biequidimensional*** *+* ***equidimensional*** $\iff$ ***biequidimensional*** *and* ***irreducible*** $\implies$ ***equidimensional***

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**Proposition 4.1.86 (Equivalent characterisations of biequidimensional)**
*Suppose $X$ is a topological space of finite dimension. Then the following are equivalent*

*a) $X$ is quasi-biequidimensional*
*b) $X$ is catenary and every irreducible component is equicodimensional*
*c) $X$ satisfies the codimension formula for $Z \subset Y$ irreducible*

$$\dim(Y) = \dim(Z) + \operatorname{codim}(Z, Y)$$

*d) $X$ satisfies c) in the case $\operatorname{codim}(Z, Y) = 1$*

*Furthermore for irreducible subsets $Z \subset Y$*

$$\operatorname{codim}(Z) = \operatorname{codim}(Z, Y) + \operatorname{codim}(Y)$$

*Proof.* This is a translation of (2.5.11) to the topological case. ∎

**Proposition 4.1.87 (Codimension Formula)**
*Suppose $X$ is a quasi-biequidimensional topological space. Then for every subset $Z \subset X$ which is irreducible and closed, and every closed subset $Y \subset Z$ the following relations hold.*

$$\dim(Z) = \dim(Y) + \operatorname{codim}(Y, Z)$$
$$\operatorname{codim}(Y) = \operatorname{codim}(Y, Z) + \operatorname{codim}(Z)$$

*Further every closed subset is also quasi-biequidimensional.*

*Proof.* This follows from (2.5.13) and (2.5.15). ∎

**Proposition 4.1.88**
*Suppose $X = \bigcup_{i \in I} U_i$. Then $\dim(X) = \sup_{i \in I} \dim(U_i)$.*

*Proof.* By (4.1.73) a chain of irreducible closed subsets of $U_i$ lifts to a chain of the same length in $X$. Therefore $\dim(U_i) \leq \dim(X)$ for all $i \in I$ whence $\sup_{i \in I} \dim(U_i) \leq \dim(X)$. Similarly given a chain of irreducible closed subsets

$$X_0 \subsetneq ... \subsetneq X_n \subseteq X$$

choose $i$ such that $X_0 \cap U_i \neq \emptyset$. Then (4.1.79) shows that this restricts to a chain of the same length in $U_i$. Therefore $\dim(X) \leq \dim(U_i) \leq \sup_{i \in I} \dim(U_i)$. ∎

**Proposition 4.1.89**
*Let $X$ be a catenary topological space and $U \subset X$ an open subset. Then $U$ is catenary.*

*Proof.* Consider $Z \subset Y \subset U$ irreducible closed subsets. Then for any saturated chain

$$Z = Z_0 \subsetneq ... \subsetneq Z_n = Y$$

we have by (4.1.73)

$$\overline{Z} = \overline{Z_0} \subsetneq ... \subsetneq \overline{Z_n} = \overline{Y}$$

a saturated chain in $X$. As $X$ is catenary we find $n = \operatorname{codim}\left(\overline{Z}, \overline{Y}\right)$. As the chain was arbitrary we find $U$ is catenary. ∎

**Lemma 4.1.90**
*Suppose that $X$ is a topological space and $x \in X$ is quasi-closed. Then $\dim\left(\overline{\{x\}}\right) = 0$.*

*Conversely suppose $Z$ is an irreducible closed set of dimension 0. Then $Z = \overline{\{z\}}$ for all $z \in Z$ and $Z$ consists of quasi-closed elements.*

*Proof.* By (4.1.67) $\overline{\{x\}}$ is irreducible. Suppose $\emptyset \neq Z \subset \overline{\{x\}}$ is closed then $y \in Z \implies x \in \overline{\{y\}} \subset Z$ whence $\overline{\{x\}} = Z$. This shows that $\dim\left(\overline{\{x\}}\right) = 0$.

Conversely if $\dim(Z) = 0$ then $\overline{\{z\}} \subset Z$ is an irreducible subset for all $z \in Z$, whence they are equal. This also shows that $z$ is quasi-closed. ∎

**Proposition 4.1.91**
*Let $X$ be a Noetherian topological space and $Z$ a closed subset. Then*

$$\dim(Z) = 0 \iff Z = \overline{\{z_1\}} \cup ... \cup \overline{\{z_n\}}$$

*for some $z_1, ..., z_n \in Z$ quasi-closed.*

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*Proof.* By (4.1.76) $Z = Z_1 \cup ... \cup Z_n$ for $Z_i$ is a decomposition into irreducible components. By (4.1.83).a) $\dim(Z_i) = 0$, so by (4.1.90) $Z_i = \overline{\{z_i\}}$.

Conversely $Z = \overline{\{z_1\}} \cup ... \cup \overline{\{z_n\}}$ is a decomposition into irreducible closed subsets which are incomparable (4.1.90). By (4.1.71) these are the irreducible components. Therefore

$$\dim(Z) \overset{(4.1.83).a)}{=} \sup_i \dim\left(\overline{\{z_i\}}\right) \overset{(4.1.90)}{=} 0$$

■

**Proposition 4.1.92 (Curves have co-finite topology)**
*Let $X$ be an irreducible, Noetherian and symmetric topological space such that $\dim(X) = 1$. Then $X_0$ has the cofinite topology.*

*Proof.* We may reduce to the case $X = X_0$ is Kolmogorov, then by (4.1.45) every point $x \in X$ is closed.

Suppose $Z \subset X$ is finite then $Z = \bigcup_{z \in Z} \{z\}$ is a finite union of closed subsets and therefore closed. Conversely if $Z \subset X$ is closed and proper then by (4.1.83).f) $\dim(Z) = 0$ and therefore $Z$ is finite by (4.1.91). ■

**Lemma 4.1.93 (Cofinite Topology is Quasi-Compact)**
*Let $X$ be a set equipped with the cofinite topology. Then $X$ is quasi-compact.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

Let $\{U_i\}_{i \in I}$ be an open cover of $X$ and choose any nonempty $U_{i_0}$ from it. Then $X \setminus U_{i_0}$ is finite by definition of the cofinite topology, say $\{x_1, ..., x_n\}$. For each $k$ choose $U_{i_k} \ni x_k$ from the cover. Then $\left\{U_{i_0}, U_{i_1}, ..., U_{i_n}\right\}$ is a finite subcover. ■

**Proposition 4.1.94**
*Let $X$ be a symmetric topological space. Then there is a correspondence*

$$\begin{aligned} X_0 &\longleftrightarrow \{W \subset X \mid \dim(W) = 0 \text{ closed and irreducible}\} \\ x &\longrightarrow \overline{\{x\}} \end{aligned}$$

## 4.1.13 Product Topology

**Definition 4.1.95 (Product Topology)**
*Let $\{X_i\}_{i \in I}$ be a family of topological spaces. The **product topology** is the topology generated by the base of open sets of the form*

$$\prod_{i \in I} U_i$$

*where $U_i \subset X_i$ is open and only finitely many are proper subsets of $X$.*

**Proposition 4.1.96**
*Let $\{X_i\}_{i \in I}$ be family of topological spaces. The family of sets given in (4.1.95) is a base. The topology generated is the smallest family of subsets for which the projection maps*

$$\pi_i : \prod_{i \in I} X_i \to X_i$$

*are continuous. Furthermore if each $X_i$ is Hausdorff (resp. Kolmogorov), then so is the product.*

**Proposition 4.1.97 (Product of Sequences)**
*Let $X_1, ..., X_m$ be a family of topological spaces and $a_n^i \to \alpha^i$ for $i = 1...m$. Then $(a_n^1, ..., a_n^m) \to (\alpha^1, ..., \alpha^m)$.*

**Proposition 4.1.98 (Diagonal is Closed)**
*Let $X$ be a topological space. Then $X$ is Hausdorff if and only if $\{(x, x) \mid x \in X\}$ is closed in $X \times X$.*

**Proposition 4.1.99**
*Let $f, g : X \to Y$ be continuous maps and $Y$ a Hausdorff space. Then*

$$\{x \in X \mid f(x) = g(x)\}$$

*is closed.*

*Proof.* The level set is $(f \times g)^{-1}(\Delta_Y)$ which by (4.1.98) is closed. ■

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### 4.1.14 Quasi-Compactness

**Definition 4.1.100 (Cluster Point of a Sequence)**
*Let $(x_n)$ be a sequence in a topological space $X$. We say that $x \in X$ is*

*a) a **cluster point** (or **accumulation point**) of $x_n$ if for every $U \in \mathcal{O}_x$ the set $\{m \in \mathbb{N} \mid x_m \in U\}$ is infinite*
*b) a **subsequential limit** of $x_n$ if there exists some subsequence $x_{n_k} \to x$*

*In general the latter implies the former, and if $X$ is first countable the concepts are equivalent as we will see.*

**Lemma 4.1.101 (Cluster Point is Subsequential Limit Point)**
*Let $X$ be a first countable topological space and $x$ a cluster point of a sequence $(x_n)$. Then there exists some subsequence $x_{n_k}$ such that $x_{n_k} \to x$.*

*Proof.* Let $(U_n)$ be a local base at $x$. By replacing it with $U_n' \coloneqq U_1 \cap ... \cap U_n$ we may assume it is without loss of generality decreasing. Define $n_0 = 0$ and $n_k$ inductively by

$$n_k \coloneqq \min\{m > n_{k-1} \mid x_m \in U_k\}$$

which is well-defined precisely because $x$ is a cluster point. Then $x_{n_k} \in U_k$ which shows that $x_{n_k} \to x$ as required. ■

**Definition 4.1.102**
*Let $X$ be a topological space. We say it is*

- ***Quasi-Compact** if every open cover of $X$ has a finite subcover*
- ***Countably Quasi-Compact** if every* countable *open cover of $X$ has a finite subcover*
- ***Sequentially Quasi-Compact** if every sequence $(x_n)$ in $X$ has a convergent subsequence*

*If in addition $X$ is Hausdorff (e.g. a metric space) we say that $X$ is compact (resp. countably compact, sequentially compact).*

**Proposition 4.1.103 (Criteria for Countable Compactness)**
*Let $X$ be a topological space. Then the following are equivalent*

*a) $X$ is countably quasi-compact*
*b) Let $\{Z_n\}_{n\in\mathbb{N}}$ be a countable collection of closed subsets such that every finite set has non-empty intersection. Then the intersection $\bigcap_{n\in\mathbb{N}} Z_n$ is non-empty*
*c) For every increasing sequence of proper open subsets*

$$U_1 \subseteq U_2 \subseteq ... \subseteq U_n \subseteq ...$$

*the union $\bigcup_{n=1}^{\infty} U_n$ is proper.*
*d) For every decreasing sequence of non-empty closed subsets*

$$Z_1 \supseteq Z_2 \supseteq ... \supseteq Z_n \supseteq ...$$

*the intersection $\bigcap_{n=1}^{\infty} Z_n$ is non-empty*

*Proof.* $a) \iff b)$ and $c) \iff d)$ are simple consequences of De Morgan's laws. For $a) \implies c)$ suppose $\bigcup_{n=1}^{\infty} U_n = X$ then by assumption $U_{n_1} \cup ... \cup U_{n_k} = X$, and by the increasing assumption $U_{n_k} = X$ which is a contradiction.

For $c) \implies a)$, suppose we have an open cover $X = \bigcup_{n=1}^{\infty} U_n$ define the increasing open cover

$$U_n' \coloneqq \bigcup_{i=1}^{n} U_i$$

by the contrapositive $U_n'$ is not proper for some $n$, which means precisely that there is a finite subcover. ■

**Proposition 4.1.104**
*Let $X$ be a topological space. Then sequentially quasi-compact $\implies$ countably quasi-compact. Under the assumption of first countability the converse holds.*

*Proof.* $\implies$) Consider a countable open cover $X = \bigcup_{n=1}^{\infty} U_n$ with no finite subcover. Then there exists $x_n \in X \setminus \bigcup_{i=1}^{n} U_i$. By assumption there is a convergent subsequence $x_{n_k} \to x$. We have $x \in U_N$ for some $N$ but $n_k \geq N \implies x_{n_k} \notin U_N$.

$\impliedby$) Consider a sequence $(x_n)$ and define

$$Z_n \coloneqq \overline{\{x_k : k > n\}}$$

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which is a decreasing sequence of non-empty closed subsets. By the countable compactness criteria (4.1.103) $Z \coloneqq \bigcap_{n=1}^{\infty} Z_n$ is non-empty. However we may show this is precisely the set of cluster points of $(x_n)$, i.e. there exists a cluster point $x \in Z$. By (4.1.101) there exists a subsequence such that $x_{n_k} \to x$ as required. ■

**Lemma 4.1.105 (Lindelof Lemma)**
*Let $X$ be a second countable topological space. Then every open cover has a countable subcover.*

*Proof.* Let $\mathcal{B}$ be a countable base and $\{U_\alpha\}_{\alpha\in\mathcal{A}}$ an open cover of $X$. Note for every $U_\alpha$ we have some subfamily $\mathcal{B}_\alpha \subset \mathcal{B}$ such that $U_\alpha = \bigcup \mathcal{B}_\alpha$. Consider $\mathcal{B}' \coloneqq \bigcup\{B_\alpha \mid \alpha \in \mathcal{A}\} \subset \mathcal{B}$. By definition for every $V \in \mathcal{B}'$ we have $V \subset U_\alpha$ for some $\alpha \in \mathcal{A}$. In other words there exists a map $f : \mathcal{B}' \to \mathcal{A}$ such that $V \subset U_{f(V)}$. We may verify that $\{U_\alpha\}_{\alpha\in \mathrm{Im}(f)}$ is a countable subcover. ■

We summarise the relationships between the concepts

**Proposition 4.1.106**
*Let $X$ be a topological space. Then*

*a) Quasi-Compact, Sequentially Quasi-Compact $\implies$ Countably Quasi-compact*
*b) First Countable and (Countably) Quasi-Compact $\implies$ Sequentially Quasi-Compact*
*c) Second Countable and Countably or Sequentially Quasi-Compact $\implies$ Quasi-Compact*

*Proof.*
a) The first implication is obvious and the second implication is (4.1.104)
b) This is (4.1.104)
c) This follows from (4.1.105) and *a*)

■

**Proposition 4.1.107 (Compact subsets of Hausdorff Spaces are Closed)**
*Let $X$ be a Hausdorff topological space and $Y \subset X$ a subset compact with respect to the subspace topology. Then $Y$ is closed.*

*Proof.* Fix $x \in X \setminus Y$. Then for every $y \in Y$ by assumption there exists disjoint open subsets $x \in U_y$ and $y \in V_y$. Then $\{V_y \cap Y \mid y \in Y\}$ is an open cover for $Y$, so there exists a finite set of points $y_1, ..., y_n$ such that $Y \subseteq V_{y_1} \cup ... \cup V_{y_n}$. Furthermore

$$U \coloneqq U_{y_1} \cap ... \cap U_{y_n}$$

is an open neighbourhood of $x$ disjoint from each $V_{y_i}$ and therefore disjoint from $Y$. By (4.1.19) we see that $x \notin \overline{Y}$, which shows $Y = \overline{Y}$, i.e. $Y$ is closed. ■

**Proposition 4.1.108 (Closed subsets of Compact Spaces are Compact)**
*Let $X$ be a quasi-compact topological space and $Y \subset X$ a closed subset. Then $Y$ is quasi-compact under the subspace topology.*

*Proof.* Let $\mathcal{U} \coloneqq \{U_\alpha\}_{\alpha\in\mathcal{A}}$ be an open cover for $Y$. Then by definition $U_\alpha = U'_\alpha \cap Y$, where $U'_\alpha \subset X$ are open. Evidently

$$\mathcal{U}' : \{U'_\alpha\}_{\alpha\in\mathcal{A}} \cup \{X \setminus Y\}$$

is an open cover of $X$. Therefore by assumption it has a finite subcover, which immediately yields a finite subcover of $Y$ as required. ■

**Proposition 4.1.109 (Image of a Compact Space is Compact)**
*Let $f : X \to Y$ be a continuous map of topological spaces where $X$ is quasi-compact. Then the image $f(X)$ is quasi-compact.*

*Further if $Y$ is Hausdorff then $f$ is an open map (i.e. $f(U)$ is open for every open $U \subset X$). If in addition $f$ is injective then it is a homeomorphism onto its image.*

*Proof.* We may assume without loss of generality that $f(X) = Y$. Let $\{U_\alpha\}_{\alpha\in\mathcal{A}}$ be an open cover of $Y$. In other words $f(X) \subseteq \bigcup_{\alpha\in\mathcal{A}} U_\alpha \implies X \subseteq \bigcup_{\alpha\in\mathcal{A}} f^{-1}(U_\alpha)$. By assumption we have a finite subcover

$$X \subseteq f^{-1}\left(U_{\alpha_1}\right) \cup ... \cup f^{-1}\left(U_{\alpha_n}\right)$$

which implies

$$f(X) \subseteq U_{\alpha_1} \cup ... \cup U_{\alpha_n}$$

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since $f$ is assumed to be surjective, and we see that $f(X)$ is compact. ∎

Analogously we have similar results for sequentially compact spaces, which may be conceptually simpler.

**Proposition 4.1.110**
*Let $X$ be a first countable Kolmogorov space and $Y \subset X$ a subset of a sequentially compact with respect to the subspace topology. Then $Y$ is closed.*

*Proof.* Consider $x \in \overline{Y}$ then by (4.1.19) there is some sequence $x_n \to x$. Further by assumption there is some subsequence $x_{n_k} \to y \in Y$. By uniqueness of limits (4.1.51) we have $x = y$, and in particular $\overline{Y} = Y$ is closed. ∎

**Proposition 4.1.111**
*Let $X$ be a sequentially compact topological space, and $Y \subset X$ a closed subset. Then $Y$ is sequentially compact.*

*Proof.* Let $x_n \in Y$ be a sequence. By assumption it has a convergent subsequence $x_{n_k} \to x$ where $x \in X$. By (4.1.18) $x$ is an adherent point of $Y$ and therefore lies in $Y$ by (4.1.19). ∎

**Proposition 4.1.112**
*Let $f : X \to Y$ be a continuous map of topological spaces where $X$ is sequentially compact. Then the image $f(X)$ is sequentially compact.*

*Proof.* Let $y_n = f(x_n)$ be a sequence in $f(X)$. Then by assumption $x_{n_k} \to x$, and therefore by (4.1.50) $y_{n_k} \to f(x)$. ∎

**Proposition 4.1.113**
*Let $X$ be a topological space with an open cover $X = \bigcup_{i=1}^{n} U_i$ such that $U_i$ is quasi-compact under the subspace topology. Then $X$ is quasi-compact.*

*Proof.* Let $X = \bigcup_{\alpha \in \mathcal{A}} V_\alpha$ then $U_i = \bigcup_{\alpha \in \mathcal{A}} (V_\alpha \cap U_i)$. By assumption there is some finite subset $\mathcal{A}_i \subset \mathcal{A}$ such that $U_i = \bigcup_{\alpha \in \mathcal{A}_i} (V_\alpha \cap U_i)$. Then

$$X = \bigcup_{\alpha \in \mathcal{A}_1 \cup \ldots \cup \mathcal{A}_n} V_\alpha$$

Therefore $X$ is quasi-compact. ∎

**Proposition 4.1.114 (Noetherian $\iff$ Hereditarily Quasi-Compact)**
*Let $X$ be a topological space. Then $X$ is Noetherian if and only if every open subset is quasi-compact.*

*Proof.* $\implies$) By (4.1.77) every open subset is Noetherian, so it is enough to show that $X$ is quasi-compact. By definition the lattice of open subsets is Noetherian and by (2.1.63) this holds iff every family of open subsets has a maximal element.

Suppose that we have an open cover $X = \bigcup_{\alpha \in \mathcal{A}} U_\alpha$. Let $\mathcal{F}$ be the family of open subsets expressible as a finite union of sets $U_\alpha$. By assumption it has a maximal element

$$U := U_{\alpha_1} \cup \ldots \cup U_{\alpha_n}$$

Suppose that $U \subsetneq X$ is proper then choose $x \in X \setminus U$. By hypothesis there is some $\alpha_{n+1}$ such that $x \in U_{\alpha_{n+1}}$. Then $U \subsetneq U \cup U_{\alpha_{n+1}} \in \mathcal{F}$, contradicting maximality. Therefore $U = X$ and $X$ is quasi-compact as required.

$\impliedby$) Omitted. ∎

**Proposition 4.1.115**
*Let $X, Y$ be quasi-compact topological spaces. Then $X \times Y$ is quasi-compact.*

*Proof.* Let $\{U_\lambda\}_{\lambda \in \Lambda}$ be an open cover of $X \times Y$. Therefore for each $(x, y)$ there is a $\lambda(x, y)$ such that $(x, y) \in U_{\lambda(x,y)}$. By definition of the product topology (4.1.95) we may find open sets $V_{(x,y)} \subset X$ and $W_{(x,y)} \subset Y$ such that $(x, y) \in V_{(x,y)} \times W_{(x,y)} \subset U_{\lambda(x,y)}$.

Fix $x$ then the family $\left\{W_{(x,y)}\right\}_{y \in Y}$ is an open cover of $Y$. So we may find a finite subset $J_x \subset Y$ such that $\left\{W_{(x,y)}\right\}_{y \in J_x}$ covers $Y$.

Consider the open sets

$$V'_x := \bigcap_{y \in J_x} V_{(x,y)}$$

which evidently cover $X$. Then there is a finite subset $I \subset X$ such that $\{V'_x\}_{x \in I}$ covers $X$.

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Then we claim that the family $\left\{U_{\lambda(x,y)}\right\}_{x\in I, y\in J_x}$ is a finite subcover of $X\times Y$. For given $(x',y')\in X\times Y$ there is an $x\in I$ such that $x'\in V_x'$, and $y\in J_x$ such that $y'\in W_{(x,y)}$. Evidently $x'\in V_{(x,y)}$, whence $(x',y')\in U_{\lambda(x,y)}$ as required.■

### 4.1.15 Connectedness

**Definition 4.1.116**
*We say a topological space $X$ is **disconnected** if $X=U\cup V$ for two non-empty, disjoint open subsets $U,V$. Otherwise we say $X$ is connected.*

**Proposition 4.1.117**
*Let $X$ be a topological space. The following are equivalent*

*a) $X$ is connected*
*b) The only subsets which are both open and closed are $X$ and $\emptyset$*

**Proposition 4.1.118**
*Let $f:X\to Y$ be a continuous map and $X$ connected. Then $f(X)$ is also connected.*

### 4.1.16 Order Topology

**Definition 4.1.119 (Order Topology)**
*Let $(X,\leq)$ be a totally ordered set. Then the **order topology** is the topology generated by the subbase of half-open intervals*

$$\{y\in X\mid y<x\}\text{ and }\{y\in X\mid y>x\}$$

*for $x\in X$.*

**Proposition 4.1.120 (Base for order topology)**
*Let $(X,\leq)$ be a totally ordered set. Then the sets of the form*

$$\{y\in X\mid y<x\},\{y\in X\mid y>x\}\text{ and }\{y\in X\mid x<y<z\}$$

*for $x,z\in X$ constitute a base for the order topology.*

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## 4.2 Sheaves

Sheaves are typically used to organise sets of functions which are defined by a local differentiability condition (e.g. holomorphic, smooth, etc.). However in some cases it may be convenient to take an abstract category-theoretic approach as the proofs become more conceptual and naturally lead to further generalisations where the topological space is replaced by an arbitrary category. However for the sheafification construction we take (the standard) concrete approach as this is simpler and nicely coincides with the notion of "étale space".

In general we assume that the value category $\mathcal{C}$ admits arbitrary products and equalisers (and therefore also limits, and in particular terminal objects), as well as filtered colimits / direct limits. In some cases we may also require it to be algebraic.

**Definition 4.2.1 (Abstract Presheaf)**
*Let $X$ be a topological space, $\mathcal{B}$ a base for the topology and $\mathcal{C}$ a category. A $\mathcal{B}$-**presheaf** on $X$ is a functor*

$$\mathcal{F} : (\mathcal{B}, \supseteq) \longrightarrow \mathcal{C}$$

*from the poset category of basic open subsets to $\mathcal{C}$ ordered by reverse inclusion for which*

$$\mathcal{F}(\emptyset) = \star \text{ is a terminal object in } \mathcal{C}$$

*This means for every pair of open subsets $U, V \in \mathcal{B}$ such that $V \subseteq U$ there exist morphisms $\rho_{UV} : \mathcal{F}(U) \to \mathcal{F}(V)$ which satisfy the following properties*

*a) $\rho_{UU} = 1_{\mathcal{F}(U)}$*
*b) $\rho_{VW} \circ \rho_{UV} = \rho_{UW}$ for any basic open subset $W \subset \mathcal{B}$*

*A morphism of presheaves is precisely a natural transformation. That is $\alpha : \mathcal{F} \to \mathcal{G}$ is a family of morphisms $\alpha_U : \mathcal{F}(U) \to \mathcal{G}(U)$ for which*

$$\rho_{UV} \circ \alpha_U = \alpha_V \circ \rho_{UV}$$

*This determines a category $\mathbf{PSh}(\mathcal{B}; \mathcal{C})$. We may omit the value category $\mathcal{C}$ if there are no additional hypotheses.*

**Definition 4.2.2 (Abstract Sheaf)**
*Let $X$ be a topological space, $\mathcal{B}$ a base for the topology closed under finite intersections and $\mathcal{F}$ a $\mathcal{B}$-presheaf.*

*For every family of non-empty basic open subsets $\{U_i\}_{i \in I}$ with $U = \bigcup_{i \in I} U_i$, consider the sequence of morphisms*

$$\mathcal{F}(U) \xrightarrow{\ e\ } \prod_{i \in I} \mathcal{F}(U_i) \underset{d_1}{\overset{d_0}{\rightrightarrows}} \prod_{(i,j) \in I \times I} \mathcal{F}(U_i \cap U_j)$$

*where $e = \left(\rho_{UU_i}\right)_{i \in I}$, $d_0 = \left(\rho_{U_i(U_i \cap U_j)} \circ \pi_i\right)_{(i,j) \in I \times I}$ and $d_1 = \left(\rho_{U_j(U_i \cap U_j)} \circ \pi_j\right)_{(i,j) \in I \times I}$.*

*We say that $\mathcal{F}$ is a $\mathcal{B}$-**sheaf** if $e$ is an equaliser for $(d_0, d_1)$ for all such families of open subsets.*

*This determines a full subcategory $\mathbf{Sh}(\mathcal{B}; \mathcal{C})$ of $\mathbf{PSh}(\mathcal{B}; \mathcal{C})$.*

*When $\mathcal{B}$ is just the collection of all open subsets of $X$ then we say $\mathcal{F}$ is a **sheaf**, and we denote the category by $\mathbf{Sh}(X; \mathcal{C})$.*

It may be convenient to write the sheaf condition as the existence of a limit.

**Lemma 4.2.3 (Sheaf Condition as a Limit)**
*Let $\mathcal{F}$ be a presheaf on $X$ and $U = \bigcup_{i \in I} U_i$ a cover; write $U_{ij} := U_i \cap U_j$ for $(i,j) \in I \times I$, and let $d_0, d_1 : \prod_{i \in I} \mathcal{F}(U_i) \to \prod_{(i,j) \in I \times I} \mathcal{F}(U_{ij})$ be as in (4.2.2).*

*Let $I^{(1)}$ be the small category with objects $I \sqcup (I \times I)$ and, for each $(i,j) \in I \times I$, morphisms $u_{ij} : i \to (i,j)$ and $v_{ij} : j \to (i,j)$ (together with identities, and no further morphisms), and let $D_{\mathcal{F}} : I^{(1)} \to \mathcal{C}$ be the diagram*

$$D_{\mathcal{F}}(i) := \mathcal{F}(U_i), \quad D_{\mathcal{F}}((i,j)) := \mathcal{F}(U_{ij}), \quad D_{\mathcal{F}}(u_{ij}) := \rho_{U_i U_{ij}}, \quad D_{\mathcal{F}}(v_{ij}) := \rho_{U_j U_{ij}}$$

*Then, as $\mathcal{C}$ has arbitrary products, the contravariant functors $\mathcal{C}^{op} \to \mathbf{Set}$*

$$X \mapsto \left\{ \xi : X \to \prod_{i \in I} \mathcal{F}(U_i) \;\middle|\; d_0 \circ \xi = d_1 \circ \xi \right\}, \quad X \mapsto \{\text{cones over } D_{\mathcal{F}} \text{ with apex } X\}$$

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*(both via precomposition) are naturally isomorphic, via* $\xi \mapsto (\pi_i \circ \xi)_{i\in I}$*, extended to* $(i,j) \in I \times I$ *by* $D_{\mathcal{F}}(u_{ij}) \circ \pi_i \circ \xi$*. Consequently* $D_{\mathcal{F}}$ *has* $\left(\mathcal{F}(U), \{\rho_{UU_i}\}_{i\in I}, \{\rho_{UU_{ij}}\}_{(i,j)\in I\times I}\right)$ *as a limit over* $I^{(1)}$ *if and only if* $\mathcal{F}$ *satisfies the sheaf condition for this cover (4.2.2).*

*Proof.* Given $\xi$ with $d_0 \circ \xi = d_1 \circ \xi$, set $\xi_i := \pi_i \circ \xi$ and $\xi_{(i,j)} := D_{\mathcal{F}}(u_{ij}) \circ \xi_i$; then also $\xi_{(i,j)} = D_{\mathcal{F}}(v_{ij}) \circ \xi_j$, since both sides equal $\pi_{(i,j)} \circ d_0 \circ \xi = \pi_{(i,j)} \circ d_1 \circ \xi$ by definition of $d_0, d_1$. So $\left(X, \{\xi_k\}_{k\in I^{(1)}}\right)$ is a cone over $D_{\mathcal{F}}$, the only nontrivial conditions being exactly this equality at $u_{ij}, v_{ij}$.

Conversely, given a cone $\left(X, \{\xi_k\}_{k\in I^{(1)}}\right)$, bundling $(\xi_i)_{i\in I}$ via the universal property of the product (2.6.81) gives $\xi := (\xi_i)_i : X \to \prod_{i\in I} \mathcal{F}(U_i)$ with $\pi_i \circ \xi = \xi_i$; then for all $(i,j)$, $\pi_{(i,j)} \circ d_0 \circ \xi = D_{\mathcal{F}}(u_{ij}) \circ \xi_i = \xi_{(i,j)} = D_{\mathcal{F}}(v_{ij}) \circ \xi_j = \pi_{(i,j)} \circ d_1 \circ \xi$, using the cone conditions at $u_{ij}, v_{ij}$, so $d_0 \circ \xi = d_1 \circ \xi$.

These two constructions are mutually inverse, directly from $\pi_i \circ \xi = \xi_i$ and the universal property of the product, and are evidently compatible with precomposition, i.e. natural in $X$: for $\phi : X' \to X$, both sides simply precompose $\xi$, resp. every $\xi_k$, by $\phi$. So $D_{\mathcal{F}}$ has a limit over $I^{(1)}$ iff $(d_0, d_1)$ has an equaliser, and the two agree. ∎

**Proposition 4.2.4 (Sheaf Conditions for Concrete Categories)**
*Let* $X$ *be a topological space,* $\mathcal{B}$ *a base for the topology closed under finite intersections and* $\mathcal{C}$ *is an algebraic category (2.6.185). Then a* $\mathcal{B}$*-presheaf* $\mathcal{F}$ *is a sheaf if and only if it satisfies the following conditions, for every open cover* $U = \bigcup_{i\in I} U_i$ *with* $U_i$ *non-empty*

*a)* $\sigma, \tau \in \mathcal{F}(U)$ *such that* $\rho_{UU_i}(\sigma) = \rho_{UU_i}(\tau)$ *for all* $i \in I$ *implies* $\sigma = \tau$
*b)* $\sigma_i \in \mathcal{F}(U_i)$ *such that* $\rho_{U_i(U_i\cap U_j)}(\sigma_i) = \rho_{U_j(U_i\cap U_j)}(\sigma_j)$ *for all* $(i,j) \in I \times I$ *implies there exists* $\sigma \in \mathcal{F}(U)$ *such that* $\rho_{UU_i}(\sigma) = \sigma_i$.

*Proof.* Suppose that $\mathcal{F}$ is a sheaf in the abstract sense. By assumption the forgetful functor $\mathcal{C} \to \mathbf{Set}$ preserves limits and so by (2.6.198) $e$ is injective and

$$\mathrm{Im}(e) = \left\{(\sigma_i)_{i\in I} \;\middle|\; \sigma_i|_{U_i\cap U_j} = \sigma_j|_{U_i\cap U_j}\right\}$$

which gives both concrete sheaf conditions.

Conversely this condition ensures the sequence is an equaliser in **Set**, and so by hypothesis is so in $\mathcal{C}$. ∎

## 4.2.1 Stalks

**Definition 4.2.5 (Stalk of a presheaf)**
*Let* $X$ *be a topological space,* $\mathcal{B}$ *a base closed under finite intersections and* $\mathcal{F}$ *a* $\mathcal{B}$*-presheaf.*

*Let* $Z$ *be an irreducible subset of* $X$*, then the family of open subsets* $U \in \mathcal{B}$ *meeting* $Z$ *is closed under finite intersections (4.1.60) and therefore forms a directed preorder (2.6.124) ordered by reverse inclusion.*

*Therefore we may define the* ***stalk*** *at* $Z$ *by the direct limit (2.6.126)*

$$\mathcal{F}_Z := \varinjlim_{U\in\mathcal{B},\, U\cap Z\neq\emptyset} \mathcal{F}(U)$$

*This means there is a family of morphisms* $\{\rho_{UZ} : \mathcal{F}(U) \to \mathcal{F}_Z\}$ *such that*

*a) all* $V \in \mathcal{B}$ *with* $V \subset U$ *intersecting* $Z$ *satisfy* $\rho_{VZ} \circ \rho_{UV} = \rho_{UZ}$*,*
*b) any collection of morphisms* $\alpha_U : \mathcal{F}(U) \to \mathcal{A}$ *satisfying* $\alpha_U = \alpha_V \circ \rho_{UV}$ *factors uniquely through a morphism* $\alpha_Z : \mathcal{F}_Z \to \mathcal{A}$

*When* $Z = \{x\}$ *we denote this simply by* $\mathcal{F}_x$*.*

**Proposition 4.2.6 (Explicit form of Stalk)**
*Consider the situation (4.2.5) where the value category* $\mathcal{C}$ *is algebraic (2.6.185). Then*

$$U(\mathcal{F}_Z) = (U \circ \mathcal{F})_Z$$

*and so the stalk* $\mathcal{F}_Z$ *may be constructed explicitly as follows*

$$\mathcal{F}_Z := \{(U,\sigma) \mid U \in \mathcal{B}, \sigma \in \mathcal{F}(U)\}/\sim$$
$$(U,\sigma) \sim (V,\tau) \iff \exists W \in \mathcal{B} \text{ s.t. } W \cap Z \neq \emptyset \text{ and } \sigma|_W = \tau|_W$$

*with morphisms*

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$$\begin{aligned}\rho_{UZ} : \mathcal{F}(U) &\to \mathcal{F}_Z \\ \sigma &\to [(U, \sigma)]\end{aligned}$$

*In particular every element of the stalk $\sigma_Z \in \mathcal{F}_Z$ is the image of some section $\sigma \in \mathcal{F}(U)$ under $\rho_{UZ}$ for some $U \in \mathcal{B}$.*

*Proof.* By (2.6.190), the forgetful functor $U : \mathcal{C} \to \mathbf{Set}$ creates directed colimits, and so the result follows from (2.6.129). ∎

**Proposition 4.2.7 (Functoriality of Stalks)**
*Let $X$ be a topological space with base $\mathcal{B}$ closed under finite intersections, $\phi : \mathcal{F} \to \mathcal{G}$ a morphism of $\mathcal{B}$-presheaves and $Z \subset X$ an irreducible subset. Then there exists a unique morphism $\phi_Z : \mathcal{F}_Z \to \mathcal{G}_Z$ ensuring the following diagram commute for all open subsets $U$ meeting $Z$*

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\phi_U} & \mathcal{G}(U) \\ {\scriptstyle \rho_{UZ}}\downarrow & & \downarrow{\scriptstyle \rho_{UZ}} \\ \mathcal{F}_Z & \xrightarrow{\phi_Z} & \mathcal{G}_Z \end{array}$$

*Furthermore $(\psi \circ \phi)_Z = \psi_Z \circ \phi_Z$. In the explicit case (4.2.6) then $\phi_Z$ is given by*

$$[(U, \sigma)] \to [(U, \phi_U(\sigma))]$$

*This also means that for any $A \in \mathcal{C}$ the bijection*

$$\mathrm{Mor}(\mathcal{F}_Z, A) \longrightarrow \left(\mathcal{F}|_{\mathcal{N}_Z} \downarrow A\right)$$

*is natural in $\mathcal{F}$, where $\mathcal{N}_Z$ is the category of open sets intersecting $Z$ ordered by reverse inclusion.*

**Lemma 4.2.8 (Lifting Stalks)**
*Suppose we are in the situation of (4.2.6) and $\sigma \in \mathcal{F}(U)$, $\tau \in \mathcal{F}(V)$ are sections such that $Z \cap U \cap V \neq \emptyset$.*

*a) Then $\sigma_Z = \tau_Z$ if and only if there is a neighbourhood $W \in \mathcal{B}$, $W \subseteq U \cap V$, meeting $Z$ such that $\sigma|_W = \tau|_W$.*
*b) If $\sigma_x = \tau_x$ for all $x \in U \cap V$, then there is an open cover $U \cap V = \bigcup_i U_i$ such that $\sigma|_{U_i} = \tau|_{U_i}$*
*c) If in addition $\mathcal{F}$ is a sheaf then $\sigma|_{U\cap V} = \tau|_{U\cap V}$.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

By (4.2.6), $\mathcal{F}_Z = \{(W, \rho) \mid W \in \mathcal{B}, \rho \in \mathcal{F}(W)\}/\sim$ with $(W, \rho) \sim (W', \rho')$ iff there is $N \in \mathcal{B}$, $N \subseteq W \cap W'$, $N \cap Z \neq \emptyset$, with $\rho|_N = \rho'|_N$; and $\sigma_Z = [(U, \sigma)]$, $\tau_Z = [(V, \tau)]$. So $\sigma_Z = \tau_Z$ if and only if $(U, \sigma) \sim (V, \tau)$, which is exactly the stated condition, proving a).

For b), suppose $\sigma_x = \tau_x$ for all $x \in U \cap V$. For each such $x$, a) (applied with $Z := \{x\}$, an irreducible subset meeting $U \cap V$) gives $W_x \in \mathcal{B}$, $W_x \subseteq U \cap V$, $x \in W_x$, with $\sigma|_{W_x} = \tau|_{W_x}$. Then $\{W_x\}_{x \in U\cap V}$ is an open cover of $U \cap V$ with the required property.

For c), as $\mathcal{B}$ is closed under finite intersections and $U, V \in \mathcal{B}$, $U \cap V \in \mathcal{B}$, so $\sigma|_{U\cap V}, \tau|_{U\cap V} \in \mathcal{F}(U \cap V)$ are defined. By b) there is an open cover $U \cap V = \bigcup_i W_i$ with $\sigma|_{W_i} = \tau|_{W_i}$, that is $\rho_{(U\cap V)W_i}(\sigma|_{U\cap V}) = \rho_{(U\cap V)W_i}(\tau|_{U\cap V})$ for all $i$; as $\mathcal{F}$ is a sheaf, the separation condition (4.2.4) gives $\sigma|_{U\cap V} = \tau|_{U\cap V}$. ∎

**Proposition 4.2.9 (Stalks when restricting to a base)**
*Let $\mathcal{B}_2 \subset \mathcal{B}_1$ be bases for $X$ which generate the same topology. Suppose $\mathcal{F}$ is a $\mathcal{B}_1$-presheaf and $Z \subset X$ is an irreducible subset. Then there is a unique morphism making the following diagram commute for all $U \in \mathcal{B}_2$ intersecting $Z$*

$$\begin{array}{ccc} \mathcal{F}|_{\mathcal{B}_2}(U) & = & \mathcal{F}(U) \\ {\scriptstyle \rho_{UZ}}\downarrow & & \downarrow{\scriptstyle \rho_{UZ}} \\ \left(\mathcal{F}|_{\mathcal{B}_2}\right)_Z & \xrightarrow{\sim} & \mathcal{F}_Z \end{array}$$

*and it is an isomorphism.*

*In the situation of (4.2.6) the map is of the obvious form.*

*Proof.* Let $\mathcal{I}_2$ and $\mathcal{I}_1$ be the diagram of open subsets $U$ of $\mathcal{B}_2$ and $\mathcal{B}_1$ meeting $Z$, ordered by reverse inclusion. Then $\mathcal{I}_2$ is a full subcategory of $\mathcal{I}_1$ which is cofinal (2.6.127). So the result follows from (2.6.128)

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In the explicit case this may be shown more directly. ∎

**Lemma 4.2.10 (Pointwise Injectivity and Injectivity on Stalks)**
*Let $\mathcal{C}$ be algebraic (2.6.185) and $\phi : \mathcal{F} \to \mathcal{G}$ a morphism of presheaves such that $\phi_U$ is injective for all $U \subset X$. Then $\phi_x$ is injective for all $x \in X$.*

*Conversely, if $\phi$ is a morphism of sheaves and $\phi_x$ is injective for all $x \in X$, then $\phi_U$ is injective for all $U \subset X$.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

Fix $x \in X$ and suppose $a, b \in \mathcal{F}_x$ with $\phi_x(a) = \phi_x(b)$. By (4.2.6), $a = (\sigma_1)_x$, $b = (\sigma_2)_x$ for some $\sigma_1 \in \mathcal{F}(V_1)$, $\sigma_2 \in \mathcal{F}(V_2)$, $V_1, V_2 \in \mathcal{B}$ containing $x$; setting $V := V_1 \cap V_2 \in \mathcal{B}$, $\sigma := \sigma_1|_V$, $\tau := \sigma_2|_V$, we have $\sigma, \tau \in \mathcal{F}(V)$ with $\sigma_x = a$, $\tau_x = b$. Then

$$(\phi_V(\sigma))_x = \phi_x(\sigma_x) = \phi_x(a) = \phi_x(b) = \phi_x(\tau_x) = (\phi_V(\tau))_x$$

so by (4.2.8).a) (with both opens taken to be $V$) there is $W \in \mathcal{B}$, $W \subseteq V$, $x \in W$, with $\phi_V(\sigma)|_W = \phi_V(\tau)|_W$, that is $\phi_W(\sigma|_W) = \phi_W(\tau|_W)$. As $\phi_W$ is injective, $\sigma|_W = \tau|_W$, so $a = \sigma_x = (\sigma|_W)_x = (\tau|_W)_x = \tau_x = b$.

Conversely, fix $U \subset X$ and $\sigma, \tau \in \mathcal{F}(U)$ with $\phi_U(\sigma) = \phi_U(\tau)$. For every $x \in U$, $\phi_x(\sigma_x) = (\phi_U(\sigma))_x = (\phi_U(\tau))_x = \phi_x(\tau_x)$, so $\sigma_x = \tau_x$ as $\phi_x$ is injective. As $\mathcal{F}$ is a sheaf, (4.2.8).c) (taking both opens of the lemma to be $U$) gives $\sigma = \tau$. ∎

**Proposition 4.2.11 (Isomorphisms of Presheaves and Sheaves)**
*Let $\phi : \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves (resp. sheaves). Then the following are equivalent*

*a) $\phi$ is an isomorphism*
*b) $\phi_U$ is an isomorphism for all $U \subset X$*

*If $\mathcal{C}$ is algebraic (2.6.185), then a), b) are further equivalent to*

*c) $\phi_U$ is bijective for all $U \subset X$*

*If moreover $\mathcal{F}, \mathcal{G}$ are sheaves, a)–c) are equivalent to*

*d) $\phi_x : \mathcal{F}_x \to \mathcal{G}_x$ is an isomorphism for all $x \in X$*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

*b)* $\Longrightarrow$ *a)*: If each $\phi_U$ is an isomorphism, define $\psi_U := \phi_U^{-1}$; for $V \subseteq U$, inverting both sides of the naturality square $\phi_V \circ \rho_{UV} = \rho_{UV} \circ \phi_U$ gives $\rho_{UV} \circ \psi_U = \psi_V \circ \rho_{UV}$, so $\psi := \{\psi_U\}$ is a morphism of presheaves (resp. sheaves), and $\psi\phi = 1_{\mathcal{F}}$, $\phi\psi = 1_{\mathcal{G}}$ hold pointwise, hence as morphisms of presheaves. So $\phi$ is an isomorphism.

*a)* $\Longrightarrow$ *b)*: If $\phi$ is an isomorphism with inverse $\psi$, then $\phi_U\psi_U = 1_{\mathcal{G}(U)}$ and $\psi_U\phi_U = 1_{\mathcal{F}(U)}$ for every $U$, so $\phi_U$ is an isomorphism.

*b)* $\Longleftrightarrow$ *c)* (when $\mathcal{C}$ is algebraic): an isomorphism is in particular bijective (a functor preserves isomorphisms, and an isomorphism in **Set** is a bijection), while conversely a bijective morphism is an isomorphism in $\mathcal{C}$ as $U$ reflects isomorphisms (2.6.190) apply this pointwise at each $U$.

*b)* $\Longrightarrow$ *d)*: with $\psi := \{\psi_U\}$ as in *b)* $\Longrightarrow$ *a)* above, $\psi$ is a morphism of sheaves with $\psi\phi = 1_{\mathcal{F}}$, $\phi\psi = 1_{\mathcal{G}}$. By functoriality of stalks (4.2.7), $\psi_x \circ \phi_x = (\psi\phi)_x = 1_{\mathcal{F}_x}$ and $\phi_x \circ \psi_x = (\phi\psi)_x = 1_{\mathcal{G}_x}$, so $\phi_x$ is an isomorphism.

*d)* $\Longrightarrow$ *c)*: Suppose $\phi_x$ is an isomorphism, in particular bijective, for every $x \in X$. By (4.2.10), $\phi_U$ is injective for every $U \subset X$. For surjectivity: fix $U \subset X$ and $\tau \in \mathcal{G}(U)$. For each $x \in U$, surjectivity of $\phi_x$ gives $a_x \in \mathcal{F}_x$ with $\phi_x(a_x) = \tau_x$; write $a_x = (\rho_x)_x$ for some $\rho_x \in \mathcal{F}(W_x)$, $W_x \in \mathcal{B}$, $x \in W_x \subseteq U$ (shrinking $W_x$ if necessary). Then $\left(\phi_{W_x}(\rho_x)\right)_x = \phi_x\left((\rho_x)_x\right) = \phi_x(a_x) = \tau_x = \left(\tau|_{W_x}\right)_x$, so by (4.2.8).a) there is $W_{x'} \in \mathcal{B}$, $x \in W_{x'} \subseteq W_x$, with $\phi_{W_x}(\rho_x)|_{W_{x'}} = \tau|_{W_{x'}}$; write $\sigma_x := \rho_x|_{W_{x'}} \in \mathcal{F}(W_{x'})$, so that $\phi_{W_{x'}}(\sigma_x) = \tau|_{W_{x'}}$ exactly.

The $\{W_{x'}\}_{x \in U}$ cover $U$; for $x, y \in U$, on $W_{x'} \cap W_{y'} \in \mathcal{B}$,

$$\phi_{W_{x'} \cap W_{y'}}\left(\sigma_x|_{W_{x'} \cap W_{y'}}\right) = \phi_{W_{x'}}(\sigma_x)|_{W_{x'} \cap W_{y'}} = \tau|_{W_{x'} \cap W_{y'}} = \phi_{W_{y'}}(\sigma_y)|_{W_{x'} \cap W_{y'}} = \phi_{W_{x'} \cap W_{y'}}\left(\sigma_y|_{W_{x'} \cap W_{y'}}\right)$$

so as $\phi_{W_{x'} \cap W_{y'}}$ is injective, $\sigma_x|_{W_{x'} \cap W_{y'}} = \sigma_y|_{W_{x'} \cap W_{y'}}$. As $\mathcal{F}$ is a sheaf, the gluing condition (4.2.4) gives $\sigma \in \mathcal{F}(U)$ with $\sigma|_{W_{x'}} = \sigma_x$ for all $x$. Then $\phi_U(\sigma)|_{W_{x'}} = \phi_{W_{x'}}\left(\sigma|_{W_{x'}}\right) = \phi_{W_{x'}}(\sigma_x) = \tau|_{W_{x'}}$ for every $x \in U$, so as $\mathcal{G}$ is a sheaf, the separation condition (4.2.4) gives $\phi_U(\sigma) = \tau$. So $\phi_U$ is surjective; being also injective, $\phi_U$ is bijective, that is c). ∎

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The following somewhat tedious result shows that morphisms are uniquely determined by morphisms on basic open sets.

**Proposition 4.2.12**
*Let $X$ be a topological space, $\mathcal{B}$ a base for the topology closed under finite intersections. Then the functor*

$$(-)|_{\mathcal{B}} : \mathbf{Sh}(X;\mathcal{C}) \to \mathbf{Sh}(\mathcal{B};\mathcal{C})$$

*is full and faithful.*

*Proof.* Let $\phi, \psi : \mathcal{F} \to \mathcal{G}$ be morphisms which agree on basic open subsets $U \in \mathcal{B}$. Consider an arbitrary open subset $U \subset X$ then by definition $U = \bigcup_{i\in I} U_i$ for $U_i \in \mathcal{B}$. By assumption there is a commutative diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{e} & \prod_{i\in I}\mathcal{F}(U_i) \\ {\scriptstyle\psi_U}\downarrow\downarrow{\scriptstyle\phi_U} & & \downarrow{\scriptstyle\prod_{i\in I}\phi|_{U_i} = \prod_{i\in I}\psi|_{U_i}} \\ \mathcal{G}(U) & \xrightarrow{e} & \prod_{i\in I}\mathcal{G}(U_i) \end{array}$$

which shows $e \circ \phi_U = e \circ \psi_U$. But $e$ is a monomorphism (2.6.136) (being part of an equaliser) so $\phi_U = \psi_U$. Therefore the functor is faithful.

Let $\phi : \mathcal{F}|_{\mathcal{B}} \to \mathcal{G}|_{\mathcal{B}}$ be a morphism of $\mathcal{B}$-sheaves. For an open set $U$ choose a basic open cover $U = \bigcup_{i\in I} U_i$ then we have a commutative diagram

$$\begin{array}{ccccc} \mathcal{F}(U) & \xrightarrow{e} & \prod_{i\in I}\mathcal{F}(U_i) & \rightrightarrows & \prod_{(i,j)\in I\times I}\mathcal{F}(U_i\cap U_j) \\ \downarrow{\scriptstyle\tilde{\phi}_U} & & \downarrow{\scriptstyle\prod_{i\in I}\phi_{U_i}} & & \downarrow{\scriptstyle\prod_{(i,j)\in I\times I}\phi_{U_i\cap U_j}} \\ \mathcal{G}(U) & \xrightarrow{e} & \prod_{i\in I}\mathcal{G}(U_i) & \rightrightarrows & \prod_{(i,j)\in I\times I}\mathcal{G}(U_i\cap U_j) \end{array}$$

where for the right hand square we must pick the top arrows or bottom arrows. Chasing the diagram shows that $\left(\prod_{i\in I}\phi_{U_i}\right)\circ e$ is equalised by the bottom two arrows, and so the factorisation exists by the universal property and $\tilde{\phi}_U$ the unique morphism which makes the left square commute, i.e. such that $\phi_{U_i}\circ\rho_{UU_i} = \rho_{UU_i}\circ\tilde{\phi}_U$ for all $i \in I$.

We claim that for any basic open subset $W \subset U$ the following diagram commutes

$$\begin{array}{ccccc} \mathcal{F}(U) & \longrightarrow & \mathcal{F}(W) & \xrightarrow{e} & \prod_{i\in I}\mathcal{F}(W\cap U_i) \\ \downarrow{\scriptstyle\tilde{\phi}_U} & & \downarrow{\scriptstyle\phi_W} & & \downarrow{\scriptstyle\prod_{i\in I}\phi_{W\cap U_i}} \\ \mathcal{G}(U) & \longrightarrow & \mathcal{G}(W) & \xrightarrow{e} & \prod_{i\in I}\mathcal{G}(W\cap U_i) \end{array}$$

and $\tilde{\phi}_U$ is the unique morphism such that this commutes for all $W \in \mathcal{B}$. The right hand square commutes by assumption. As $e$ is a monomorphism it is enough to show that

$$\left(\prod_{i\in I}\phi_{W\cap U_i}\right)\circ e\circ\rho_{UW} = e\circ\phi_W\circ\rho_{UW} = e\circ\rho_{UW}\circ\tilde{\phi}_U$$

First observe we have a commutative diagram

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{e} & \prod_{i\in I}\mathcal{F}(U_i) \\ {\scriptstyle\rho_{UW}}\downarrow & & \downarrow{\scriptstyle\prod_{i\in I}\rho_{U_i(W\cap U_i)}} \\ \mathcal{F}(W) & \xrightarrow{e} & \prod_{i\in I}\mathcal{F}(W\cap U_i) \end{array}$$

which follows because both directions are equal to the diagonal $\left(\rho_{U(W\cap U_i)}\right)_{i\in I}$. Therefore we may combine the two diagrams to find

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$$\begin{aligned}
e \circ \phi_W \circ \rho_{UW} &= \left(\prod_{i\in I} \phi_{W\cap U_i}\right) \circ e \circ \rho_{UW} \\
&= \left(\prod_{i\in I} \phi_{W\cap U_i}\right) \circ \left(\prod_{i\in I} \rho_{U_i(W\cap U_i)}\right) \circ e \\
&= \left(\prod_{i\in I} \phi_{W\cap U_i} \circ \rho_{U_i(W\cap U_i)}\right) \circ e \\
&= \left(\prod_{i\in I} \rho_{U_i(W\cap U_i)} \circ \phi_{U_i}\right) \circ e \\
&= \left(\prod_{i\in I} \rho_{U_i(W\cap U_i)}\right) \circ \left(\prod_{i\in I} \phi_{U_i}\right) \circ e \\
&= \left(\prod_{i\in I} \rho_{U_i(W\cap U_i)}\right) \circ e \circ \tilde{\phi}_U \\
&= e \circ \rho_{UW} \circ \tilde{\phi}_U
\end{aligned}$$

As $e$ is a monomorphism (2.6.136) then $\phi_W \circ \rho_{UW} = \rho_{UW} \circ \tilde{\phi}_U$ as required. Then $\tilde{\phi}_U$ is the unique morphism such that this holds for all $W \in \mathcal{B}$ and $W \subset U$, because it is already uniquely determined by $W = U_i$. This also shows that $\tilde{\phi}_W = \phi_W$.

In particular $\tilde{\phi}_U$ does not depend on the choice of cover $\{U_i\}$ of $U$: if $\tilde{\phi}'_U$ is constructed from a different basic cover of $U$, it satisfies the same property $\phi_W \circ \rho_{UW} = \rho_{UW} \circ \tilde{\phi}'_U$ for all $W \in \mathcal{B}$, $W \subset U$; taking $W$ to range over the original cover $\{U_i\}$ shows $\tilde{\phi}'_U$ also satisfies the equalising property defining $\tilde{\phi}_U$, so $\tilde{\phi}'_U = \tilde{\phi}_U$ by the uniqueness established above.

Finally for an arbitrary open subset $V = \bigcup_{i\in I} W_i \subset U$, $W_i \in \mathcal{B}$, we wish to show we have a commutative diagram

$$\begin{array}{ccccc}
\mathcal{F}(U) & \longrightarrow & \mathcal{F}(V) & \xrightarrow{e} & \prod_{i\in I} \mathcal{F}(W_i) \\
\downarrow{\scriptstyle \tilde{\phi}_U} & & \downarrow{\scriptstyle \tilde{\phi}_V} & & \downarrow{\scriptstyle \prod_{i\in I} \phi_{W_i}} \\
\mathcal{G}(U) & \longrightarrow & \mathcal{G}(V) & \xrightarrow{e} & \prod_{i\in I} \mathcal{G}(W_i)
\end{array}$$

We've already shown that the large square and right square commute. This shows that

$$e \circ \tilde{\phi}_V \circ \rho_{UV} = e \circ \rho_{UV} \circ \tilde{\phi}_U$$

and as $e$ is a monomorphism (2.6.136) the left square also commutes.

Therefore $\tilde{\phi}$ is a morphism of sheaves and $\tilde{\phi}|_{\mathcal{B}} = \phi$. Therefore the functor is full.

Alternatively, when $\mathcal{C}$ is algebraic (2.6.185), this can be shown more directly by working with elements via the forgetful functor $U : \mathcal{C} \to \mathbf{Set}$, citing (4.2.4) throughout with the base taken to be all of $\mathcal{O}(X)$ (a valid choice of base, since $\mathcal{F}, \mathcal{G}$ are sheaves on all of $X$ (4.2.2)).

*Drafted by Claude Sonnet, reviewed by David Rufino.*

**Faithful.** Let $\phi, \psi : \mathcal{F} \to \mathcal{G}$ agree on $U \in \mathcal{B}$, and fix an arbitrary open $V \subset X$ and $\sigma \in \mathcal{F}(V)$. Choose, for each $x \in V$, a basic open $U_x \in \mathcal{B}$ with $x \in U_x \subseteq V$. Then

$$\rho_{VU_x}(\phi_V(\sigma)) = \phi_{U_x}\left(\rho_{VU_x}(\sigma)\right) = \psi_{U_x}\left(\rho_{VU_x}(\sigma)\right) = \rho_{VU_x}(\psi_V(\sigma))$$

using naturality of $\phi, \psi$ and that they agree on $U_x \in \mathcal{B}$. As $\{U_x\}_{x\in V}$ is a cover of $V$, separation for $\mathcal{G}$ gives $\phi_V(\sigma) = \psi_V(\sigma)$. As $\sigma, V$ were arbitrary, $\phi = \psi$.

**Full.** Let $\phi : \mathcal{F}|_{\mathcal{B}} \to \mathcal{G}|_{\mathcal{B}}$ be a morphism of $\mathcal{B}$-sheaves; by (4.2.7) this determines stalk maps $\phi_x : \mathcal{F}_x \to \mathcal{G}_x$ for every $x \in X$.

*Construction.* Fix an open $V \subset X$ and $\sigma \in \mathcal{F}(V)$. Choose, for each $x \in V$, a basic open $U_x \in \mathcal{B}$ with $x \in U_x \subseteq V$, and set $\tau_x := \phi_{U_x}\left(\sigma|_{U_x}\right) \in \mathcal{G}(U_x)$. For $x, y \in V$, on $U_x \cap U_y \in \mathcal{B}$,

$$\tau_x|_{U_x\cap U_y} = \phi_{U_x\cap U_y}\left(\sigma|_{U_x\cap U_y}\right) = \tau_y|_{U_x\cap U_y}$$

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using naturality of $\phi$ and that $\big(\sigma|_{U_x}\big)|_{U_x\cap U_y} = \sigma|_{U_x\cap U_y} = \big(\sigma|_{U_y}\big)|_{U_x\cap U_y}$. So by the gluing condition for $\mathcal{G}$, applied to the cover $\{U_x\}_{x\in V}$ of $V$, there is $\tau\in\mathcal{G}(V)$ with $\tau|_{U_x}=\tau_x$ for all $x$; define $\tilde{\phi}_V(\sigma) \coloneqq \tau$. By construction and functoriality of stalks (4.2.7), $\tilde{\phi}_V(\sigma)$ satisfies the property

$$(\star)\qquad \big(\tilde{\phi}_V(\sigma)\big)_x = \big(\tau|_{U_x}\big)_x = (\tau_x)_x = \phi_x\Big(\big(\sigma|_{U_x}\big)_x\Big) = \phi_x(\sigma_x) \qquad \forall x\in V$$

*Well-definedness.* Suppose $\tau,\tau'\in\mathcal{G}(V)$ both satisfy $(\star)$ for the same $\sigma$, that is $\tau_x=\phi_x(\sigma_x)=\tau'_x$ for all $x\in V$. For each $x\in V$, choose $W\in\mathcal{B}$ with $x\in W\subseteq V$; then $(\tau|_W)_x=\tau_x=\tau'_x=(\tau'|_W)_x$, so by (4.2.8).a), applied to the sections $\tau|_W,\tau'|_W\in\mathcal{G}(W)$, there is $W_x\in\mathcal{B}$, $x\in W_x\subseteq W$, with $\tau|_{W_x}=\tau'|_{W_x}$. As $\{W_x\}_{x\in V}$ covers $V$, separation for $\mathcal{G}$ gives $\tau=\tau'$. So $\tilde{\phi}_V(\sigma)$ does not depend on the choice of cover $\{U_x\}$, and is the unique element of $\mathcal{G}(V)$ satisfying $(\star)$.

*$\tilde{\phi}$ restricts to $\phi$ on $\mathcal{B}$.* For $U\in\mathcal{B}$ and $\sigma\in\mathcal{F}(U)$, $\phi_U(\sigma)$ itself satisfies $(\phi_U(\sigma))_x=\phi_x(\sigma_x)$ for $x\in U$ (4.2.7), so by uniqueness $\tilde{\phi}_U(\sigma)=\phi_U(\sigma)$.

*$\tilde{\phi}$ is a morphism of presheaves.* For $V\subseteq U$ and $\sigma\in\mathcal{F}(U)$, both $\tilde{\phi}_V(\sigma|_V)$ and $\tilde{\phi}_U(\sigma)|_V$ satisfy $(\star)$ for $\sigma|_V$: for $x\in V$, $\big(\tilde{\phi}_V(\sigma|_V)\big)_x=\phi_x\big((\sigma|_V)_x\big)=\phi_x(\sigma_x)=\big(\tilde{\phi}_U(\sigma)\big)_x=\big(\tilde{\phi}_U(\sigma)|_V\big)_x$. By uniqueness, $\tilde{\phi}_V(\sigma|_V)=\tilde{\phi}_U(\sigma)|_V$, so $\tilde{\phi}$ is a morphism of sheaves with $\tilde{\phi}|_{\mathcal{B}}=\phi$. Therefore the functor is full. ∎

## 4.2.2 Sheafification

**Definition 4.2.13 (Sheafification)**
*Let $X$ be a topological space, $\mathcal{B}$ a basis of open sets for $X$ closed under finite intersections and $\mathcal{F}$ a $\mathcal{B}$-presheaf (with values in a category $\mathcal{C}$). Then we say that a pair $(\eta_{\mathcal{F}},\mathcal{F}^+)$ is a **sheafification** of $\mathcal{F}$ if*

*a) $\mathcal{F}^+$ is a sheaf (with values in a category $\mathcal{C}$)*
*b) $\eta_{\mathcal{F}}:\mathcal{F}\to\mathcal{F}^+|_{\mathcal{B}}$ is a morphism of $\mathcal{B}$-presheaves*
*c) for every sheaf $\mathcal{G}$ there is a bijection*

$$\begin{aligned}(-)^\sharp : \mathrm{Mor}(\mathcal{F}^+,\mathcal{G}) &\xrightarrow{\sim} \mathrm{Mor}(\mathcal{F},\mathcal{G}|_{\mathcal{B}})\\ \phi &\to \ \ \phi|_{\mathcal{B}}\circ\eta_{\mathcal{F}}\end{aligned}$$

*which is natural in $\mathcal{G}$, with the inverse map denoted by $(-)^\flat$.*

*In other words $\big(\mathcal{F}^+,(-)^\sharp\big)$ represents the functor $\mathcal{G}\to\mathrm{Mor}(\mathcal{F},\mathcal{G}|_{\mathcal{B}})$.*

*In this case we denote by $\varepsilon_{\mathcal{G}}:(\mathcal{G}|_{\mathcal{B}})^+\to\mathcal{G}$ the unique morphism which maps to $1_{\mathcal{G}|_{\mathcal{B}}}$ under the bijection.*

**Proposition 4.2.14 (Sheafification is Functorial)**
*Using the notation of (4.2.13) suppose there exists a sheafification $(\eta_{\mathcal{F}},\mathcal{F}^+)$ for all pre-sheaves $\mathcal{F}$. Then this can be extended to a functor*

$$(-)^+ : \mathbf{PSh}(X;\mathcal{B})\to\mathbf{Sh}(X)$$

*uniquely determined by either the condition $(-)^+$ is left-adjoint to $(-)|_{\mathcal{B}}$ or that the following diagram commutes*

$$\begin{array}{ccc}\mathcal{F}_1 & \xrightarrow{\ \phi\ } & \mathcal{F}_2\\ {\scriptstyle\eta_{\mathcal{F}_1}}\downarrow & & \downarrow{\scriptstyle\eta_{\mathcal{F}_2}}\\ \mathcal{F}_1^+|_{\mathcal{B}} & \xrightarrow[\phi^+|_{\mathcal{B}}]{\ \ \ \ \ } & \mathcal{F}_2^+|_{\mathcal{B}}\end{array}$$

*for all morphisms $\phi:\mathcal{F}_1\to\mathcal{F}_2$. More precisely we have the relationship*

$$\phi^+ = \big(\eta_{\mathcal{F}_2}\circ\phi\big)^\flat$$

*In particular $(\psi\circ\phi)^+=\psi^+\circ\phi^+$ for all morphisms $\psi:\mathcal{F}_2\to\mathcal{F}_3$ and $\eta_{\mathcal{F}}$ is the unit of the adjunction and $\varepsilon_{\mathcal{G}}$ the counit.*

*Furthermore $(-)|_{\mathcal{B}}:\mathbf{Sh}(X)\to\mathbf{PSh}(X;\mathcal{B})$ is fully faithful, and $\varepsilon$ is a natural isomorphism.*

*Proof.* See (2.6.74).

By (4.2.12), $\mathbf{Sh}(X)\to\mathbf{Sh}(\mathcal{B};\mathcal{C})$ is fully faithful; composing with the (trivially fully faithful) inclusion $\mathbf{Sh}(\mathcal{B};\mathcal{C})\hookrightarrow\mathbf{PSh}(X;\mathcal{B})$ shows $(-)|_{\mathcal{B}}:\mathbf{Sh}(X)\to\mathbf{PSh}(X;\mathcal{B})$ is fully faithful, so by (2.6.72), $\varepsilon$ is a natural isomorphism. ∎

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**Proposition 4.2.15**
*Using the notation of (4.2.13) suppose there exists a sheafification $(\eta_{\mathcal{F}}, \mathcal{F}^+)$ for $\mathcal{F}$. Then for every $x \in X$ there is a unique morphism making the following diagram commute for all $U \in \mathcal{B}$ containing $x$*

$$\begin{array}{ccc} \mathcal{F}(U) & \xrightarrow{\eta_U} & \mathcal{F}^+(U) \\ \downarrow & & \downarrow \\ \mathcal{F}_x & \dashrightarrow & (\mathcal{F}^+)_x \end{array}$$

*Proof.* By (4.2.7) and (4.2.9) there is a commutative diagram for all $U \in \mathcal{B}$ and $x \in U$

$$\begin{array}{ccccc} \mathcal{F}(U) & \xrightarrow{\eta_U} & \mathcal{F}^+(U) & = & \mathcal{F}^+(U) \\ \downarrow & & \downarrow & & \downarrow \\ \mathcal{F}_x & \longrightarrow & (\mathcal{F}^+|_{\mathcal{B}})_x & \xrightarrow{\sim} & \mathcal{F}^+_x \end{array}$$

where the bottom left arrow is uniquely determined. This shows that the required map exists and is uniquely determined according to this condition. ∎

**Proposition 4.2.16 (Sheafification)**
*Let $\mathcal{C}$ be an algebraic category (2.6.185) then the sheafification (4.2.13) exists.*

*More precisely define*

$$\mathcal{F}^+(U) := \left\{ s : U \to \coprod_{x \in U} \mathcal{F}_x \;\middle|\; U = \bigcup_{i \in I} U_i, U_i \in \mathcal{B}, \sigma_i \in \mathcal{F}(U_i), s(x) = (\sigma_i)_x \quad \forall x \in U_i \right\}$$

*Then this is evidently a sheaf and the adjunction $\phi \to \phi^\sharp$ is the unique presheaf morphism such that*

$$\phi^\sharp_U(\sigma)_x = \phi\left((\sigma_x)_{x \in U}\right)_x \quad \forall U \in \mathcal{B}, \sigma \in \mathcal{F}(U), x \in U$$

*and the inverse adjunction $\psi^\flat \leftarrow \psi$ is the unique sheaf morphism such that*

$$\psi^\flat(s)(x) = \psi_x(s(x)) \quad \forall x \in U$$

*(using (4.2.9) to identify $(\mathcal{G}|_{\mathcal{B}})_x$ and $\mathcal{G}_x$). The unit of the adjunction is given by*

$$(\eta_{\mathcal{F}})_U(\sigma) = (\sigma_x)_{x \in U}$$

*and for a morphism of $\mathcal{B}$-sheaves $\phi : \mathcal{F} \to \mathcal{G}$ the morphism $\phi^+$ defined in (4.2.14) is simply*

$$\phi^+(s)(x) = \phi_x(s(x)).$$

*Furthermore $\eta_{\mathcal{F}}$ is an isomorphism if and only if $\mathcal{F}$ is a $\mathcal{B}$-sheaf.*

*Proof.* We first demonstrate this when $\mathcal{C} = \mathbf{Set}$. The sheaf conditions are easy to verify. There is an obvious unit morphism

$$\begin{aligned} \eta_{\mathcal{F}}(U) : \mathcal{F}(U) &\to \mathcal{F}^+(U) \\ \sigma &\to x \to \sigma_x \end{aligned}$$

which is a natural transformation by (4.2.7). According to (2.6.74) it is enough to ensure the following map is a bijection and natural in $\mathcal{G}$

$$\begin{aligned} \mathrm{Mor}(\mathcal{F}^+, \mathcal{G}) &\to \mathrm{Mor}(\mathcal{F}, \mathcal{G}|_{\mathcal{B}}) \\ \phi &\to \phi|_{\mathcal{B}} \circ \eta_{\mathcal{F}} \end{aligned}$$

Naturality in $\mathcal{G}$ follows formally. Suppose $\phi, \psi : \mathcal{F}^+ \to \mathcal{G}$ are morphisms such that $\phi|_{\mathcal{B}} \circ \eta_{\mathcal{F}} = \psi|_{\mathcal{B}} \circ \eta_{\mathcal{F}}$. For a given $s \in \mathcal{F}^+(U)$ we wish to show that $\phi(s) = \psi(s)$. Suppose that $s$ is determined by sections $\sigma_i \in \mathcal{F}(U_i)$, that is to say $s|_{U_i} = \eta_{\mathcal{F}}(\sigma_i)$. Therefore by hypothesis $\phi(s)|_{U_i} = \phi\left(s|_{U_i}\right) = \psi\left(s|_{U_i}\right) = \psi(s)|_{U_i}$. By the concrete sheaf conditions (4.2.4) on $\mathcal{G}$ we conclude that $\phi(s) = \psi(s)$, and so the given map is injective.

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Conversely given $\phi : \mathcal{F} \to \mathcal{G}|_{\mathcal{B}}$, we wish to construct a morphism of sheaves $\phi^\flat : \mathcal{F}^+(U) \to \mathcal{G}(U)$ such that $\phi^\flat\big((\sigma_x)_{x\in U}\big) = \phi(\sigma)$ for all $\sigma \in \mathcal{F}(U)$ and $U \in \mathcal{B}$. We claim that there exists a family of maps $\big\{\phi^\flat_U : \mathcal{F}^+ \to \mathcal{G}\big\}$ such that $\phi^\flat_U(s)_x = \phi_x(s(x))$ for all $s \in \mathcal{F}^+(U)$. Such a family is unique by (4.2.8).

To show existence consider a section $s \in \mathcal{F}^+(U)$, then by definition $s(x) = \big((\sigma_i)_x\big)$ for some $\sigma_i \in \mathcal{F}(U_i)$, $U_i \in \mathcal{B}$ and all $x \in U_i$. Define $\tau_i := \phi(\sigma_i)$ then by (4.2.7) $(\tau_i)_x = \phi_x\big((\sigma_i)_x\big) = \phi_x(s(x))$. This shows that $(\tau_i)_x = \big(\tau_j\big)_x$ for all $x \in U_i \cap U_j$ so by (4.2.8) $\tau_i|_{U_i\cap U_j} = \tau_j|_{U_i\cap U_j}$. By (4.2.4) there exists $\tau \in \mathcal{G}(U)$ such that $\tau|_{U_i} = \tau_i$. This yields a map $\phi^\flat_U$ satisfying $\phi^\flat_U(s)_x = \tau_x = (\tau_i)_x = \phi(\sigma_i)_x = \phi_x\big((\sigma_i)_x\big) = \phi_x(s(x))$ as required.

It remains to show that $\phi^\flat$ is a morphism of sheaves, that is the following diagram is commutative for all open sets $V \subset U$

$$\begin{array}{ccc}
\mathcal{F}^+(U) & \xrightarrow{\phi^\flat_U} & \mathcal{G}(U) \\
\downarrow{\scriptstyle \rho_{UV}} & & \downarrow{\scriptstyle \rho_{UV}} \\
\mathcal{F}^+(V) & \xrightarrow{\phi^\flat_V} & \mathcal{G}(V) \\
\downarrow & & \downarrow{\scriptstyle \rho_x} \\
\mathcal{F}_x & \xrightarrow{\phi_x} & \mathcal{G}_x
\end{array}$$

The large and bottom squares commute by construction. Therefore for $s \in \mathcal{F}^+(U)$ we have $\big(\phi^\flat_U(s)|_V\big)_x = \phi_x(s|_V\ (x)) = \phi^\flat_V(s|_V)_x$. As this holds for all $x \in V$ we find from (4.2.8) that $\phi^\flat_U(s)|_V = \phi^\flat_V(s|_V)$ as required.

As $(-)^\sharp$ is injective it is sufficient by (2.6.36) to show that it has a right inverse, specifically $\big(\psi^\flat\big)^\sharp = \psi$. However $\psi(\sigma)_x = \psi^\flat\big((\sigma_{x'})_{x'\in U}\big)(x) = \big(\psi^\flat\big)^\sharp(\sigma)_x$ by construction so we are done by (4.2.8).

Evidently when $\eta_{\mathcal{F}}$ is an isomorphism then $\mathcal{F}$ is a $\mathcal{B}$-sheaf. We claim that the converse holds, namely suppose $\mathcal{F}$ is a $\mathcal{B}$-sheaf, then we require to show $(\eta_{\mathcal{F}})_U$ is injective and surjective. It is injective by (4.2.8). Similarly it is surjective for suppose $s \in \mathcal{F}^+(U)$ for $U \in \mathcal{B}$ and $s(x) = (\sigma_i)_x$ for all $x \in U_i$ and $\sigma_i \in \mathcal{F}(U_i)$. Then by the same result we have $(\sigma_i)|_{U_i\cap U_j} = \big(\sigma_j\big)|_{U_i\cap U_j}$, whence by (4.2.4) these lift to a section $\sigma \in \mathcal{F}(U)$ such that $\sigma|_{U_i} = \sigma_i$, and in particular $\sigma_x = (\sigma_i)_x = s(x)$ for all $x \in U_i$. Therefore we conclude the map $(\eta_{\mathcal{F}})_U$ is surjective and therefore an isomorphism in **Set**, which means $\eta_{\mathcal{F}}$ is an isomorphism of sheaves.

We now turn to the case of a general algebraic category $\mathcal{C}$, determined by $(\Omega, E)$ (2.6.185) write $U : \mathcal{C} \to \mathbf{Set}$ for the forgetful functor.

*The remainder of this proof is drafted by Claude Sonnet 5, reviewed by David Rufino.*

By (2.6.191), for $U \in \mathcal{B}$ the product $\prod_{x\in U} \mathcal{F}_x$ exists in $\mathcal{C}$, with each $\omega \in \Omega$ acting coordinatewise.

**$\mathcal{F}^+(U)$ is a subalgebra of $\prod_{x\in U} \mathcal{F}_x$.** Given $\omega \in \Omega$ and $s^{(1)}, \dots, s^{(n_\omega)} \in \mathcal{F}^+(U)$, and $x \in U$: choose for each $r = 1, \dots, n_\omega$ a representative $\sigma^{(r)} \in \mathcal{F}(V_r)$, $V_r \in \mathcal{B}$, $x \in V_r$, with $s^{(r)}(y) = \big(\sigma^{(r)}\big)_y$ for all $y \in V_r$. As $n_\omega$ is finite and $\mathcal{B}$ is closed under finite intersections, $N := \bigcap_r V_r \in \mathcal{B}$, and, by functoriality of stalks (4.2.7), $\tau := \omega_{\mathcal{F}(N)}\big(\sigma^{(1)}|_N, \dots, \sigma^{(n_\omega)}|_N\big) \in \mathcal{F}(N)$ has $\tau_y = \omega_{\mathcal{F}_y}\big(s^{(1)}(y), \dots, s^{(n_\omega)}(y)\big)$ for every $y \in N$. So the germ-function $y \mapsto \omega_{\mathcal{F}_y}\big(s^{(1)}(y), \dots, s^{(n_\omega)}(y)\big)$ — which is exactly $\omega$ applied coordinatewise to $s^{(1)}, \dots, s^{(n_\omega)}$ in $\prod_{x\in U} \mathcal{F}_x$ — is locally represented by $\tau$ near every point, hence lies in $\mathcal{F}^+(U)$. So $\mathcal{F}^+(U)$ is closed under every $\omega \in \Omega$, and by (2.6.194) is a subalgebra, in particular an object of $\mathcal{C}$, with $\omega_{\mathcal{F}^+(U)}$ given pointwise as above.

**The maps constructed above are morphisms of $\mathcal{C}$.** If $h : A \to B$ is a homomorphism and $S \subseteq U(A)$, $T \subseteq U(B)$ are subalgebras with $h(S) \subseteq T$, then $h|_S : S \to T$ is a homomorphism, as $\omega_S, \omega_T$ are literally restrictions of $\omega_A, \omega_B$. We apply this three times.

For the restriction maps: the coordinate projection $\prod_{x\in U} \mathcal{F}_x \to \prod_{x\in V} \mathcal{F}_x$ ($V \subseteq U$ in $\mathcal{B}$) is a homomorphism, carries $\mathcal{F}^+(U)$ into $\mathcal{F}^+(V)$ by definition of restriction of germ-functions, and restricts to $\rho_{UV}$.

For the unit: each stalk map $\rho_{Ux} : \mathcal{F}(U) \to \mathcal{F}_x$ is a morphism of $\mathcal{C}$ (4.2.5), so $(\rho_{Ux})_{x\in U} : \mathcal{F}(U) \to \prod_{x\in U} \mathcal{F}_x$ is a homomorphism (a map into a product is a homomorphism iff each component is); its image lies in $\mathcal{F}^+(U)$ (represented by $\sigma$ itself, via the cover $\{U\}$), and it restricts to $(\eta_{\mathcal{F}})_U$.

For $\phi^\flat$: given a morphism $\phi : \mathcal{F} \to \mathcal{G}|_{\mathcal{B}}$ of $\mathcal{C}$-presheaves with $\mathcal{G}$ a $\mathcal{B}$-sheaf, each $\phi_x : \mathcal{F}_x \to \mathcal{G}_x$ is a morphism of $\mathcal{C}$ (4.2.7), so $h := \prod_{x\in U} \phi_x : \prod_{x\in U} \mathcal{F}_x \to \prod_{x\in U} \mathcal{G}_x$ is a homomorphism. As above, $(\rho_{Ux})_{x\in U} : \mathcal{G}(U) \to \prod_{x\in U} \mathcal{G}_x$ is an injective (by (4.2.8), as $\mathcal{G}$ is a sheaf) homomorphism, so identifies $\mathcal{G}(U)$ with a subalgebra $T$ of $\prod_{x\in U} \mathcal{G}_x$ — its image, which

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is a subalgebra since the image of a homomorphism is always closed under the operations of the target. By the construction of $\phi^\flat_U$ above, $h(\mathcal{F}^+(U)) \subseteq T$: indeed for $s \in \mathcal{F}^+(U)$ we exhibited $\tau \in \mathcal{G}(U)$ with $\tau_x = \phi_x(s(x))$ for all $x \in U$. So $h$ restricts to a homomorphism $\mathcal{F}^+(U) \to T \cong \mathcal{G}(U)$, which is $\phi^\flat_U$.

Given this, every map constructed in the case $\mathcal{C} = \mathbf{Set}$ above is now a morphism of $\mathcal{C}$, and the remaining verifications — injectivity of $\phi \mapsto \phi|_{\mathcal{B}} \circ \eta_{\mathcal{F}}$, existence and commutativity of $\phi^\flat$, and the final isomorphism criterion — proceed entirely by manipulating elements of underlying sets via (4.2.8) and (4.2.4), both already stated for general algebraic categories, so the argument given there applies verbatim; in the final step, bijectivity of $(\eta_{\mathcal{F}})_U$ on underlying sets upgrades to an isomorphism of $\mathcal{C}$ since $U$ reflects isomorphisms (2.6.190). ∎

**Corollary 4.2.17**
*Let $\mathcal{C}$ be an algebraic category (2.6.185) then there is an equivalence of categories*

$$\mathbf{Sh}(X;\mathcal{B}) \underset{(-)|_{\mathcal{B}}}{\overset{(-)^+}{\rightleftarrows}} \mathbf{Sh}(X)$$

*Proof.* By the last statement of (4.2.16), and (2.6.73) the essential image of $(-)|_{\mathcal{B}}$ is the full subcategory of $\mathcal{B}$-sheaves, and the unit and counit natural transformations induce the equivalence of categories. ∎

**Lemma 4.2.18 (Sheafification Preserves Stalks)**
*Let $\mathcal{C}$ be algebraic (2.6.185), $\mathcal{F}$ a presheaf on $X$ and $x \in X$. Then $\eta_{\mathcal{F},x} : \mathcal{F}_x \to (\mathcal{F}^+)_x$ (4.2.7) is an isomorphism.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

**Surjectivity.** Let $\xi \in (\mathcal{F}^+)_x$; by (4.2.6), $\xi = s_x$ for some $s \in \mathcal{F}^+(U)$, $U \ni x$. By (4.2.16), $s(y) = (\sigma_i)_y$ for all $y \in U_i$, for some cover $U = \bigcup_i U_i$ and $\sigma_i \in \mathcal{F}(U_i)$; choosing $i$ with $x \in U_i$, this says $s|_{U_i} = \eta_{U_i}(\sigma_i)$, so

$$\xi = s_x = \left(s|_{U_i}\right)_x = \left(\eta_{U_i}(\sigma_i)\right)_x = \eta_x\left((\sigma_i)_x\right)$$

so $\xi \in \mathrm{Im}(\eta_x)$.

**Injectivity.** Suppose $\eta_x(a) = \eta_x(b)$ for $a, b \in \mathcal{F}_x$. By (4.2.6), $a = \sigma_x$, $b = \tau_x$ for some $\sigma \in \mathcal{F}(U)$, $\tau \in \mathcal{F}(V)$, $U, V \ni x$; then $\eta_x(a) = (\eta_U(\sigma))_x$ and $\eta_x(b) = (\eta_V(\tau))_x$, so by (4.2.8).a) there is $W \subseteq U \cap V$, $x \in W$, with $\eta_U(\sigma)|_W = \eta_V(\tau)|_W$ in $\mathcal{F}^+(W)$, that is $\eta_W(\sigma|_W) = \eta_W(\tau|_W)$. As elements of $\mathcal{F}^+(W)$ are functions $W \to \coprod_{y \in W} \mathcal{F}_y$ (4.2.16), this equality holds in particular at $y = x$:

$$(\sigma|_W)_x = \eta_W(\sigma|_W)(x) = \eta_W(\tau|_W)(x) = (\tau|_W)_x$$

that is $\sigma_x = \tau_x$, so $a = b$. ∎

## 4.2.3 Inverse Image Functor

**Definition 4.2.19 (Inverse Image Sheaf)**
*Let $f : X \to Y$ be a continuous map and $\mathcal{G}$ a presheaf on $Y$. We define as follows the **inverse image sheaf***

$$(f^{-1}\mathcal{G})(U) := \left\{ s : U \to \coprod_{x \in U} \mathcal{G}_{f(x)} \;\middle|\; \forall x \in U, \exists V \in \mathcal{O}_{f(x)} \text{ and } \sigma \in \mathcal{G}(V) \text{ s.t. } s(y) = \sigma_y \text{ for all } y \in f^{-1}(V) \right\}$$

*By definition it is a sheaf. Furthermore we see there is a canonical isomorphism*

$$\begin{aligned} (f^{-1}\mathcal{G})_x &\xrightarrow{\sim} \mathcal{G}_{f(x)} \\ (U, s) &\to \quad s(x) \end{aligned}$$

## 4.2.4 Skyscraper Sheaf

**Lemma 4.2.20**
*Consider a family of objects in $\{A_i\}_{i \in I}$ such that $i \in I_0 \implies A_i$ is terminal. Then there is a canonical isomorphism for which the following diagram commutes for all morphisms $f$*

$$\begin{array}{ccc} B & \xrightarrow{f} & \prod_{i \in I} A_i \\ & \searrow^{(\pi_i \circ f)_{i \in I_0}} & \downarrow \sim \\ & & \prod_{i \in I_0} A_i \end{array}$$

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*namely that given by* $(\pi_i)_{i\in I_0}$.

*Proof.* Let $\phi = (\pi_i)_{i\in I_0}$ and let $\psi : \prod_{i\in I_0} A_i \to \prod_{i\in I} A_i$ be the unique morphism such that

$$\pi_j \circ \psi = \begin{cases} \pi_j & j \in I_0 \\ !_{\prod_{i\in I_0} A_i} & j \notin I_0 \end{cases}$$

Then $\pi_i \circ \psi \circ \phi = \pi_i \circ \phi = \pi_i$ when $i \in I_0$ and it equals $!_{\prod_{i\in I_0} A_i} \circ \phi = !_{\prod_{i\in I} A_i} = \pi_i$ when $i \notin I_0$, whence $\psi \circ \phi = 1$. The reverse is similar. ∎

**Proposition 4.2.21**
*Let $A \in \mathcal{C}$ be an object, $\star$ a terminal object and $x \in X$. The following assignment*

$$i_x(A)(U) := \begin{cases} A & x \in U \\ \star & x \notin U \end{cases}$$

*determines a sheaf with restriction maps given by*

$$\rho_{UV} := \begin{cases} 1_A & x \in V \\ !_A & x \in U \setminus V \\ 1_\star & x \notin U \end{cases}$$

*This is known as the* ***skyscraper sheaf*** *at $x$.*

*Proof.* Suppose that $U = \bigcup_{i\in I} U_i$ and let $I_0 = \{i \in I \mid x \in U_i\}$ then we may show there is a commutative diagram

$$\begin{array}{ccccc} \mathcal{F}(U) & \xrightarrow{e} & \prod_{i\in I} \mathcal{F}(U_i) & \underset{d_1}{\overset{d_0}{\rightrightarrows}} & \prod_{(i,j)\in I\times I} \mathcal{F}(U_i \cap U_j) \\ & \searrow^{e} & \downarrow \sim & & \downarrow \sim \\ & & \prod_{i\in I_0} \mathcal{F}(U_i) & \underset{d_1}{\overset{d_0}{\rightrightarrows}} & \prod_{(i,j)\in I_0\times I_0} \mathcal{F}(U_i \cap U_j) \end{array}$$

where $e = \left(\rho_{UU_i}\right)$, $d_0 = \left(\rho_{U_i(U_i\cap U_j)} \circ \pi_i\right)$ and $d_1 = \left(\rho_{U_j(U_i\cap U_j)} \circ \pi_j\right)$. Then it's enough to show that the bottom diagram is an equaliser.

For $i, j \in I_0$ we have $\mathcal{F}(U) = \mathcal{F}(U_i) = \mathcal{F}(U_i \cap U_j) = A$.

Suppose $f : B \to \prod_{i\in I_0} \mathcal{F}(U_i)$ such that $d_0 \circ f = d_1 \circ f$. Then

$$\begin{aligned} d_0 \circ f = d_1 \circ f &\implies \pi_{(i,j)} \circ d_0 \circ f = \pi_{(i,j)} \circ d_1 \circ f \quad \forall i, j \in I_0 \\ &\implies 1_A \circ \pi_i \circ f = 1_A \circ \pi_j \circ f \quad \forall i, j \in I_0 \\ &\implies \pi_i \circ f = \pi_j \circ f \quad \forall i, j \in I_0 \end{aligned}$$

Therefore define the morphism $f' : B \to \mathcal{F}(U)$ to be $\pi_j \circ f$ for some $j \in I_0$. Observing $e = (1_A)_{i\in I_0}$ we have $e \circ f' = (f')_{i\in I_0} = (\pi_j \circ f)_{i\in I_0} = (\pi_i \circ f)_{i\in I_0} = f$. ∎

**Proposition 4.2.22 (Stalk functor is left-adjoint to Sky-Scraper Sheaf)**
*Let $X$ be a topological space, $\mathcal{C}$ a category with a terminal object $\star$ and directed colimits, $A \in \mathcal{C}$ and $\mathcal{F}$ a sheaf on $X$. Then for all $x \in X$ there is a bijection*

$$\begin{aligned} \mathrm{Mor}(\mathcal{F}_x, A) &\xrightarrow{\sim} \mathrm{Mor}(\mathcal{F}, i_x(A)) \\ \phi \to \quad \phi^\sharp_U &:= \begin{cases} \phi \circ \rho_{Ux} & x \in U \\ !_{\mathcal{F}(U)} & x \notin U \end{cases} \end{aligned}$$

*which is natural in both $\mathcal{F}$ and $A$.*

*Proof.* Let $\mathcal{N}_x$ be the category of open subsets $U$ of $X$ containing $x$ ordered by reverse inclusion. Then by definition of the stalk (4.2.7) there is a bijection

$$\begin{aligned} \mathrm{Mor}(\mathcal{F}_x, A) &\xrightarrow{\sim} \left(\mathcal{F}|_{\mathcal{N}_x} \downarrow A\right) \\ \phi &\to (\phi \circ \rho_{Ux})_{U\in\mathcal{N}_x} \end{aligned}$$

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which is natural in $\mathcal{F}$ and $A$. It then remains to verify that there is a natural bijection

$$\left(\mathcal{F}|_{\mathcal{N}_x} \downarrow A\right) \xrightarrow{\sim} \mathrm{Mor}(\mathcal{F}, i_x(A))$$

$$(\Phi_U)_{U\ni x} \to \begin{cases} \Phi_U & x \in U \\ !_{\mathcal{F}(U)} & x \notin U \end{cases}$$

■

## 4.2.5 Limit Sheaf

**Proposition 4.2.23 (Limits of Presheaves and Sheaves may be constructed pointwise)** *Let $K$ be a small category and $F : K \to \mathbf{PSh}(X;\mathcal{C})$ (resp. $\mathbf{Sh}(X;\mathcal{C})$) a diagram, that is a presheaf (resp. sheaf) $F(k)$ for each $k \in K$ together with a morphism $F(\kappa) : F(k) \to F(k')$ for each arrow $\kappa : k \to k'$, functorially. Write $F(-)(U) : K \to \mathcal{C}$ for the diagram $k \mapsto F(k)(U)$, and suppose $\lim_K F(-)(U)$ exists in $\mathcal{C}$ for every $U \subset X$.*

*Then the assignment*

$$\left(\lim_K F\right)(U) \coloneqq \lim_K F(-)(U)$$

*can be made into a presheaf (resp. sheaf) uniquely such that the projections $\lambda_k : \lim_K F \to F(k)$ are morphisms of presheaves (resp. sheaves) for every $k \in K$. Furthermore $\left(\lim_K F, \{\lambda_k\}_{k\in K}\right)$ is a limit for $F$ in the category of presheaves (resp. sheaves).*

*When $\mathcal{C}$ is an algebraic category (2.6.185) this may be constructed in the obvious way*

$$\left(\lim_K F\right)(U) \coloneqq \left\{ (\sigma_k)_{k\in K} \in \prod_{k\in K} F(k)(U) \;\middle|\; F(\kappa)_U(\sigma_k) = \sigma_{k'} \text{ for every arrow } \kappa : k \to k' \text{ in } K \right\}$$

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

**Restriction maps.** Fix $V \subseteq U$, and write $\{\lambda_{k,U}\}_{k\in K}$, $\{\lambda_{k,V}\}_{k\in K}$ for the given limit cones for $F(-)(U)$, $F(-)(V)$. For every arrow $\kappa : k \to k'$ of $K$, using that $F(\kappa) : F(k) \to F(k')$ is a morphism of presheaves and the cone condition $F(\kappa)_U \circ \lambda_{k,U} = \lambda_{k',U}$,

$$F(\kappa)_V \circ \left(\rho_{UV}^{F(k)} \circ \lambda_{k,U}\right) = \rho_{UV}^{F(k')} \circ F(\kappa)_U \circ \lambda_{k,U} = \rho_{UV}^{F(k')} \circ \lambda_{k',U}$$

so $\left\{\rho_{UV}^{F(k)} \circ \lambda_{k,U}\right\}_{k\in K}$ is a cone over $F(-)(V)$ with apex $\lim_K F(-)(U)$. By the universal property of $\lim_K F(-)(V)$, there is a unique morphism $\rho_{UV} : \lim_K F(-)(U) \to \lim_K F(-)(V)$ such that $\lambda_{k,V} \circ \rho_{UV} = \rho_{UV}^{F(k)} \circ \lambda_{k,U}$ for every $k \in K$; this is precisely the condition for each $\lambda_k$ to be a morphism of presheaves.

Taking $U = V$, both $\rho_{UU}$ and $1_{\lim_K F(-)(U)}$ satisfy $\lambda_{k,U} \circ (-) = \lambda_{k,U}$ for every $k$, so $\rho_{UU} = 1_{\lim_K F(-)(U)}$ by uniqueness. For $W \subseteq V \subseteq U$, both $\rho_{VW} \circ \rho_{UV}$ and $\rho_{UW}$ satisfy, for every $k$, $\lambda_{k,W} \circ (-) = \rho_{VW}^{F(k)} \circ \rho_{UV}^{F(k)} \circ \lambda_{k,U} = \rho_{UW}^{F(k)} \circ \lambda_{k,U}$, so $\rho_{VW} \circ \rho_{UV} = \rho_{UW}$ by uniqueness. So $\lim_K F$, with these restriction maps, is a presheaf, and each $\lambda_k$ is a morphism of presheaves; uniqueness of restriction maps making every $\lambda_k$ a morphism of presheaves is immediate from the uniqueness clause of the universal property of $\lim_K F(-)(V)$.

**Limit in the category of presheaves.** Let $\mathcal{K}$ be a presheaf and $\{\mu_k : \mathcal{K} \to F(k)\}_{k\in K}$ a cone over $F$ in $\mathbf{PSh}(X;\mathcal{C})$, that is $F(\kappa) \circ \mu_k = \mu_{k'}$ for every arrow $\kappa : k \to k'$. For each $U$, $\left\{(\mu_k)_U\right\}_{k\in K}$ is a cone over $F(-)(U)$ with apex $\mathcal{K}(U)$, so the universal property of $\lim_K F(-)(U)$ gives a unique $\mu_U : \mathcal{K}(U) \to \lim_K F(-)(U)$ with $\lambda_{k,U} \circ \mu_U = (\mu_k)_U$ for every $k$. For $V \subseteq U$, both $\rho_{UV} \circ \mu_U$ and $\mu_V \circ \rho_{UV}^{\mathcal{K}}$ satisfy, for every $k$, $\lambda_{k,V} \circ (-) = \rho_{UV}^{F(k)} \circ (\mu_k)_U = (\mu_k)_V \circ \rho_{UV}^{\mathcal{K}}$ (naturality of $\mu_k$, and the defining properties of $\mu_U, \mu_V$), so $\rho_{UV} \circ \mu_U = \mu_V \circ \rho_{UV}^{\mathcal{K}}$ by uniqueness in the universal property of $\lim_K F(-)(V)$: that is, $\mu \coloneqq \{\mu_U\}$ is a morphism of presheaves, with $\lambda_k \circ \mu = \mu_k$ for every $k$ by construction. If $\mu'$ is another morphism of presheaves with $\lambda_k \circ \mu' = \mu_k$ for every $k$, then $\mu'_U$ satisfies $\lambda_{k,U} \circ \mu'_U = (\mu_k)_U$ for every $k$, so $\mu'_U = \mu_U$ by the pointwise uniqueness above, that is $\mu' = \mu$.

**Sheaf case.** Suppose $F(k)$ is a sheaf for every $k \in K$; we show $\lim_K F$ is also a sheaf. Fix $U = \bigcup_{i\in I} U_i$, and let $I^{(1)}$, $D_-$ be as in (4.2.3).

Define a bifunctor $H : K \times I^{(1)} \to \mathcal{C}$ by $H(k,-) \coloneqq D_{F(k)}$ for $k \in K$, and $H(\kappa, 1_l) \coloneqq F(\kappa)_{U_l}$ for every arrow $\kappa : k \to k'$ of $K$ and $l \in I^{(1)}$ (writing $U_l$ for $U_i$, resp. $U_i \cap U_j$, when $l = i$, resp. $l = (i,j)$); this is a bifunctor precisely because each $F(\kappa)$ is a morphism of presheaves, so commutes with the maps $D_{F(k)}(u_{ij}), D_{F(k)}(v_{ij})$ (and likewise for $F(k')$).

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Since each $F(k)$ is a sheaf, (4.2.3) gives $\lim_{I^{(1)}} H(k,-) = F(k)(U)$ for every $k \in K$; applying (2.6.99) (with the roles of $I, J$ there taken to be our $K, I^{(1)}$), this determines a functor $L : K \to \mathcal{C}$ (2.6.96) with $L(k) = F(k)(U)$ for every $k$; as $\lim_{k\in K} L(k) = \lim_K F(-)(U)$ exists by hypothesis, (2.6.99) gives that $\lim_{K\times I^{(1)}} H$ exists, with limit cone $\left(\lim_K F(-)(U), \left\{\eta_{k,l} \circ \lambda_{k,U}\right\}_{(k,l)}\right)$, where $\left(\eta_{k,l}\right)_{l\in I^{(1)}}$ is the limit cone for $F(k)(U)$ over $I^{(1)}$ furnished by (4.2.3) in particular the component at $(k,l)$, for $l \in I^{(1)}$, is $\rho^{F(k)}_{UU_l} \circ \lambda_{k,U}$.

Applying (2.6.99) directly to $H$ (whose symmetric hypothesis, that $\lim_{k\in K} H(k,l)$ exists for every $l \in I^{(1)}$, is exactly $\lim_K F(-)(U_l) = D_{\lim_K F}(l)$ existing by hypothesis) to the cone on $\lim_K F(-)(U)$ constructed above: writing $R(l) \coloneqq \lim_{k\in K} H(k,l) = D_{\lim_K F}(l)$, $\lim_{I^{(1)}} R$ exists, with limit cone component at $l \in I^{(1)}$ the unique morphism $\sigma_l : \lim_K F(-)(U) \to \lim_K F(-)(U_l)$ satisfying $\lambda_{k,U_l} \circ \sigma_l = \rho^{F(k)}_{UU_l} \circ \lambda_{k,U}$ for every $k \in K$. This is exactly the defining property of the restriction map $\rho_{UU_l}$ of $\lim_K F$ established above, so $\sigma_l = \rho_{UU_l}$ by uniqueness.

So $D_{\lim_K F}$ has $\left(\lim_K F(-)(U), \left\{\rho_{UU_l}\right\}_{l\in I^{(1)}}\right)$ as a limit over $I^{(1)}$, and by (4.2.3) this says exactly that $\lim_K F$ satisfies the sheaf condition for this cover, as required.

Alternatively, when $\mathcal{C}$ is algebraic (2.6.185), this can be shown more explicitly, using the concrete construction noted above. By (2.6.85) and (2.6.190) (the forgetful functor $U : \mathcal{C} \to \mathbf{Set}$ creates limits), $\lim_K F(-)(U)$ is exactly $\left\{(\sigma_k)_{k\in K} \in \prod_{k\in K} F(k)(U) \;\middle|\; F(\kappa)_U(\sigma_k) = \sigma_{k'}\ \forall \kappa : k \to k'\right\}$ for every $U$, with $\lambda_{k,U}$ the coordinate projections, and $\rho_{UV}$ acting coordinatewise (restricting each $\sigma_k$).

By (4.2.4) it suffices to verify the two concrete sheaf conditions for $\lim_K F$ directly, using that each $F(k)$ satisfies them.

**Separation.** Suppose $(\sigma_k)_k, (\sigma'_k)_k \in \left(\lim_K F\right)(U)$ satisfy $\rho_{UU_i}\left((\sigma_k)_k\right) = \rho_{UU_i}\left((\sigma'_k)_k\right)$ for all $i \in I$. Since restriction acts coordinatewise, $\rho^{F(k)}_{UU_i}(\sigma_k) = \rho^{F(k)}_{UU_i}(\sigma'_k)$ for every $k \in K$ and $i \in I$, so $\sigma_k = \sigma'_k$ by separation for $F(k)$; as this holds for every $k$, $(\sigma_k)_k = (\sigma'_k)_k$.

**Gluing.** Suppose $\left(\sigma_{k,i}\right)_k \in \left(\lim_K F\right)(U_i)$ for $i \in I$ satisfy $\rho_{U_i(U_i\cap U_j)}\left(\left(\sigma_{k,i}\right)_k\right) = \rho_{U_j(U_i\cap U_j)}\left(\left(\sigma_{k,j}\right)_k\right)$ for all $(i,j) \in I \times I$. Fix $k \in K$; since restriction acts coordinatewise, $\rho^{F(k)}_{U_i(U_i\cap U_j)}(\sigma_{k,i}) = \rho^{F(k)}_{U_j(U_i\cap U_j)}(\sigma_{k,j})$ for all $(i,j)$, so gluing for $F(k)$ gives $\sigma_k \in F(k)(U)$ with $\rho^{F(k)}_{UU_i}(\sigma_k) = \sigma_{k,i}$ for all $i$. For an arrow $\kappa : k \to k'$ of $K$, and every $i \in I$,

$$\rho^{F(k')}_{UU_i}(F(\kappa)_U(\sigma_k)) = F(\kappa)_{U_i}\left(\rho^{F(k)}_{UU_i}(\sigma_k)\right) = F(\kappa)_{U_i}(\sigma_{k,i}) = \sigma_{k',i} = \rho^{F(k')}_{UU_i}(\sigma_{k'})$$

using naturality of $F(\kappa)$, the defining property of $\sigma_k$, that $\left(\sigma_{k,i}\right)_k \in \left(\lim_K F\right)(U_i)$, and the defining property of $\sigma_{k'}$. As $\{U_i\}$ covers $U$, separation for $F(k')$ gives $F(\kappa)_U(\sigma_k) = \sigma_{k'}$. As $\kappa$ was arbitrary, $(\sigma_k)_k \in \left(\lim_K F\right)(U)$, with $\rho_{UU_i}\left((\sigma_k)_k\right) = \left(\sigma_{k,i}\right)_k$ for all $i$, as required. ∎

## 4.2.6 Equaliser Sheaf

**Proposition 4.2.24 (Equaliser Sheaf may be constructed pointwise)**
*Let $\alpha, \beta : \mathcal{F} \rightrightarrows \mathcal{G}$ be morphisms of presheaves (resp. sheaves) with a family of pointwise equalisers $\{e_U : \mathrm{eq}(\alpha_U, \beta_U) \to \mathcal{F}(U)\}_{U\subset X}$. Then the assignment*

$$\mathrm{eq}(\alpha,\beta)(U) \coloneqq \mathrm{eq}(\alpha_U, \beta_U)$$

*can be made into a presheaf (resp. sheaf) uniquely such that*

$$e : \mathrm{eq}(\alpha,\beta) \to \mathcal{F}$$

*is a morphism of presheaves (resp. sheaves). Further $e$ is the equaliser of $(\alpha,\beta)$ in the category of presheaves (resp. sheaves).*

*When $\mathcal{C}$ is an algebraic category this may be constructed in the obvious way*

$$\mathrm{eq}(\alpha,\beta)(U) \coloneqq \{\sigma \in \mathcal{F}(U) \mid \alpha_U(\sigma) = \beta_U(\sigma)\}$$

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

Let $\mathbf{2} = \{\emptyset \rightrightarrows 1\}$ and let $F : \mathbf{2} \to \mathbf{PSh}(X;\mathcal{C})$ (resp. $\mathbf{Sh}(X;\mathcal{C})$) be the diagram $F(\emptyset) \coloneqq \mathcal{F}$, $F(1) \coloneqq \mathcal{G}$, sending the two morphisms $\emptyset \rightrightarrows 1$ to $\alpha, \beta$ respectively. Then $F(-)(U) : \mathbf{2} \to \mathcal{C}$ is exactly the pair $\alpha_U, \beta_U : \mathcal{F}(U) \rightrightarrows \mathcal{G}(U)$, whose limit is $\mathrm{eq}(\alpha_U, \beta_U)$ (2.6.135), which exists by hypothesis for every $U$. By (4.2.23), the assignment $U \mapsto \mathrm{eq}(\alpha_U, \beta_U)$ can therefore be made into a presheaf (resp. sheaf) uniquely such that the projection $e$ to $\mathcal{F}$ is a morphism of presheaves

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(resp. sheaves), and $\mathrm{eq}(\alpha,\beta) \coloneqq \lim_{\mathbf{2}} F$, with this projection $e$, is a limit for $F$ in the category of presheaves (resp. sheaves), that is the equaliser of $(\alpha,\beta)$ there (2.6.135), as required.

Alternatively, when $\mathcal{C}$ is algebraic (2.6.185), this follows directly from the algebraic case of (4.2.23): writing $(\sigma_\emptyset,\sigma_1) \in \mathcal{F}(U)\times\mathcal{G}(U)$ for the coordinates indexed by $\mathbf{2} = \{\emptyset \rightrightarrows 1\}$, the compatibility condition $\alpha_U(\sigma_\emptyset) = \sigma_1 = \beta_U(\sigma_\emptyset)$ pins $\sigma_1$ down in terms of $\sigma_\emptyset$, so under the projection $(\sigma_\emptyset,\sigma_1)\mapsto\sigma_\emptyset$ this recovers exactly $\mathrm{eq}(\alpha,\beta)(U) = \{\sigma\in\mathcal{F}(U) \mid \alpha_U(\sigma) = \beta_U(\sigma)\}$ noted above. ∎

**Corollary 4.2.25 (Kernel may be constructed pointwise)**
*Let $\mathcal{C}$ be a pointed category and $\alpha:\mathcal{F}\to\mathcal{G}$ a morphism of presheaves (resp. sheaves). Suppose for all $U\subset X$ we have a kernel $(i_U,\ker(\alpha_U))$ of $\alpha_U$.*

*Then we may define $\ker(\alpha)$ as follows*

$$\ker(\alpha)(U) \coloneqq \ker(\alpha_U)$$

*and this may be made into a presheaf (resp. sheaf) uniquely such that $i$ is a morphism of presheaves (resp. sheaves). Furthermore $(i,\ker(\alpha))$ is a kernel in the category of presheaves (resp. sheaves).*

*When $\mathcal{C}$ is an algebraic category (2.6.185) this may be constructed in the obvious way namely*

$$\ker(\alpha)(U) \coloneqq \{\sigma\in\mathcal{F}(U) \mid \alpha_U(\sigma) = 0\}$$

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

By (2.6.151), $(i_U,\ker(\alpha_U))$ is a kernel for $\alpha_U$ if and only if $i_U$ is an equaliser for $(\alpha_U,0_U)$, where $0:\mathcal{F}\to\mathcal{G}$ is the zero morphism of presheaves (computed pointwise, since $\mathbf{PSh}(X;\mathcal{C})$, resp. $\mathbf{Sh}(X;\mathcal{C})$, is pointed with zero morphisms computed pointwise). So the hypothesis gives a family of pointwise equalisers $\{i_U:\mathrm{eq}(\alpha_U,0_U)\to\mathcal{F}(U)\}_{U\subset X}$ for $(\alpha,0)$, and applying (4.2.24) to this pair, $\mathrm{eq}(\alpha,0)(U) \coloneqq \mathrm{eq}(\alpha_U,0_U) = \ker(\alpha_U) = \ker(\alpha)(U)$ can be made into a presheaf (resp. sheaf) uniquely such that $i$ is a morphism of presheaves (resp. sheaves), and $i$ is the equaliser of $(\alpha,0)$ in the category of presheaves (resp. sheaves); by (2.6.151) again, this is exactly $(i,\ker(\alpha))$ being a kernel for $\alpha$.

When $\mathcal{C}$ is algebraic, the algebraic case of (4.2.24), applied to $(\alpha,0)$, gives $\mathrm{eq}(\alpha,0)(U) = \{\sigma\in\mathcal{F}(U) \mid \alpha_U(\sigma) = 0_U(\sigma)\} = \{\sigma\in\mathcal{F}(U) \mid \alpha_U(\sigma) = 0\}$, that is $\ker(\alpha)(U) = \{\sigma\in\mathcal{F}(U) \mid \alpha_U(\sigma) = 0\}$, as stated. ∎

## 4.2.7 Product Presheaf

**Proposition 4.2.26 (Product Presheaf may be constructed pointwise)**
*Let $\{\mathcal{F}_i\}_{i\in I}$ be a family of presheaves (resp. sheaves) with a family of pointwise products $\left\{\left(\prod_{i\in I}\mathcal{F}_i(U),\{\pi_{i,U}\}_{i\in I}\right)\right\}_{U\subset X}$. Then the assignment*

$$\left(\prod_{i\in I}\mathcal{F}_i\right)(U) \coloneqq \prod_{i\in I}\mathcal{F}_i(U)$$

*can be made into a presheaf (resp. sheaf) uniquely such that $\pi_i:\prod_{i\in I}\mathcal{F}_i\to\mathcal{F}_i$ is a morphism of presheaves (resp. sheaves) for every $i\in I$. Furthermore $\left(\prod_{i\in I}\mathcal{F}_i,\{\pi_i\}_{i\in I}\right)$ is a product of $\{\mathcal{F}_i\}_{i\in I}$ in the category of presheaves (resp. sheaves).*

*When $\mathcal{C}$ is algebraic (2.6.185) this may be constructed in the obvious way*

$$\left(\prod_{i\in I}\mathcal{F}_i\right)(U) \coloneqq \left\{(\sigma_i)_{i\in I} \;\middle|\; \sigma_i\in\mathcal{F}_i(U)\right\}$$

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

Regard $I$ as a discrete category and let $F:I\to\mathbf{PSh}(X;\mathcal{C})$ (resp. $\mathbf{Sh}(X;\mathcal{C})$) be the diagram $F(i) \coloneqq \mathcal{F}_i$ (a diagram since $I$ is discrete, so there is nothing further to specify). Then $F(-)(U):I\to\mathcal{C}$ is exactly the discrete family $\{\mathcal{F}_i(U)\}_{i\in I}$, whose limit is $\prod_{i\in I}\mathcal{F}_i(U)$ (2.6.81), which exists by hypothesis for every $U$. By (4.2.23), the assignment $U\mapsto\prod_{i\in I}\mathcal{F}_i(U)$ can therefore be made into a presheaf (resp. sheaf) uniquely such that each projection $\pi_i$ is a morphism of presheaves (resp. sheaves), and $\left(\prod_{i\in I}\mathcal{F}_i,\{\pi_i\}_{i\in I}\right)$ is a limit for $F$ in the category of presheaves (resp. sheaves), that is a product of $\{\mathcal{F}_i\}_{i\in I}$ there (2.6.81), as required.

When $\mathcal{C}$ is algebraic, this is exactly the algebraic case of (4.2.23) specialised to the discrete category $I$, where (there being no arrows) the compatibility condition is vacuous, leaving the unconstrained tuples $(\sigma_i)_{i\in I}\in\prod_{i\in I}\mathcal{F}_i(U)$ noted above. ∎

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### 4.2.8 Kernel Pair Sheaf

**Corollary 4.2.27 (Kernel Pair Presheaf may be constructed pointwise)**
*Let $\mu : \mathcal{H} \to \mathcal{G}$ be a morphism of presheaves (resp. sheaves), and suppose for every $U \subset X$ there is a kernel pair $(Z_U, p_{1,U}, p_{2,U})$ of $\mu_U$.*

*Then we may define $(\mathcal{H} \times_{\mathcal{G}} \mathcal{H})(U) \coloneqq Z_U$ and this may be made into a presheaf (resp. sheaf) uniquely such that $p_1, p_2$ are morphisms of presheaves (resp. sheaves). Furthermore $(\mathcal{H} \times_{\mathcal{G}} \mathcal{H}, p_1, p_2)$ is a kernel pair of $\mu$ in the category of presheaves (resp. sheaves).*

*When $\mathcal{C}$ is algebraic (2.6.185) this may be constructed in the obvious way*

$$(\mathcal{H} \times_{\mathcal{G}} \mathcal{H})(U) \coloneqq \{(\sigma_1, \sigma_2) \in \mathcal{H}(U) \times \mathcal{H}(U) \mid \mu_U(\sigma_1) = \mu_U(\sigma_2)\}$$

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

Let $F : \mathbf{\Lambda} \to \mathbf{PSh}(X; \mathcal{C})$ (resp. $\mathbf{Sh}(X; \mathcal{C})$) be the diagram $F(a) \coloneqq \mathcal{H}$, $F(b) \coloneqq \mathcal{H}$, $F(c) \coloneqq \mathcal{G}$, sending both morphisms $a \to c$, $b \to c$ to $\mu$ (2.6.145). Then $F(-)(U) : \mathbf{\Lambda} \to \mathcal{C}$ is exactly the cospan $\mu_U : \mathcal{H}(U) \to \mathcal{G}(U) \leftarrow \mathcal{H}(U) : \mu_U$, whose limit is the pullback $(Z_U, p_{1,U}, p_{2,U})$ of $(\mu_U, \mu_U)$, that is a kernel pair of $\mu_U$, which exists by hypothesis for every $U$. By (4.2.23), the assignment $U \mapsto Z_U$ can therefore be made into a presheaf (resp. sheaf) uniquely such that the projections $p_1, p_2$ are morphisms of presheaves (resp. sheaves), and $(\mathcal{H} \times_{\mathcal{G}} \mathcal{H}, p_1, p_2) \coloneqq \left(\lim_{\mathbf{\Lambda}} F, p_1, p_2\right)$ is a limit for $F$ in the category of presheaves (resp. sheaves), that is a pullback of $(\mu, \mu)$, i.e. a kernel pair of $\mu$, there, as required.

When $\mathcal{C}$ is algebraic, this is the algebraic case of (4.2.23) specialised to $\mathbf{\Lambda}$: writing $(\sigma_a, \sigma_b, \sigma_c)$ for the coordinates, the compatibility condition $\mu_U(\sigma_a) = \sigma_c = \mu_U(\sigma_b)$ pins $\sigma_c$ down in terms of $\sigma_a, \sigma_b$, so under the projection $(\sigma_a, \sigma_b, \sigma_c) \mapsto (\sigma_a, \sigma_b)$ this recovers exactly $\{(\sigma_1, \sigma_2) \in \mathcal{H}(U) \times \mathcal{H}(U) \mid \mu_U(\sigma_1) = \mu_U(\sigma_2)\}$ noted above. ∎

**Proposition 4.2.28 (Monomorphisms of Presheaves and Sheaves)**
*Let $\phi : \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves (resp. sheaves). Then the following are equivalent*

*a) $\phi$ is a monomorphism*
*b) $\phi_U$ is monic in $\mathcal{C}$ for all $U \subset X$*

*If $\mathcal{C}$ is algebraic (2.6.185), then a), b) are further equivalent to*

*c) $\phi_U$ is injective for all $U \subset X$*

*If moreover $\mathcal{F}, \mathcal{G}$ are sheaves, a)–c) are equivalent to*

*d) $\phi_x$ is injective for all $x \in X$*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

*b) $\implies$ a)*: Suppose $\alpha, \beta : \mathcal{H} \to \mathcal{F}$ satisfy $\phi\alpha = \phi\beta$. Composition of morphisms of presheaves is computed pointwise, so $\phi_U \alpha_U = \phi_U \beta_U$ for every $U$; as $\phi_U$ is monic, $\alpha_U = \beta_U$ for every $U$, that is $\alpha = \beta$.

*a) $\implies$ b)*: Since $\mathcal{C}$ has products and equalisers, by (2.6.146) $\phi_U$ has a kernel pair for every $U$, so by (4.2.27) these assemble into a kernel pair $(\mathcal{F} \times_{\mathcal{G}} \mathcal{F}, p_1, p_2)$ of $\phi$ in the category of presheaves (resp. sheaves), with $U$-components $\left(\mathcal{F}(U) \times_{\mathcal{G}(U)} \mathcal{F}(U), p_{1,U}, p_{2,U}\right)$ a kernel pair of $\phi_U$. As $\phi$ is monic, (2.6.148) gives $p_1 = p_2$; since morphisms of presheaves are equal iff equal pointwise, $p_{1,U} = p_{2,U}$ for every $U$, so by (2.6.148) again (applied in $\mathcal{C}$), $\phi_U$ is monic for every $U$.

*b) $\iff$ c)* (when $\mathcal{C}$ is algebraic): immediate from (2.6.195) (monic iff injective), applied at each $U$.

*c) $\implies$ d)*: (4.2.10).

*d) $\implies$ c)*: (4.2.10). ∎

**Corollary 4.2.29 (Monomorphism of Sheaves is an Isomorphism iff Surjective on Stalks)**
*Let $\mathcal{C}$ be algebraic (2.6.185) and $m : \mathcal{F} \to \mathcal{G}$ a monomorphism of sheaves. Then $m$ is an isomorphism if and only if $m_x$ is surjective for every $x \in X$.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

By (4.2.28), $m_U$ is monic, hence injective (2.6.195) (as $\mathcal{C}$ is algebraic), for every $U \subset X$, so by (4.2.10), $m_x$ is injective for every $x \in X$.

As $\mathcal{C}$ is algebraic, $U$ reflects isomorphisms (2.6.190), so an injective morphism of $\mathcal{C}$ is an isomorphism if and only if it is surjective; applying this to $m_x$ (already known injective) for each $x$, $m_x$ is an isomorphism for every $x \in X$ if and only if $m_x$ is surjective for every $x \in X$. The result now follows from (4.2.11) ($a$) $\iff$ $d$)). ∎

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**Definition 4.2.30 (Subpresheaf, Subsheaf)**
*Let $\mathcal{C}$ be algebraic (2.6.185) and $\mathcal{F}$ a presheaf on $X$. A **subpresheaf** of $\mathcal{F}$ is an assignment, to every open $U \subset X$, of a subalgebra $\mathcal{G}(U) \subseteq U(\mathcal{F}(U))$ (2.6.193), such that $\rho^{\mathcal{F}}_{UV}(\mathcal{G}(U)) \subseteq \mathcal{G}(V)$ for every $V \subseteq U$.*

*By (2.6.194), each $\mathcal{G}(U)$ carries a unique $\mathcal{C}$-algebra structure making the inclusion $\iota_U : \mathcal{G}(U) \hookrightarrow \mathcal{F}(U)$ a morphism of $\mathcal{C}$; since $\rho^{\mathcal{F}}_{UV}$ is a homomorphism carrying $\mathcal{G}(U)$ into $\mathcal{G}(V)$, it restricts to a morphism $\rho^{\mathcal{G}}_{UV} : \mathcal{G}(U) \to \mathcal{G}(V)$ of $\mathcal{C}$, making $\mathcal{G}$ into a presheaf and $\iota := \{\iota_U\} : \mathcal{G} \to \mathcal{F}$ into a morphism of presheaves.*

*We say $\mathcal{G}$ is a **subsheaf** of $\mathcal{F}$ if moreover $\mathcal{G}$, with this presheaf structure, is a sheaf.*

**Remark 4.2.31**
*Each $\iota_U$ is injective, hence monic (2.6.195), so $\iota$ itself is monic (4.2.28). Conversely, if $\iota : \mathcal{G} \to \mathcal{F}$ is a monomorphism of presheaves (resp. sheaves), then each $\iota_U$ is injective (4.2.28) and (2.6.195), and its image $U(\iota_U)(U(\mathcal{G}(U))) \subseteq U(\mathcal{F}(U))$ is a subalgebra (being the image of a homomorphism), so $\iota$ identifies $\mathcal{G}$ with a subpresheaf (resp. subsheaf) of $\mathcal{F}$.*

**Proposition 4.2.32 (Sheafification of a Subpresheaf)**
*Let $\mathcal{C}$ be algebraic (2.6.185) and $\iota : \mathcal{F} \to \mathcal{G}$ a monomorphism of presheaves with $\mathcal{G}$ a sheaf. Define*

$$\mathcal{F}^+(U) := \left\{ \tau \in U(\mathcal{G}(U)) \;\middle|\; U = \bigcup_{i \in I} U_i \text{ for some open cover, with } \tau|_{U_i} \in U\big(\iota_{U_i}\big)(U(\mathcal{F}(U_i))) \text{ for every } i \right\}$$

*Then $\mathcal{F}^+$ is a subsheaf of $\mathcal{G}$, and the corestriction $\iota' : \mathcal{F} \to \mathcal{F}^+$ of $\iota$ is a sheafification of $\mathcal{F}$ (4.2.13) (justifying, by (2.6.64), the use of the notation $\mathcal{F}^+$ for it, as usual).*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

**$\mathcal{F}^+$ is a subsheaf.** For $\omega \in \Omega$ and $\tau_1, ..., \tau_{n_\omega} \in \mathcal{F}^+(U)$, choose covers $U = \bigcup_i U_i^{(r)}$ witnessing $\tau_r \in \mathcal{F}^+(U)$; the common refinement $\left\{U_{i_1}^{(1)} \cap \cdots \cap U_{i_{n_\omega}}^{(n_\omega)}\right\}$ is again a cover of $U$, and on each piece $W$ of it, $\tau_r|_W = \iota_W(\sigma_r)$ for some $\sigma_r \in \mathcal{F}(W)$ (restricting $\tau_r|_{U_{i_r}^{(r)}} = \iota_{U_{i_r}^{(r)}}(\sigma_r)$ further, as $\iota$ is a morphism of presheaves), so

$$\omega_{\mathcal{G}(U)}\big(\tau_1, ..., \tau_{n_\omega}\big)|_W = \omega_{\mathcal{G}(W)}(\tau_1|_W, ...) = \omega_{\mathcal{G}(W)}(\iota_W(\sigma_1), ...) = \iota_W\big(\omega_{\mathcal{F}(W)}(\sigma_1, ...)\big) \in \iota_W(\mathcal{F}(W))$$

As this holds for every $W$ in the cover, $\omega_{\mathcal{G}(U)}\big(\tau_1, ..., \tau_{n_\omega}\big) \in \mathcal{F}^+(U)$, so $\mathcal{F}^+(U)$ is a subalgebra of $\mathcal{G}(U)$ (2.6.193). Restricting a witnessing cover for $\tau \in \mathcal{F}^+(U)$ to $V \subseteq U$ shows $\rho^{\mathcal{G}}_{UV}(\mathcal{F}^+(U)) \subseteq \mathcal{F}^+(V)$, so $\mathcal{F}^+$ is a subpresheaf of $\mathcal{G}$ (4.2.30).

Separation for $\mathcal{F}^+$ is immediate, since $\mathcal{F}^+(U) \subseteq U(\mathcal{G}(U))$ and $\mathcal{G}$ is separated. For gluing, let $\tau_i \in \mathcal{F}^+(U_i)$ for a cover $U = \bigcup_i U_i$, agreeing on overlaps; as $\mathcal{G}$ is a sheaf there is $\tau \in \mathcal{G}(U)$ with $\tau|_{U_i} = \tau_i$. Each $\tau_i$ has a witnessing cover $U_i = \bigcup_k U_{i,k}$ with $\tau_i|_{U_{i,k}} \in \iota_{U_{i,k}}\big(\mathcal{F}\big(U_{i,k}\big)\big)$; then $\big\{U_{i,k}\big\}_{i,k}$ is a cover of $U$ with $\tau|_{U_{i,k}} = \tau_i|_{U_{i,k}} \in \iota_{U_{i,k}}\big(\mathcal{F}\big(U_{i,k}\big)\big)$, so $\tau \in \mathcal{F}^+(U)$. So $\mathcal{F}^+$ is a subsheaf of $\mathcal{G}$ (4.2.30). (None of this used the germ-function construction of (4.2.16).)

**$\iota'$ satisfies the universal property.** We show directly that for every sheaf $\mathcal{K}$,

$$(-)^\sharp : \mathrm{Mor}(\mathcal{F}^+, \mathcal{K}) \to \mathrm{Mor}(\mathcal{F}, \mathcal{K}), \quad \phi \mapsto \phi \circ \iota'$$

is a bijection; it is then manifestly natural in $\mathcal{K}$, being given by post-composition with the fixed $\iota'$, so this establishes (4.2.13) for $(\iota', \mathcal{F}^+)$.

*Construction of $\psi^\flat$.* Let $\psi : \mathcal{F} \to \mathcal{K}$ and $\tau \in \mathcal{F}^+(U)$, with witnessing cover $U = \bigcup_i U_i$, $\tau|_{U_i} = \iota_{U_i}(\sigma_i)$ for (unique, as $\iota$ is monic) $\sigma_i \in \mathcal{F}(U_i)$. On overlaps, $\iota_{U_i \cap U_j}\big(\sigma_i|_{U_i \cap U_j}\big) = \tau|_{U_i \cap U_j} = \iota_{U_i \cap U_j}\big(\sigma_j|_{U_i \cap U_j}\big)$, so as $\iota$ is monic, $\sigma_i|_{U_i \cap U_j} = \sigma_j|_{U_i \cap U_j}$, whence $\psi_{U_i}(\sigma_i)|_{U_i \cap U_j} = \psi_{U_j}(\sigma_j)|_{U_i \cap U_j}$. As $\mathcal{K}$ is a sheaf there is therefore a unique $\psi^\flat_U(\tau) \in \mathcal{K}(U)$ with $\psi^\flat_U(\tau)|_{U_i} = \psi_{U_i}(\sigma_i)$ for every $i$.

*Independence of the witnessing cover.* If $U = \bigcup_j V_j$, $\tau|_{V_j} = \iota_{V_j}(\rho_j)$ is another witnessing cover, then on $U_i \cap V_j$, $\iota_{U_i \cap V_j}\big(\sigma_i|_{U_i \cap V_j}\big) = \tau|_{U_i \cap V_j} = \iota_{U_i \cap V_j}\big(\rho_j|_{U_i \cap V_j}\big)$, so (as $\iota$ monic) $\sigma_i|_{U_i \cap V_j} = \rho_j|_{U_i \cap V_j}$, whence $\psi_{U_i}(\sigma_i)|_{U_i \cap V_j} = \psi_{V_j}(\rho_j)|_{U_i \cap V_j}$; as $\big\{U_i \cap V_j\big\}_{i,j}$ covers $U$, the two candidate values for $\psi^\flat_U(\tau)$ agree on a common refinement, so are equal by separation of $\mathcal{K}$.

*$\psi^\flat_U$ is a homomorphism.* Given $\tau_1, ..., \tau_{n_\omega} \in \mathcal{F}^+(U)$, using the common refinement cover $\{W\}$ from the first paragraph (with local representatives $\sigma_r \in \mathcal{F}(W)$ for $\tau_r$), independence of witnessing cover gives $\psi^\flat_U(\tau_r)|_W = \psi_W(\sigma_r)$; as this same cover witnesses $\omega_{\mathcal{F}^+(U)}\big(\tau_1, ..., \tau_{n_\omega}\big) \in \mathcal{F}^+(U)$ with local representatives $\omega_{\mathcal{F}(W)}(\sigma_1, ...)$ (shown above), independence of witnessing cover also gives $\psi^\flat_U\big(\omega_{\mathcal{F}^+(U)}(\tau_1, ...)\big)|_W = \psi_W\big(\omega_{\mathcal{F}(W)}(\sigma_1, ...)\big) = \omega_{\mathcal{K}(W)}(\psi_W(\sigma_1), ...) = \omega_{\mathcal{K}(U)}\big(\psi^\flat_U(\tau_1), ...\big)|_W$ for every $W$, so by separation of $\mathcal{K}$, $\psi^\flat_U\big(\omega_{\mathcal{F}^+(U)}\big(\tau_1, ..., \tau_{n_\omega}\big)\big) = \omega_{\mathcal{K}(U)}\big(\psi^\flat_U(\tau_1), ..., \psi^\flat_U\big(\tau_{n_\omega}\big)\big)$.

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*$\psi^\flat$ is a morphism of presheaves.* For $V \subseteq U$ and $\tau \in \mathcal{F}^+(U)$ with witnessing cover $\{U_i\}$, $\{U_i \cap V\}$ witnesses $\tau|_V$, so $\psi^\flat_V(\tau|_V)|_{U_i\cap V} = \psi_{U_i\cap V}\left(\sigma_i|_{U_i\cap V}\right) = \psi_{U_i}(\sigma_i)|_{U_i\cap V} = \psi^\flat_U(\tau)|_{U_i\cap V}$ for every $i$, so $\psi^\flat_V(\tau|_V) = \psi^\flat_U(\tau)|_V$ by separation of $\mathcal{K}$.

$\psi^{\flat\sharp} = \psi$. For $\sigma \in \mathcal{F}(U)$, $\iota'_U(\sigma) = \iota_U(\sigma) \in \mathcal{F}^+(U)$ is witnessed by the trivial cover $\{U\}$ with representative $\sigma$ itself, so $\psi^\flat_U(\iota'_U(\sigma)) = \psi_U(\sigma)$, that is $\psi^\flat \circ \iota' = \psi$.

*Uniqueness.* If $\phi : \mathcal{F}^+ \to \mathcal{K}$ also satisfies $\phi \circ \iota' = \psi$, then for $\tau \in \mathcal{F}^+(U)$ with witnessing cover $\{U_i\}$, $\phi_U(\tau)|_{U_i} = \phi_{U_i}\left(\tau|_{U_i}\right) = \phi_{U_i}\left(\iota'_{U_i}(\sigma_i)\right) = \psi_{U_i}(\sigma_i) = \psi^\flat_U(\tau)|_{U_i}$ for every $i$, so $\phi_U(\tau) = \psi^\flat_U(\tau)$ by separation of $\mathcal{K}$; as $U, \tau$ were arbitrary, $\phi = \psi^\flat$.

So $(-)^\sharp$ is a bijection with inverse $\psi \mapsto \psi^\flat$, for every sheaf $\mathcal{K}$: $(\iota', \mathcal{F}^+)$ is a sheafification of $\mathcal{F}$ (4.2.13). ∎

## 4.2.9 Coequaliser Sheaf

**Proposition 4.2.33 (Coequaliser Presheaf)**
*Let $\alpha, \beta : \mathcal{F} \rightrightarrows \mathcal{G}$ be morphisms of presheaves with a family of pointwise coequalisers $\{p_U : \mathcal{G}(U) \to \operatorname{coeq}(\alpha_U, \beta_U)\}_{U\subset X}$. Then we may define*

$$\operatorname{coeq}(\alpha, \beta)(U) \coloneqq \operatorname{coeq}(\alpha_U, \beta_U)$$

*and this may be made into a presheaf uniquely such that $p$ is a morphism of presheaves. Furthermore $(p, \operatorname{coeq}(\alpha, \beta))$ is a coequaliser in the category of presheaves.*

*In particular if $\mathcal{C}$ has all coequalisers then so does $\mathbf{PSh}(X; \mathcal{C})$.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

We require to show that for all open subsets $V \subset U$ there is a unique restriction morphism $\rho_{UV} : \operatorname{coeq}(\alpha_U, \beta_U) \to \operatorname{coeq}(\alpha_V, \beta_V)$ ensuring the following diagram commutes

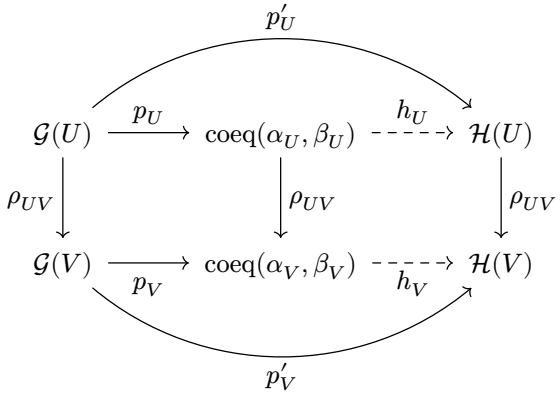
$$\begin{array}{ccc} \mathcal{G}(U) & \xrightarrow{p_U} & \operatorname{coeq}(\alpha_U, \beta_U) \\ {\scriptstyle \rho_{UV}}\downarrow & & \downarrow{\scriptstyle \rho_{UV}} \\ \mathcal{G}(V) & \xrightarrow[p_V]{} & \operatorname{coeq}(\alpha_V, \beta_V) \end{array}$$

We have that

$$p_V \circ \rho_{UV} \circ \alpha_U = p_V \circ \alpha_V \circ \rho_{UV} = p_V \circ \beta_V \circ \rho_{UV} = p_V \circ \rho_{UV} \circ \beta_U$$

whence by the universal property applied to $\operatorname{coeq}(\alpha_U, \beta_U)$ there exists a unique factorisation of $p_V \circ \rho_{UV}$ through $p_U$. The uniqueness then shows that this satisfies the conditions to be a presheaf ($\rho_{UU} = 1$ and $\rho_{VW} \circ \rho_{UV} = \rho_{UW}$ both follow from uniqueness of the factorisation of $p_W \circ \rho_{UW}$ through $p_U$), and $p$ is a morphism of presheaves.

We now show that $p$ is a coequaliser of $(\alpha, \beta)$ in the category of presheaves. Evidently $p \circ \alpha = p \circ \beta$, since this holds pointwise by hypothesis. Suppose that $p' : \mathcal{G} \to \mathcal{H}$ is a morphism of presheaves with $p' \circ \alpha = p' \circ \beta$, then $p'_U \circ \alpha_U = p'_U \circ \beta_U$ whence there is a unique factorisation $h_U$ such that $h_U \circ p_U = p'_U$.




The large square and left-hand square commute, which we may use to show that

$$h_V \circ \rho_{UV} \circ p_U = h_V \circ p_V \circ \rho_{UV} = p'_V \circ \rho_{UV} = \rho_{UV} \circ p'_U = \rho_{UV} \circ h_U \circ p_U$$

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As $p_U$ is a coequaliser it is an epimorphism (2.6.136), whence $h_V \circ \rho_{UV} = \rho_{UV} \circ h_U$, which shows that the right-hand square commutes. This means precisely that $h$ is a morphism of presheaves, and by construction $h \circ p = p'$.

Finally suppose $h'$ is another morphism of presheaves with $h' \circ p = p'$. Then for every $U$, $h'_U \circ p_U = p'_U = h_U \circ p_U$, so by the uniqueness of the pointwise factorisation $h'_U = h_U$, that is $h' = h$.

For the final statement, if $\mathcal{C}$ has all coequalisers then the family of pointwise coequalisers $\{p_U : \mathcal{G}(U) \to \mathrm{coeq}(\alpha_U, \beta_U)\}_{U \subset X}$ exists for every pair $\alpha, \beta : \mathcal{F} \rightrightarrows \mathcal{G}$ of morphisms of presheaves, so the above applies to every such pair and $\mathbf{PSh}(X;\mathcal{C})$ has all coequalisers. ∎

**Proposition 4.2.34**
*Let $\alpha, \beta : \mathcal{F} \rightrightarrows \mathcal{G}$ be morphisms of sheaves and $p : \mathcal{G} \to \mathrm{coeq}^{\mathrm{psh}}(\alpha, \beta)$ be a coequaliser in $\mathbf{PSh}(X;\mathcal{C})$. Then*

$$\eta \circ p : \mathcal{G} \to \mathrm{coeq}(\alpha, \beta) := \left(\mathrm{coeq}^{\mathrm{psh}}(\alpha, \beta)\right)^+$$

*is a coequaliser in $\mathbf{Sh}(X;\mathcal{C})$.*

*In particular if $\mathcal{C}$ has all coequalisers then so does $\mathbf{Sh}(X;\mathcal{C})$.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

By (4.2.13) and (4.2.14), $(-)^+$ is left adjoint to the inclusion $\mathbf{Sh}(X;\mathcal{C}) \hookrightarrow \mathbf{PSh}(X;\mathcal{C})$ with unit $\eta$, and this inclusion is (trivially) fully faithful, since a morphism of sheaves is by definition just a morphism of the underlying presheaves. Hence this is an instance of (2.6.144), applied to this reflective subcategory with the coequaliser $p$ of $(\alpha, \beta)$ in $\mathbf{PSh}(X;\mathcal{C})$.

For the final statement, if $\mathcal{C}$ has all coequalisers then by (4.2.33) $\mathbf{PSh}(X;\mathcal{C})$ has all coequalisers, so the hypotheses of this proposition are satisfied for every pair $(\alpha, \beta)$ of morphisms of sheaves, and $\mathbf{Sh}(X;\mathcal{C})$ has all coequalisers. ∎

**Definition 4.2.35 (Cokernel Presheaf, Cokernel Sheaf)**
*Let $\mathcal{C}$ be pointed and algebraic (2.6.185) and $\alpha : \mathcal{F} \to \mathcal{G}$ a morphism of presheaves. Define the **cokernel presheaf** of $\alpha$ by*

$$\mathrm{coker}^{\mathrm{psh}}(\alpha) := \mathrm{coeq}^{\mathrm{psh}}(\alpha, 0), \quad q_U(\sigma) := \sigma \bmod \alpha_U(\mathcal{F}(U)) \quad (4.2.33)$$

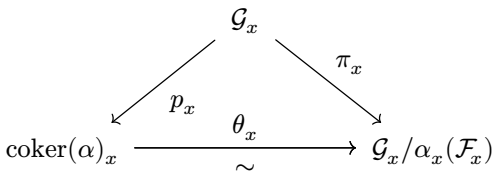
*with $q : \mathcal{G} \to \mathrm{coker}^{\mathrm{psh}}(\alpha)$ the coequaliser projection of $\alpha$ and the zero morphism in $\mathbf{PSh}(X;\mathcal{C})$. When $\mathcal{F}, \mathcal{G}$ are sheaves, define the **cokernel sheaf** of $\alpha$ by*

$$\mathrm{coker}(\alpha)(U) := \left\{ \tau : U \to \coprod_{x \in U} \mathcal{G}_x / \alpha_x(\mathcal{F}_x) \;\middle|\; U = \bigcup_i U_i \text{ for some open cover, and } \sigma_i \in \mathcal{G}(U_i) \right.$$
$$\left. \text{such that } \tau(x) = \sigma_{i,x} \bmod \alpha_x(\mathcal{F}_x) \text{ for all } x \in U_i \right\}$$

*with restriction maps given by restricting functions, and the **cokernel map** $p : \mathcal{G} \to \mathrm{coker}(\alpha)$ by $p_U(\sigma)(x) := \sigma_x \bmod \alpha_x(\mathcal{F}_x)$.*

**Remark 4.2.36**
*When $\alpha$ is (in particular) a monomorphism, each $\alpha_U$ is injective (4.2.28), identifying $\mathcal{F}$ with a subsheaf of $\mathcal{G}$; in this case we also write $\mathcal{G}/\mathcal{F} := \mathrm{coker}(\alpha)$.*

**Lemma 4.2.37 (Cokernel is a Sheaf, and Commutes with Stalks)**
*Let $\mathcal{C}$ be pointed and algebraic, $\alpha : \mathcal{F} \to \mathcal{G}$ a morphism of sheaves, and $p$ as in (4.2.35). Then $\mathrm{coker}(\alpha)$ is a sheaf, $(\mathrm{coker}(\alpha), p)$ is a coequaliser of $(\alpha, 0)$ in $\mathbf{Sh}(X;\mathcal{C})$, and, for $x \in X$ there is a unique morphism $\theta_x : \mathrm{coker}(\alpha)_x \to \mathcal{G}_x/\alpha_x(\mathcal{F}_x)$ making*




*commute, and $\theta_x$ is (moreover) an isomorphism.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

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**An auxiliary sheafification-based model.** By (4.2.34), $p^0 := \eta \circ q : \mathcal{G} \to \operatorname{coker}^0(\alpha) := \left(\operatorname{coker}^{\mathrm{psh}}(\alpha)\right)^+$ is a coequaliser of $(\alpha, 0)$ in $\mathbf{Sh}(X;\mathcal{C})$, with $q, \operatorname{coker}^{\mathrm{psh}}(\alpha)$ as in (4.2.35) and $\eta$ the sheafification unit (4.2.16) this auxiliary model is used only within this proof, to relate the definition above to the general existence of coequalisers.

**$\eta_x$ is an isomorphism.** (4.2.18).

**$q_x$ is surjective with kernel $\alpha_x(\mathcal{F}_x)$.** Since $\operatorname{coker}^{\mathrm{psh}}(\alpha)(U) = \mathcal{G}(U)/\alpha_U(\mathcal{F}(U))$ (4.2.33), by definition of stalk $\left(\operatorname{coker}^{\mathrm{psh}}(\alpha)\right)_x = \varinjlim_{U \ni x} \mathcal{G}(U)/\alpha_U(\mathcal{F}(U))$, with $q_x(\sigma_x)$ the class of $\sigma$ in this direct limit for $\sigma \in \mathcal{G}(U)$, $U \ni x$.

*Surjective*: every element of the direct limit is represented by the class of some $\sigma \in \mathcal{G}(U)$, $U \ni x$, that is $q_x(\sigma_x)$.

*Kernel* $\supseteq \alpha_x(\mathcal{F}_x)$: for $\rho \in \mathcal{F}_x$, represented by $\tau \in \mathcal{F}(U)$, $\alpha_x(\rho) = (\alpha_U(\tau))_x$; as $\alpha_U(\tau) \in \alpha_U(\mathcal{F}(U))$, its class in $\mathcal{G}(U)/\alpha_U(\mathcal{F}(U))$ is 0, so $q_x(\alpha_x(\rho)) = 0$.

*Kernel* $\subseteq \alpha_x(\mathcal{F}_x)$: suppose $\sigma \in \mathcal{G}(U)$, $U \ni x$, with $q_x(\sigma_x) = 0$, that is, the class of $\sigma$ becomes 0 in the direct limit. By (4.2.6) applied to the presheaf $U \mapsto \mathcal{G}(U)/\alpha_U(\mathcal{F}(U))$, this means there is $W \subseteq U$, $x \in W$, with $\sigma|_W \in \alpha_W(\mathcal{F}(W))$, that is $\sigma|_W = \alpha_W(\tau)$ for some $\tau \in \mathcal{F}(W)$; then $\sigma_x = (\sigma|_W)_x = (\alpha_W(\tau))_x = \alpha_x(\tau_x) \in \alpha_x(\mathcal{F}_x)$.

**Existence and uniqueness of $\vartheta^0_x$ with $\vartheta^0_x \circ q_x = \pi_x$.** Since $q_x$ is surjective with kernel $\alpha_x(\mathcal{F}_x)$, it is (up to the canonical comparison isomorphism) the quotient map $\mathcal{G}_x \to \mathcal{G}_x/\alpha_x(\mathcal{F}_x)$, hence satisfies the corresponding universal property: any morphism out of $\mathcal{G}_x$ vanishing on $\ker(q_x)$ factors uniquely through $q_x$. As $\pi_x$ vanishes on $\ker(\pi_x) = \alpha_x(\mathcal{F}_x) = \ker(q_x)$, this gives a unique $\vartheta^0_x : \left(\operatorname{coker}^{\mathrm{psh}}(\alpha)\right)_x \to \mathcal{G}_x/\alpha_x(\mathcal{F}_x)$ with $\vartheta^0_x \circ q_x = \pi_x$; uniqueness alone accounts for this, and says nothing yet about $\vartheta^0_x$ being an isomorphism.

**$\vartheta^0_x$ is an isomorphism.** This is a separate check. *Surjective*: given $z \in \mathcal{G}_x/\alpha_x(\mathcal{F}_x)$, as $\pi_x$ is surjective $z = \pi_x(g)$ for some $g \in \mathcal{G}_x$, and $z = \vartheta^0_x(q_x(g))$. *Injective*: every element of $\left(\operatorname{coker}^{\mathrm{psh}}(\alpha)\right)_x$ is $q_x(g)$ for some $g$, since $q_x$ is surjective; if $\vartheta^0_x(q_x(g)) = 0$ then $\pi_x(g) = 0$, that is $g \in \ker(\pi_x) = \ker(q_x)$, so $q_x(g) = 0$.

**$\vartheta_x := \vartheta^0_x \circ \eta_x^{-1}$ is an isomorphism $\operatorname{coker}^0(\alpha)_x \to \mathcal{G}_x/\alpha_x(\mathcal{F}_x)$ with $\vartheta_x \circ p^0_x = \pi_x$.** It is an isomorphism, being a composite of the isomorphisms $\eta_x^{-1}$ and $\vartheta^0_x$. As $p^0_x = \eta_x \circ q_x$,

$$\vartheta_x \circ p^0_x = \vartheta^0_x \circ \eta_x^{-1} \circ \eta_x \circ q_x = \vartheta^0_x \circ q_x = \pi_x$$

**Relating $\operatorname{coker}^0(\alpha)$ to $\operatorname{coker}(\alpha)$.** By (4.2.16), elements of $\operatorname{coker}^0(\alpha)(U)$ are functions $\tau^0 : U \to \coprod_{x \in U} \operatorname{coker}^0(\alpha)_x$ such that, for some open cover $U = \bigcup_i U_i$ and $\sigma_i \in \mathcal{G}(U_i)$, $\tau^0(x) = p^0_{U_i}(\sigma_i)_x$ for $x \in U_i$. Using the isomorphisms $\vartheta_x$, define

$$\vartheta : \coprod_{x \in X} \operatorname{coker}^0(\alpha)_x \longrightarrow \coprod_{x \in X} \mathcal{G}_x/\alpha_x(\mathcal{F}_x), \quad \vartheta|_{\operatorname{coker}^0(\alpha)_x} := \vartheta_x$$

a bijection, being a disjoint union of the bijections $\vartheta_x$. Set $\Psi_U(\tau^0) := \vartheta \circ \tau^0$ for $\tau^0 \in \operatorname{coker}^0(\alpha)(U)$.

*$\Psi_U$ is a well-defined bijection onto* $\operatorname{coker}(\alpha)(U)$. As $\vartheta$ is a bijection carrying each fibre $\operatorname{coker}^0(\alpha)_x$ to $\mathcal{G}_x/\alpha_x(\mathcal{F}_x)$, post-composition by $\vartheta$ is a bijection from functions $U \to \coprod_{x \in U} \operatorname{coker}^0(\alpha)_x$ to functions $U \to \coprod_{x \in U} \mathcal{G}_x/\alpha_x(\mathcal{F}_x)$, carrying the representing data $(U_i, \sigma_i)$ for $\tau^0$ to representing data for $\Psi_U(\tau^0)$: $\vartheta\left(p^0_{U_i}(\sigma_i)_x\right) = \vartheta_x\left(p^0_{U_i}(\sigma_i)_x\right) = \sigma_{i,x} \bmod \alpha_x(\mathcal{F}_x)$, so $\Psi_U(\tau^0)(x) = \sigma_{i,x} \bmod \alpha_x(\mathcal{F}_x)$ for $x \in U_i$, that is, exactly membership in $\operatorname{coker}(\alpha)(U)$ (4.2.35). Hence $\Psi_U$ restricts to a bijection $\operatorname{coker}^0(\alpha)(U) \to \operatorname{coker}(\alpha)(U)$.

*$\Psi$ is an isomorphism of presheaves, and* $\operatorname{coker}(\alpha)$ *is a sheaf.* For $V \subseteq U$ and $\tau^0 \in \operatorname{coker}^0(\alpha)(U)$, $\Psi_V(\tau^0|_V) = \vartheta \circ (\tau^0|_V) = (\vartheta \circ \tau^0)|_V = \Psi_U(\tau^0)|_V$, since restricting a function and then composing with $\vartheta$ agrees with composing then restricting; hence $\Psi$ commutes with restriction, and is a natural transformation, bijective at every $U$ — an isomorphism of presheaves $\operatorname{coker}^0(\alpha) \cong \operatorname{coker}(\alpha)$. As $\operatorname{coker}^0(\alpha)$ is a sheaf and the sheaf condition transports across an isomorphism of presheaves, $\operatorname{coker}(\alpha)$ is also a sheaf.

$\Psi \circ p^0 = p$. For $\sigma \in \mathcal{G}(U)$, $x \in U$, $\Psi_U(p^0_U(\sigma))(x) = \vartheta_x(p^0_U(\sigma)_x) = \sigma_x \bmod \alpha_x(\mathcal{F}_x) = p_U(\sigma)(x)$, again using $\vartheta_x \circ p^0_x = \pi_x$.

**$(\operatorname{coker}(\alpha), p)$ is a coequaliser of $(\alpha, 0)$.** Since $p^0 \circ \alpha = p^0 \circ 0$ (as $p^0$ coequalises $(\alpha, 0)$) and $p = \Psi \circ p^0$, also $p \circ \alpha = \Psi \circ p^0 \circ \alpha = \Psi \circ p^0 \circ 0 = p \circ 0$, so $(\operatorname{coker}(\alpha), p)$ is a co-cone for $(\alpha, 0)$; and $\Psi$, an isomorphism of presheaves between sheaves, is also an isomorphism of $\mathbf{Sh}(X;\mathcal{C})$ (this inclusion into $\mathbf{PSh}(X;\mathcal{C})$ being fully faithful (4.2.34)). As $(\operatorname{coker}^0(\alpha), p^0)$ is a coequaliser of $(\alpha, 0)$, that is, a colimit of the corresponding diagram (2.6.141), (2.6.90) shows $(\operatorname{coker}(\alpha), p)$ is also such a colimit, that is, a coequaliser of $(\alpha, 0)$.

**Conclusion.** Set $\theta_x := \vartheta_x \circ \Psi_x^{-1}$, an isomorphism, being a composite of the isomorphisms $\vartheta_x$ and $\Psi_x^{-1}$. Since $p_x = \Psi_x \circ p^0_x$ (4.2.7),

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$$\theta_x \circ p_x = \vartheta_x \circ \Psi_x^{-1} \circ \Psi_x \circ p_x^0 = \vartheta_x \circ p_x^0 = \pi_x$$

and $\theta_x$ is unique with this property, since $p_x = \Psi_x \circ p_x^0$ is surjective (a composite of the surjections $\Psi_x$, an isomorphism, and $p_x^0 = \eta_x \circ q_x$), hence epic, so at most one morphism out of $\operatorname{coker}(\alpha)_x$ can satisfy $\theta_x \circ p_x = \pi_x$. ∎

## 4.2.10 Image Sheaf

Unlike kernels and equalisers (4.2.24), the image is not itself a limit — it is initial in the category of factorisations through a monomorphism (2.6.168) — so the argument via (4.2.3) and (2.6.99) used there does not apply, and the pointwise image presheaf need not satisfy the sheaf condition even when $\mathcal{F}, \mathcal{G}$ are sheaves; instead an image sheaf is obtained by sheafifying the image presheaf (4.2.40). We do however first record the following easier pointwise construction of an image presheaf, whenever $\mathcal{C}$ has regular epi-mono factorisations.

**Lemma 4.2.38 (Regular Epimorphisms of Presheaves may be Determined Pointwise)**
*Let $e : \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves such that $e_U$ is a regular epimorphism in $\mathcal{C}$ for every $U \subset X$. Then $e$ is a regular epimorphism of $\mathbf{PSh}(X; \mathcal{C})$.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

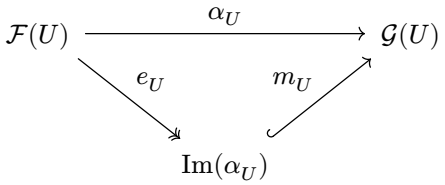
Since $\mathcal{C}$ has products and equalisers, by (2.6.146) $e_U$ has a kernel pair for every $U$, so by (4.2.27) these assemble into a kernel pair $(\mathcal{F} \times_{\mathcal{G}} \mathcal{F}, p_1, p_2)$ of $e$ in $\mathbf{PSh}(X; \mathcal{C})$, with $U$-components $\left(\mathcal{F}(U) \times_{\mathcal{G}(U)} \mathcal{F}(U), p_{1,U}, p_{2,U}\right)$ a kernel pair of $e_U$.

Since $e_U$ is regular, (2.6.150) shows $e_U$ is the coequaliser of $\left(p_{1,U}, p_{2,U}\right)$, for every $U$. So applying (4.2.33) to the pair of morphisms of presheaves $(p_1, p_2) : \mathcal{F} \times_{\mathcal{G}} \mathcal{F} \rightrightarrows \mathcal{F}$, this pointwise family of coequalisers assembles into a presheaf $\operatorname{coeq}(p_1, p_2)$ with restriction maps uniquely determined by making the induced map $q := \{e_U\}$ a morphism of presheaves, and $(q, \operatorname{coeq}(p_1, p_2))$ is a coequaliser of $(p_1, p_2)$ in $\mathbf{PSh}(X; \mathcal{C})$.

The presheaf $\mathcal{G}$ also has $U$-component $\mathcal{G}(U) = \operatorname{coeq}(p_1, p_2)(U)$, with restriction maps uniquely determined by the same condition, that $e$ is a morphism of presheaves; by this uniqueness, $\mathcal{G} = \operatorname{coeq}(p_1, p_2)$ as presheaves and $e = q$. Hence $e$ is a coequaliser of presheaves, i.e. a regular epimorphism of $\mathbf{PSh}(X; \mathcal{C})$. ∎

**Proposition 4.2.39 (Image Presheaf may be constructed pointwise)**
*If $\mathcal{C}$ has regular epi-mono factorisations for every morphism, e.g.* **Set** *(2.6.176) or an Abelian category (2.7.7), then $\mathbf{PSh}(X; \mathcal{C})$ has regular epi-mono factorisations for every morphism.*

*More precisely, let $\alpha : \mathcal{F} \to \mathcal{G}$ be a morphism of presheaves with a family of pointwise regular epi-mono factorisations $\{\alpha_U = m_U \circ e_U\}_{U \subset X}$*



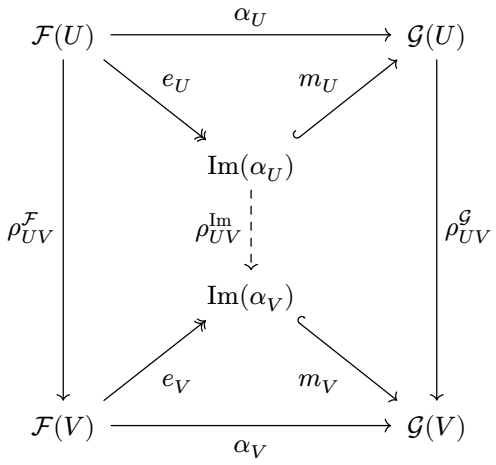

*Then the assignment $\operatorname{Im}(\alpha)(U) := \operatorname{Im}(\alpha_U)$ can be made into a presheaf uniquely such that $m_\alpha : \operatorname{Im}(\alpha) \to \mathcal{G}$, $(m_\alpha)_U := m_U$, and $e_\alpha : \mathcal{F} \to \operatorname{Im}(\alpha)$, $(e_\alpha)_U := e_U$, are morphisms of presheaves, and $m_\alpha \circ e_\alpha = \alpha$; furthermore this is a regular epi-mono factorisation of $\alpha$ in $\mathbf{PSh}(X; \mathcal{C})$, and hence $(\operatorname{Im}(\alpha), e_\alpha, m_\alpha)$ is an image of $\alpha$ there (2.6.175).*

*If moreover $\mathcal{C}$ is algebraic (2.6.185), the pointwise factorisations may be taken as in (2.6.197), namely $\operatorname{Im}(\alpha_U) := U(\alpha_U)(U(\mathcal{F}(U))) \subseteq U(\mathcal{G}(U))$; then $\operatorname{Im}(\alpha)$ is simply the subpresheaf of $\mathcal{G}$ of sections pointwise in the image of $\alpha$ (4.2.30), with restriction maps inherited from $\mathcal{G}$ and $m_\alpha$ the inclusion.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

**Restriction maps.** Fix $V \subseteq U$. Since $\alpha$ is a morphism of presheaves, $\rho^{\mathcal{G}}_{UV} \circ \alpha_U = \alpha_V \circ \rho^{\mathcal{F}}_{UV}$, so applying (2.6.178) to the regular epi-mono factorisations $\alpha_U = m_U \circ e_U$, $\alpha_V = m_V \circ e_V$ with $a := \rho^{\mathcal{F}}_{UV}$, $b := \rho^{\mathcal{G}}_{UV}$ gives a unique morphism $\rho^{\operatorname{Im}}_{UV} : \operatorname{Im}(\alpha_U) \to \operatorname{Im}(\alpha_V)$ making the following diagram commute

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namely

$$\rho^{\mathrm{Im}}_{UV} \circ e_U = e_V \circ \rho^{\mathcal{F}}_{UV} \quad \text{and} \quad m_V \circ \rho^{\mathrm{Im}}_{UV} = \rho^{\mathcal{G}}_{UV} \circ m_U$$

The first identity is precisely the condition for $e_\alpha$ to be a morphism of presheaves, and the second for $m_\alpha$ to be one.

**$\mathrm{Im}(\alpha)$ is a presheaf.** By the functoriality clause of (2.6.178) (taking $a = 1_{\mathcal{F}(U)}$, $b = 1_{\mathcal{G}(U)}$), $\rho^{\mathrm{Im}}_{UU} = 1_{\mathrm{Im}(\alpha_U)}$; and for $W \subseteq V \subseteq U$, taking $a := \rho^{\mathcal{F}}_{UV}$, $b := \rho^{\mathcal{G}}_{UV}$, $c := \rho^{\mathcal{F}}_{VW}$, $d := \rho^{\mathcal{G}}_{VW}$, the composition law gives $\rho^{\mathrm{Im}}_{VW} \circ \rho^{\mathrm{Im}}_{UV} = \rho^{\mathrm{Im}}_{UW}$ (using $\rho^{\mathcal{F}}_{VW} \circ \rho^{\mathcal{F}}_{UV} = \rho^{\mathcal{F}}_{UW}$, and likewise for $\mathcal{G}$). So $\mathrm{Im}(\alpha)$, with these restriction maps, is a presheaf, and $e_\alpha$, $m_\alpha$ are morphisms of presheaves. Uniqueness of restriction maps making $e_\alpha$, $m_\alpha$ into morphisms of presheaves is immediate from the uniqueness clause of (2.6.178), since these two identities are exactly what is required of $e_\alpha, m_\alpha$.

Finally $(m_\alpha \circ e_\alpha)_U = m_U \circ e_U = \alpha_U$ for every $U$, so $m_\alpha \circ e_\alpha = \alpha$.

**Regularity.** By (4.2.38) (using that $\mathcal{C}$ has products and equalisers, and each $e_U$ is regular), $e_\alpha$ is a regular epimorphism of $\mathbf{PSh}(X;\mathcal{C})$. Since $m_U$ is monic in $\mathcal{C}$ for every $U$, $m_\alpha$ is monic in $\mathbf{PSh}(X;\mathcal{C})$ (4.2.28). So $\alpha = m_\alpha \circ e_\alpha$ is a regular epi-mono factorisation of $\alpha$ in $\mathbf{PSh}(X;\mathcal{C})$ (2.6.174), and hence also an image of $\alpha$ there (2.6.175).

For the first statement, if $\mathcal{C}$ has regular epi-mono factorisations for every morphism then every $\alpha_U : \mathcal{F}(U) \to \mathcal{G}(U)$ has one, giving such a family for every morphism $\alpha$ of presheaves, so the above applies to every such $\alpha$ and $\mathbf{PSh}(X;\mathcal{C})$ has regular epi-mono factorisations for every morphism.

**The algebraic case.** If $\mathcal{C}$ is algebraic, taking $\mathrm{Im}(\alpha_U) := U(\alpha_U)(U(\mathcal{F}(U)))$ as in (2.6.197), the restriction maps constructed above may be identified with those inherited from $\mathcal{G}$: since $\alpha$ is a morphism of presheaves, for $\sigma \in \mathcal{F}(U)$, $\rho^{\mathcal{G}}_{UV}(\alpha_U(\sigma)) = \alpha_V\big(\rho^{\mathcal{F}}_{UV}(\sigma)\big) \in \mathrm{Im}(\alpha_V)$, so $\rho^{\mathcal{G}}_{UV}(\mathrm{Im}(\alpha_U)) \subseteq \mathrm{Im}(\alpha_V)$; by the uniqueness clause of (2.6.178), this restriction of $\rho^{\mathcal{G}}_{UV}$ is exactly $\rho^{\mathrm{Im}}_{UV}$. So in this case $\mathrm{Im}(\alpha)$ is simply the subpresheaf of $\mathcal{G}$ of sections pointwise in the image of $\alpha$ (4.2.30), with $e_\alpha$ the (pointwise surjective) corestriction of $\alpha$ and $m_\alpha$ the inclusion — without needing (2.6.178) to construct the restriction maps in the first place. ∎

**Proposition 4.2.40 (Image Sheaf)**
*Let $\mathcal{C}$ be algebraic (2.6.185), hence have regular epi-mono factorisations for every morphism (2.6.197). Then $\mathbf{Sh}(X;\mathcal{C})$ has regular epi-mono factorisations for every morphism.*

*More precisely, let $\alpha : \mathcal{F} \to \mathcal{G}$ be a morphism of sheaves, and define*

$$\mathrm{Im}(\alpha)(U) := \left\{ \tau \in U(\mathcal{G}(U)) \;\middle|\; U = \bigcup_{i\in I} U_i \text{ for some open cover, with } \tau|_{U_i} \in U\big(\alpha_{U_i}\big)(U(\mathcal{F}(U_i))) \text{ for every } i \right\}$$

*the subsheaf of $\mathcal{G}$ of sections locally in the (pointwise) image of $\alpha$ (4.2.32). Then*

$$e : \mathcal{F} \longrightarrow \mathrm{Im}(\alpha), e_U(\sigma) := \alpha_U(\sigma) \quad \text{and} \quad m : \mathrm{Im}(\alpha) \hookrightarrow \mathcal{G} \text{ the inclusion}$$

*are morphisms of sheaves with $m \circ e = \alpha$, and this is a regular epi-mono factorisation of $\alpha$ in $\mathbf{Sh}(X;\mathcal{C})$; hence $(\mathrm{Im}(\alpha), e, m)$ is an image of $\alpha$ there (2.6.175).*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

**Setup.** Let $\big(\mathrm{Im}^{\mathrm{psh}}(\alpha), e_\alpha, m_\alpha\big)$ be the image of $\alpha$ in $\mathbf{PSh}(X;\mathcal{C})$, so $\alpha = m_\alpha \circ e_\alpha$ is a regular epi-mono factorisation there (4.2.39), with $\mathrm{Im}^{\mathrm{psh}}(\alpha)(U) = \mathrm{Im}(\alpha_U) \subseteq \mathcal{G}(U)$ and $(m_\alpha)_U$ the inclusion. In particular $m_\alpha$ is a monomorphism of presheaves (4.2.28), and "locally in the image of $m_\alpha$" is the same condition as "locally in the image of $\alpha$", so applying

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(4.2.32) to $m_\alpha$: $\mathrm{Im}(\alpha)$ (as defined above) is a subsheaf of $\mathcal{G}$, and the corestriction $\iota' : \mathrm{Im}^{\mathrm{psh}}(\alpha) \to \mathrm{Im}(\alpha)$ of $m_\alpha$ is a sheafification of $\mathrm{Im}^{\mathrm{psh}}(\alpha)$. Since $\iota'$ is a corestriction of the injective map $(m_\alpha)_U$ for every $U$, $\iota'$ is itself monic (4.2.28).

**Factorisation.** For $\sigma \in \mathcal{F}(U)$, $\iota'_U\big((e_\alpha)_U(\sigma)\big) = (m_\alpha)_U\big((e_\alpha)_U(\sigma)\big) = \alpha_U(\sigma)$, so $e = \iota' \circ e_\alpha$; as $\mathcal{F}$ and $\mathrm{Im}(\alpha)$ are sheaves and $\iota', e_\alpha$ are morphisms of presheaves, $e$ is a morphism of sheaves. Writing $j : \mathrm{Im}(\alpha) \hookrightarrow \mathcal{G}$ for the inclusion, $j \circ \iota' = m_\alpha$, so

$$m \circ e = j \circ \iota' \circ e_\alpha = m_\alpha \circ e_\alpha = \alpha$$

**Monicity.** $m$ is the inclusion of the subsheaf $\mathrm{Im}(\alpha)$ into $\mathcal{G}$, hence monic by the remark following (4.2.30).

**Regularity.** By (4.2.27), $e_\alpha$ has a kernel pair $\big(\mathcal{F} \times_{\mathrm{Im}^{\mathrm{psh}}(\alpha)} \mathcal{F}, p_1, p_2\big)$ in $\mathbf{PSh}(X;\mathcal{C})$, with $U$-components $\big\{(\sigma,\tau) \in \mathcal{F}(U)^2 \;\big|\; (e_\alpha)_U(\sigma) = (e_\alpha)_U(\tau)\big\}$; as each $(e_\alpha)_U = e_U$ is regular (4.2.38) exhibits $e_\alpha$ as its coequaliser there.

As $\iota'$ is monic, $(e_\alpha)_U(\sigma) = (e_\alpha)_U(\tau)$ if and only if $\iota'_U\big((e_\alpha)_U(\sigma)\big) = \iota'_U\big((e_\alpha)_U(\tau)\big)$, that is $e_U(\sigma) = e_U(\tau)$, so the $U$-components of the kernel pairs of $e_\alpha$ and of $e$ agree, with the same projections $p_1, p_2$ into $\mathcal{F}$; by the uniqueness clause of (4.2.27), the two kernel-pair presheaves coincide. Since $\mathcal{F}$ and $\mathrm{Im}(\alpha)$ are both sheaves, applying (4.2.27) to $e$ directly, in its sheaf case, shows this common kernel pair is itself a sheaf, with $p_1, p_2$ morphisms of sheaves.

It remains to show $e$ is a coequaliser of $(p_1, p_2)$ in $\mathbf{Sh}(X;\mathcal{C})$. Let $q' : \mathcal{F} \to \mathcal{K}$ be a morphism of sheaves with $q' \circ p_1 = q' \circ p_2$. As $e_\alpha$ is a coequaliser of $(p_1, p_2)$ in $\mathbf{PSh}(X;\mathcal{C})$, there is a unique morphism of presheaves $\tilde{q} : \mathrm{Im}^{\mathrm{psh}}(\alpha) \to \mathcal{K}$ with $\tilde{q} \circ e_\alpha = q'$. Since $\mathcal{K}$ is a sheaf, the bijection $(-)^\sharp : \mathrm{Mor}(\mathrm{Im}(\alpha), \mathcal{K}) \to \mathrm{Mor}\big(\mathrm{Im}^{\mathrm{psh}}(\alpha), \mathcal{K}\big)$, $\phi \mapsto \phi \circ \iota'$, from the universal property of $\iota'$ (4.2.32) gives a unique $\bar{q} : \mathrm{Im}(\alpha) \to \mathcal{K}$ with $\bar{q} \circ \iota' = \tilde{q}$. Then

$$\bar{q} \circ e = \bar{q} \circ \iota' \circ e_\alpha = \tilde{q} \circ e_\alpha = q'$$

For uniqueness, if $\bar{q}' : \mathrm{Im}(\alpha) \to \mathcal{K}$ also satisfies $\bar{q}' \circ e = q'$, then $(\bar{q}' \circ \iota') \circ e_\alpha = \bar{q}' \circ e = q' = \tilde{q} \circ e_\alpha$, and as $e_\alpha$ is epic (2.6.136), $\bar{q}' \circ \iota' = \tilde{q}$; by uniqueness in the bijection $(-)^\sharp$, $\bar{q}' = \bar{q}$.

So $(p_1, p_2) : \mathcal{F} \times_{\mathrm{Im}^{\mathrm{psh}}(\alpha)} \mathcal{F} \rightrightarrows \mathcal{F}$ are morphisms of $\mathbf{Sh}(X;\mathcal{C})$ with coequaliser $e$ there. In particular $e$ is a regular epimorphism of $\mathbf{Sh}(X;\mathcal{C})$, so $\alpha = m \circ e$ is a regular epi-mono factorisation of $\alpha$ in $\mathbf{Sh}(X;\mathcal{C})$ (2.6.174), and hence $(\mathrm{Im}(\alpha), e, m)$ is an image of $\alpha$ there (2.6.175). ∎

**Remark 4.2.41**
*As established in the **Setup** above, $\iota' : \mathrm{Im}^{\mathrm{psh}}(\alpha) \to \mathrm{Im}(\alpha)$ is a sheafification of the pointwise image presheaf $\mathrm{Im}^{\mathrm{psh}}(\alpha)$ (4.2.39): this recovers the abstract description of the image sheaf, promised at the start of this subsection, as obtained by sheafifying the image presheaf, from the concrete construction above.*

**Corollary 4.2.42 (Stalk Criterion for Surjectivity)**
*Let $\mathcal{C}$ be algebraic (2.6.185), $\alpha : \mathcal{F} \to \mathcal{G}$ a morphism of sheaves, and $\alpha = m \circ e$ a regular epi-mono factorisation of $\alpha$ in $\mathbf{Sh}(X;\mathcal{C})$ (4.2.40). Then $m$ is an isomorphism if and only if $\alpha_x$ is surjective for every $x \in X$.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

**The canonical factorisation.** Write $\alpha = m_0 \circ e_0$ for the canonical factorisation of (4.2.40), where $e_0 = \eta \circ e_\alpha$, with $\big(\mathrm{Im}^{\mathrm{psh}}(\alpha), e_\alpha, m_\alpha\big)$ the image of $\alpha$ in $\mathbf{PSh}(X;\mathcal{C})$, built from pointwise regular epi-mono factorisations $\{\alpha_U = m_U \circ e_U\}_{U \subset X}$ (4.2.39), and $\eta : \mathrm{Im}^{\mathrm{psh}}(\alpha) \to \mathrm{Im}(\alpha) =: \big(\mathrm{Im}^{\mathrm{psh}}(\alpha)\big)^+$ the corestriction of $m_\alpha$, a sheafification of $\mathrm{Im}^{\mathrm{psh}}(\alpha)$ (4.2.32), i.e. the unit of the sheafification as defined in (4.2.13).

Fix $x \in X$. As $\mathcal{C}$ is algebraic and $e_U$ is a regular epimorphism of $\mathcal{C}$ for every $U$, (2.6.196) shows $e_U$ is surjective for every $U$. Given $\xi \in \mathrm{Im}^{\mathrm{psh}}(\alpha)_x$, by (4.2.6) $\xi = \iota_x$ for some $\iota \in \mathrm{Im}^{\mathrm{psh}}(\alpha)(U) = \mathrm{Im}(\alpha_U)$, $U \ni x$; as $e_U$ is surjective, $\iota = e_U(\sigma)$ for some $\sigma \in \mathcal{F}(U)$, so $\xi = \iota_x = (e_U(\sigma))_x = (e_\alpha)_x(\sigma_x)$. So $(e_\alpha)_x$ is surjective.

By (4.2.18), $\eta_x$ is an isomorphism, in particular surjective. Hence $(e_0)_x = \eta_x \circ (e_\alpha)_x$, being a composite of surjections, is surjective.

Since $\alpha_x = (m_0)_x \circ (e_0)_x$ (4.2.7) with $(e_0)_x$ surjective, $\alpha_x$ is surjective if and only if $(m_0)_x$ is surjective: if $(m_0)_x$ is surjective this is immediate, as a composite of surjections; conversely if $\alpha_x$ is surjective then $\mathrm{Im}\big((m_0)_x\big) \supseteq \mathrm{Im}(\alpha_x) = \mathcal{G}_x$. So by (4.2.29), $m_0$ is an isomorphism if and only if $\alpha_x$ is surjective for every $x \in X$.

**General regular epi-mono factorisations.** Let $\alpha = m \circ e$ be any regular epi-mono factorisation of $\alpha$ in $\mathbf{Sh}(X;\mathcal{C})$. By (2.6.179), applied to this and the canonical factorisation $\alpha = m_0 \circ e_0$, there is an isomorphism $\phi$ with $m_0 \circ \phi = m$. So $m$ is an isomorphism if and only if $m_0$ is (composing with the isomorphism $\phi$ on the right preserves being an isomorphism, in either direction), which by the above holds if and only if $\alpha_x$ is surjective for every $x \in X$. ∎

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### 4.2.11 Exactness

**Definition 4.2.43 (Exact Sequence of Sheaves)**
*Let $\mathcal{C}$ be pointed and algebraic; then $\mathbf{Sh}(X;\mathcal{C})$ is evidently pointed (zero morphisms computed pointwise, the constant zero presheaf being a zero object), with a regular epi-mono factorisation (4.2.40) for every morphism, so (2.6.201) applies with $\mathbf{Sh}(X;\mathcal{C})$ in place of $\mathcal{C}$: a pair of composable morphisms of sheaves $\mathcal{F} \xrightarrow{\alpha} \mathcal{G} \xrightarrow{\beta} \mathcal{H}$ is **exact at $\mathcal{G}$** if $\mathrm{Im}(\alpha) = \ker(\beta)$, that is $(\mathrm{Im}(\alpha), m)$ is (literally) a kernel of $\beta$, for any (equivalently, by the same argument, every) choice of image factorisation $\alpha = m \circ e$ — in particular we may as well take the canonical construction of (4.2.40).*

**Proposition 4.2.44 (Concrete Criterion for Exactness of Sheaves)**
*Let $\mathcal{C}$ be pointed and algebraic and $\mathcal{F} \xrightarrow{\alpha} \mathcal{G} \xrightarrow{\beta} \mathcal{H}$ a pair of composable morphisms of sheaves. Regard $\mathrm{Im}(\alpha)$ (4.2.40) and $\ker(\beta)$ (4.2.25) as the canonical constructions, both literal subsheaves of $\mathcal{G}$ since $\mathcal{C}$ is algebraic. Then the following are equivalent*

*a) the pair is exact at $\mathcal{G}$ (4.2.43) (for any choice of image factorisation of $\alpha$)*
*b) $\mathrm{Im}(\alpha)(U) = \ker(\beta)(U)$, as subsets of $\mathcal{G}(U)$, for every open $U \subset X$*
*c) $\beta \circ \alpha = 0$, and every section of $\ker(\beta)(U)$ is locally in the image of $\alpha$, for every open $U \subset X$*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

Write $\alpha = m \circ e$ for the canonical image factorisation (4.2.40) and $i : \ker(\beta) \hookrightarrow \mathcal{G}$ for the canonical kernel (4.2.25).

*a) $\iff$ b)* **Reduction to the canonical representatives.** By (4.2.43) (applied with $\mathbf{Sh}(X;\mathcal{C})$ in place of $\mathcal{C}$), exactness at $\mathcal{G}$ does not depend on the choice of image factorisation of $\alpha$, so it suffices to treat the canonical representative $m$: the pair is exact at $\mathcal{G}$ if and only if $(\mathrm{Im}(\alpha), m)$ is a kernel of $\beta$. Since $(\ker(\beta), i)$ is also a kernel of $\beta$, by (2.6.80) this holds if and only if there is an isomorphism $\theta : \mathrm{Im}(\alpha) \to \ker(\beta)$ with $i \circ \theta = m$.

($\Longleftarrow$) If $\mathrm{Im}(\alpha)(U) = \ker(\beta)(U)$ for every $U$, then $m = i$ as morphisms of sheaves, both being the inclusion of this common subsheaf into $\mathcal{G}$, with the (unique, (2.6.194)) induced structure making it a morphism. So $\theta := 1_{\mathrm{Im}(\alpha)}$ satisfies $i \circ \theta = i = m$.

($\Longrightarrow$) Suppose there is an isomorphism $\theta : \mathrm{Im}(\alpha) \to \ker(\beta)$ with $i \circ \theta = m$. Fix $U \subset X$ and $\sigma \in \mathrm{Im}(\alpha)(U)$; since $m_U$ is the inclusion of $\mathrm{Im}(\alpha)(U)$ into $\mathcal{G}(U)$, $i_U(\theta_U(\sigma)) = m_U(\sigma) = \sigma$, and $i_U$ is also an inclusion, so $\theta_U(\sigma) = \sigma$; hence $\sigma \in \ker(\beta)(U)$, giving $\mathrm{Im}(\alpha)(U) \subseteq \ker(\beta)(U)$. As $\theta$ is an isomorphism, $\theta_U$ is a bijection onto $\ker(\beta)(U)$, so every element of $\ker(\beta)(U)$ equals $\theta_U(\sigma) = \sigma$ for some $\sigma \in \mathrm{Im}(\alpha)(U)$, giving $\ker(\beta)(U) \subseteq \mathrm{Im}(\alpha)(U)$. So $\mathrm{Im}(\alpha)(U) = \ker(\beta)(U)$ for every $U$.

*b) $\iff$ c)* Condition *b)* is, for every $U$, the conjunction of $\mathrm{Im}(\alpha)(U) \subseteq \ker(\beta)(U)$ and $\ker(\beta)(U) \subseteq \mathrm{Im}(\alpha)(U)$; the latter is precisely "every section of $\ker(\beta)(U)$ is locally in the image of $\alpha$" (4.2.40), so it remains to show $\mathrm{Im}(\alpha)(U) \subseteq \ker(\beta)(U)$ for every $U$ is equivalent to $\beta \circ \alpha = 0$.

($\Longleftarrow$) Suppose $\beta \circ \alpha = 0$. For $\tau \in \mathrm{Im}(\alpha)(U)$, take a witnessing cover $U = \bigcup_j U_j$ with $\tau|_{U_j} = \alpha_{U_j}(\sigma_j)$ for some $\sigma_j \in \mathcal{F}(U_j)$; then $\beta_U(\tau)|_{U_j} = \beta_{U_j}(\tau|_{U_j}) = \beta_{U_j}(\alpha_{U_j}(\sigma_j)) = (\beta \circ \alpha)_{U_j}(\sigma_j) = 0$ for every $j$, so by separation of $\mathcal{H}$ (4.2.4), $\beta_U(\tau) = 0$, i.e. $\tau \in \ker(\beta)(U)$.

($\Longrightarrow$) Suppose $\mathrm{Im}(\alpha)(U) \subseteq \ker(\beta)(U)$ for every $U$. For any $U$ and $\sigma \in \mathcal{F}(U)$, $\alpha_U(\sigma) \in \mathrm{Im}(\alpha)(U)$ (witnessed by the trivial cover $\{U\}$), so $\alpha_U(\sigma) \in \ker(\beta)(U)$, that is $\beta_U(\alpha_U(\sigma)) = 0$. As $U, \sigma$ were arbitrary, $\beta \circ \alpha = 0$. ∎

**Lemma 4.2.45 (Subsheaves are Determined by Stalks)**
*Let $\mathcal{C}$ be algebraic and $\mathcal{G}_1, \mathcal{G}_2$ be subsheaves of a sheaf $\mathcal{G}$ (4.2.30). Then $\mathcal{G}_1(U) = \mathcal{G}_2(U)$, as subsets of $\mathcal{G}(U)$, for every open $U \subset X$, if and only if $(\mathcal{G}_1)_x = (\mathcal{G}_2)_x$, as subsets of $\mathcal{G}_x$ (via the canonical, injective (4.2.28) stalk maps of the inclusions), for every $x \in X$.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

($\Longrightarrow$) Immediate: if $\mathcal{G}_1(U) = \mathcal{G}_2(U)$ for every $U$, then by (4.2.6) every element of $(\mathcal{G}_1)_x$ is $\sigma_x$ for some $\sigma \in \mathcal{G}_1(U) = \mathcal{G}_2(U)$, $U \ni x$, hence lies in $(\mathcal{G}_2)_x$, and symmetrically.

($\Longleftarrow$) Suppose $(\mathcal{G}_1)_x = (\mathcal{G}_2)_x$ for every $x \in X$. Fix $U \subset X$ and $\sigma \in \mathcal{G}_1(U) \subseteq \mathcal{G}(U)$; we show $\sigma \in \mathcal{G}_2(U)$ (the reverse inclusion, and hence $\mathcal{G}_1(U) = \mathcal{G}_2(U)$, then follows by symmetry).

For $x \in U$, $\sigma_x \in (\mathcal{G}_1)_x = (\mathcal{G}_2)_x$, so by (4.2.6) $\sigma_x = \tau_x$ for some $\tau \in \mathcal{G}_2(V_x) \subseteq \mathcal{G}(V_x)$, $x \in V_x$; by (4.2.8).a) (applied in $\mathcal{G}$, as $\sigma, \tau$ agree at the stalk $x$) there is $W_x \subseteq U \cap V_x$, $x \in W_x$, with $\sigma|_{W_x} = \tau|_{W_x}$ in $\mathcal{G}(W_x)$; as $\mathcal{G}_2$ is a subpresheaf, $\tau|_{W_x} \in \mathcal{G}_2(W_x)$, so $\sigma|_{W_x} \in \mathcal{G}_2(W_x)$.

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So $\{W_x\}_{x\in U}$ is an open cover of $U$ with $\sigma|_{W_x}\in\mathcal{G}_2(W_x)$ for every $x$; these restrictions agree pairwise on overlaps (all restricting from the common $\sigma$), so as $\mathcal{G}_2$ is a sheaf, gluing gives $\rho\in\mathcal{G}_2(U)$ with $\rho|_{W_x}=\sigma|_{W_x}$ for every $x$. Regarding $\rho$ as an element of $\mathcal{G}(U)$ (via the inclusion $\mathcal{G}_2\hookrightarrow\mathcal{G}$), $\rho$ and $\sigma$ agree on the cover $\{W_x\}$ of $U$, so by separation of $\mathcal{G}$ (4.2.4), $\rho=\sigma$; hence $\sigma\in\mathcal{G}_2(U)$. ∎

**Lemma 4.2.46 (Kernels and Stalks Commute)**
*Let $\mathcal{C}$ be pointed and algebraic, $\beta:\mathcal{G}\to\mathcal{H}$ a morphism of sheaves, and $\ker(\beta)(U)=\{\sigma\in\mathcal{G}(U)\mid\beta_U(\sigma)=0\}$ its kernel (4.2.25), regarded as a subsheaf of $\mathcal{G}$. Then for every $x\in X$, the canonical stalk inclusion identifies $\ker(\beta)_x$ with the subset $\ker(\beta_x):=\{\xi\in\mathcal{G}_x\mid\beta_x(\xi)=0\}$ of $\mathcal{G}_x$.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

Write $i:\ker(\beta)\hookrightarrow\mathcal{G}$ for the inclusion. By (2.6.151) and (2.6.136), $i$ is monic, so by (4.2.28).d) (as $\mathcal{C}$ is algebraic and $\ker(\beta),\mathcal{G}$ are sheaves), $i_x$ is injective for every $x\in X$.

Since $\beta\circ i=0$ (2.6.151), stalk-functoriality (4.2.7) gives $\beta_x\circ i_x=(\beta\circ i)_x=0_x=0$ (the stalk of the zero presheaf, valued 0 at every open with identity restrictions, is 0, so the stalk functor sends the zero morphism to the zero morphism). So $i_x(\eta)\in\ker(\beta_x)$ for every $\eta\in\ker(\beta)_x$.

Conversely let $\xi\in\ker(\beta_x)$, i.e. $\xi\in\mathcal{G}_x$ with $\beta_x(\xi)=0$. By (4.2.6), $\xi=\sigma_x$ for some $\sigma\in\mathcal{G}(U)$, $U\in\mathcal{B}$, $x\in U$. Then $(\beta_U(\sigma))_x=\beta_x(\sigma_x)=\beta_x(\xi)=0=0_x$, so by (4.2.8).a) there is $W\in\mathcal{B}$, $W\subseteq U$, $x\in W$, with $\beta_U(\sigma)|_W=0|_W$, that is $\beta_W(\sigma|_W)=0$, so $\sigma|_W\in\ker(\beta)(W)$. Then $i_x\big((\sigma|_W)_x\big)=(i_W(\sigma|_W))_x=(\sigma|_W)_x=\sigma_x=\xi$, so $\xi\in\mathrm{Im}(i_x)$.

So $i_x$ is injective with image exactly $\ker(\beta_x)$, that is, identifies $\ker(\beta)_x$ with this subset of $\mathcal{G}_x$. ∎

**Lemma 4.2.47 (Images and Stalks Commute)**
*Let $\mathcal{C}$ be algebraic, $\alpha:\mathcal{F}\to\mathcal{G}$ a morphism of sheaves, and $\alpha=m\circ e$ its image factorisation (4.2.40). Then for every $x\in X$, the canonical stalk inclusion identifies $\mathrm{Im}(\alpha)_x$ with the subset $\mathrm{Im}(\alpha_x):=\alpha_x(\mathcal{F}_x)$ of $\mathcal{G}_x$.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

Retain the notation from the proof of (4.2.42): writing $e=\eta\circ e_\alpha$ with $\big(\mathrm{Im}^{\mathrm{psh}}(\alpha),e_\alpha,m_\alpha\big)$ the image of $\alpha$ in $\mathbf{PSh}(X;\mathcal{C})$ (4.2.39) and $\eta$ the sheafification unit exhibiting $\mathrm{Im}(\alpha)=\big(\mathrm{Im}^{\mathrm{psh}}(\alpha)\big)^+$, it is shown there that $(e_\alpha)_x$ is surjective (each $e_U$ being regular, hence surjective, (2.6.196)) and $\eta_x$ is an isomorphism (4.2.18), so $e_x=\eta_x\circ(e_\alpha)_x$ is surjective. Also $m_x$ is injective by (4.2.28).d), as $m$ is a monomorphism of sheaves.

Since $m_x\circ e_x=\alpha_x$ and $e_x$ is surjective onto $\mathrm{Im}(\alpha)_x$,

$$m_x(\mathrm{Im}(\alpha)_x)=m_x(e_x(\mathcal{F}_x))=\alpha_x(\mathcal{F}_x)=\mathrm{Im}(\alpha_x)$$

so, $m_x$ being injective, it identifies $\mathrm{Im}(\alpha)_x$ with the subset $\mathrm{Im}(\alpha_x)$ of $\mathcal{G}_x$. ∎

**Proposition 4.2.48 (Stalk Criterion for Exactness)**
*Let $\mathcal{C}$ be pointed and algebraic and $\mathcal{F}\xrightarrow{\alpha}\mathcal{G}\xrightarrow{\beta}\mathcal{H}$ a pair of composable morphisms of sheaves. Then this pair is exact at $\mathcal{G}$ (4.2.43) if and only if, for every $x\in X$, $\mathrm{Im}(\alpha_x)=\ker(\beta_x)$, as subsets of $\mathcal{G}_x$, that is, the stalk pair $\mathcal{F}_x\xrightarrow{\alpha_x}\mathcal{G}_x\xrightarrow{\beta_x}\mathcal{H}_x$ is exact at $\mathcal{G}_x$ in $\mathcal{C}$ (in the sense of (4.2.43)), applied verbatim in $\mathcal{C}$).*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

By (4.2.44), the pair is exact at $\mathcal{G}$ if and only if $\mathrm{Im}(\alpha)(U)=\ker(\beta)(U)$ for every $U\subset X$ (using the canonical constructions of $\mathrm{Im}(\alpha),\ker(\beta)$). By (4.2.45) applied to these subsheaves of $\mathcal{G}$, this holds if and only if $\mathrm{Im}(\alpha)_x=\ker(\beta)_x$, as subsets of $\mathcal{G}_x$, for every $x\in X$. By (4.2.47) and (4.2.46), this is exactly the condition $\mathrm{Im}(\alpha_x)=\ker(\beta_x)$, as subsets of $\mathcal{G}_x$, for every $x\in X$. ∎

## 4.2.12 Constant Sheaves

**Definition 4.2.49**
*Let $A\in\mathcal{C}$ and $X$ a topological space. Define the **constant presheaf** as follows*

$$\mathcal{A}^{\mathrm{pre}}(U):=A$$

*with restriction maps the identity.*

**Proposition 4.2.50**
*Let $X$ be an irreducible topological space then the constant presheaf $\mathcal{A}^{\mathrm{pre}}(U)=A$ is a sheaf, which we denote by $\underline{A}$.*

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*Proof.* Suppose that $U = \bigcup_{i \in I} U_i$ where $U_i$ are non-empty, then as $U$ is irreducible (4.1.70), by definition every $U_i \cap U_j \neq \emptyset$. Therefore for sections on $U_i$ are only compatible if they are equal, and so there is an obvious extension to $U$. ∎

**Proposition 4.2.51 (Quotient of Constant Sheaf)**
*Let $\mathcal{F}$ be a subsheaf of the constant sheaf $\underline{A}$, define the quotient presheaf as follows*

$$(\underline{A}/\mathcal{F})(U) \coloneqq \left\{ \sigma : U \to \coprod_{x \in U} A/\mathcal{F}_x \;\middle|\; \sigma(x) \in A/\mathcal{F}_x \right\}$$

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## 4.3 Space with Functions

**Definition 4.3.1 (Space with functions)**
*Let $X$ be a topological space and $\mathcal{O}_X$ be a subsheaf of the $\overline{k}$-valued functions on $X$ (which is a sheaf of $k$-algebras). Then we say $(X, \mathcal{O}_X)$ is a **space with functions**. Additionally it's **local** if it satisfies the condition*

*a) For every $f \in \mathcal{O}_X(U)$ we have $D(f) := \{f \neq 0\}$ is open and $\frac{1}{f} \in \mathcal{O}_X(D(f))$.*

*Note that the sheaf restriction maps $\rho_{UV} : \mathcal{O}_X(U) \to \mathcal{O}_X(V)$ correspond simply to restrictions of functions.*

*A local space with functions is a locally ringed space over $k$ (4.3.5).*

*A **regular morphism** or **regular map** $(X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ is a continuous function $\phi : X \to Y$ such that*

$$g \in \mathcal{O}_Y(U) \implies g \circ \phi \in \mathcal{O}_X(\phi^{-1}(U))$$

*This determines a morphism of sheaves*

$$\phi^\sharp : \mathcal{O}_Y \longrightarrow \phi_\star \mathcal{O}_X$$

*We say a regular morphism is **dominant** if $\phi(X)$ is dense in $Y$.*

*An isomorphism is a homeomorphism and induces an isomorphism of sheaves, or equivalently has a two-sided regular inverse.*

**Proposition 4.3.2**
*The composition of regular maps is well-defined and associative.*

### 4.3.1 Local Rings

**Definition 4.3.3 (Local Ring)**
*Let $X$ be a local space with functions and $W \subset X$ an irreducible subset. Then we define the **local ring** at $W$ to be (see (4.2.5))*

$$\mathcal{O}_{X,W} := \varinjlim_{U \cap W \neq \emptyset} \mathcal{O}_X(U)$$

*If $\overline{W} = \overline{W'}$ then $\mathcal{O}_{X,W} = \mathcal{O}_{X,W'}$, so for example $\mathcal{O}_{X,W} = \mathcal{O}_{X,\overline{W}}$.*

*It is a local ring with unique maximal ideal*

$$\mathfrak{m}_{X,W} := \{[(V, \sigma)] \mid V \cap W \neq \emptyset, \sigma \in \mathcal{O}_X(V), \sigma(x) = 0 \, \forall x \in V \cap W\}$$

*Define the **residue field***

$$k(\mathfrak{m}_{X,W}) := \mathcal{O}_{X,W}/\mathfrak{m}_{X,W}$$

*In the case $W = \{x\}$ we write $\mathcal{O}_{X,x}$, $\mathfrak{m}_{X,x}$ and $k(\mathfrak{m}_{X,x})$.*

*Proof.* The open sets $U$ with $U \cap W \neq \emptyset$ are directed by reverse inclusion because $W$ is irreducible, see (4.1.60).

Consider $(V, \sigma) \in \mathcal{O}_{X,W} \setminus \mathfrak{m}_{X,W}$. Then by definition $D(\sigma) \cap W \neq \emptyset$ and $(D(\sigma), \sigma^{-1})$ is an explicit inverse. ∎

**Corollary 4.3.4 (Field of Rational Functions)**
*Let $X$ be an irreducible space with functions. Then $\mathcal{O}_{X,X} =: k(X)$ is a field, which we call the **field of rational functions**.*

**Proposition 4.3.5 (Local Spaces with Functions are Locally Ringed Spaces)**
*Let $(X, \mathcal{O}_X)$ be a local space with functions. Then $(X, \mathcal{O}_X)$ is a locally ringed space over $k$, that is*

*a) $\mathcal{O}_X$ is a sheaf of $k$-algebras*
*b) for every $x \in X$ the stalk $\mathcal{O}_{X,x}$ is a local $k$-algebra with maximal ideal $\mathfrak{m}_{X,x}$, and evaluation at $x$ induces a $k$-embedding $k(\mathfrak{m}_{X,x}) \hookrightarrow \overline{k}$*
*c) every regular map $\phi : X \to Y$ of local spaces with functions induces a local homomorphism of $k$-algebras $\phi_x : \mathcal{O}_{Y,\phi(x)} \to \mathcal{O}_{X,x}$ for every $x \in X$.*

*In particular there is a faithful functor from local spaces with functions over $k$ to locally ringed spaces over $k$.*

*Proof. Drafted by Claude Sonnet, reviewed by David Rufino.*

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a) holds by definition. For b), the stalk is a local ring with maximal ideal $\mathfrak{m}_{X,x}$ by (4.3.3) applied to $W = \{x\}$. Evaluation $[(U,\sigma)] \mapsto \sigma(x)$ is a well-defined $k$-algebra homomorphism $\mathcal{O}_{X,x} \to \overline{k}$ with kernel $\mathfrak{m}_{X,x}$, so it induces the required embedding of the residue field. For c), the stalk map $\phi_x$ is the map of (4.3.7), and it is local because $g(\phi(x)) = 0$ implies $(g \circ \phi)(x) = 0$. The functor is faithful because a regular map is determined by its underlying map of sets. ■

**Remark 4.3.6 (Comparison with Schemes)**
*A local space with functions differs from an arbitrary locally ringed space over $k$ in the following ways.*

- ***Evaluation is part of the structure.*** *The residue fields come with fixed embeddings into $\overline{k}$, and a regular map must respect them. A morphism of locally ringed spaces over $k$ only gives a $k$-embedding of residue fields, which need not be compatible with these. Consequently a regular map is determined by the underlying map of sets.*
- ***Reducedness.*** *Sections are functions, so a section is determined by its values and there are no nilpotents.*
- ***Points.*** *The points of $X$ are $\overline{k}$-valued, so Galois conjugate points are topologically indistinguishable. For an affine variety $X$ the Kolmogorov quotient $X_0$ (4.1.36) corresponds to the closed points of $\operatorname{Spec} k[X]$ (6.1.11), and the irreducible closed subsets of $X$ correspond to all of its points, that is to the prime ideals of $k[X]$ (6.1.19). For an irreducible subset $W$ the local ring $\mathcal{O}_{X,W}$ is the scheme-theoretic local ring at the corresponding point (6.1.56).*

**Proposition 4.3.7 (Stalk Maps)**
*Suppose $\phi : X \to Y$ is a regular morphism and $Z \subset \overline{\phi(W)}$. Then there exists a unique (local) $k$-algebra homomorphism $\phi_W : \mathcal{O}_{Y,Z} \to \mathcal{O}_{X,W}$ such that the following diagram commutes*

$$\begin{array}{ccc} \mathcal{O}_Y(V) & \xrightarrow{\rho \circ \phi_V^\sharp} & \mathcal{O}_X(U) \\ \downarrow & & \downarrow \\ \mathcal{O}_{Y,Z} & \dashrightarrow & \mathcal{O}_{X,W} \end{array}$$

*for all open sets $V \subset Y$ for which $V \cap Z \neq \emptyset$ and some $U \subset \phi^{-1}(V)$ for which $U \cap W \neq \emptyset$. This is functorial in the sense that*

$$(\psi \circ \phi)_W = \psi_Z \circ \phi_W$$

*Proof.* If $V$ meets $Z$ then by (4.1.19) $\phi^{-1}(V)$ meets $W$. Therefore the vertical maps are well-defined. We define

$$\phi_W((V',\sigma)) := (\phi^{-1}(V'), \sigma \circ \phi)$$

and immediately verify it is well-defined and ensures the diagram commutes. ■

**Proposition 4.3.8 (Stalk Maps II)**
*Let $\phi : X \to Y$ be a regular morphism of local spaces with functions. Suppose $y \in \overline{\{\phi(x)\}}$. Then there is a unique local algebra homomorphism $\phi_x : \mathcal{O}_{Y,\phi(x)} \to \mathcal{O}_{X,x}$ making the following diagram commute*

$$\begin{array}{ccc} \mathcal{O}_Y(V) & \xrightarrow{\phi_V^\sharp} & \mathcal{O}_X(U) \\ \downarrow & & \downarrow \\ \mathcal{O}_{Y,\phi(x)} & \dashrightarrow & \mathcal{O}_{X,x} \end{array}$$

*for all open neighbourhoods $V$ of $\phi(x)$ and some open neighbourhood $U \subset \phi^{-1}(V)$ of $x$. This is functorial in the sense that*

$$(\psi \circ \phi)_x = \psi_{\phi(x)} \circ \phi_x$$

*Proof.* This follows immediately from (4.3.7). ■

**Proposition 4.3.9 (Stalk Maps Function Field)**
*Let $X, Y$ be irreducible local spaces with functions and $\phi : X \to Y$ a dominant regular morphism. Then there is a unique homomorphism $\phi_\star : k(Y) \hookrightarrow k(X)$ making the following diagram commute*

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$$\begin{array}{ccc} \mathcal{O}_Y(V) & \xrightarrow{\phi_V^\sharp} & \mathcal{O}_X(U) \\ \downarrow & & \downarrow \\ k(Y) & \underset{\phi_\star}{\hookrightarrow} & k(X) \end{array}$$

*for all non-empty open subsets $V \subset Y$ and some non-empty open subset $U \subset \phi^{-1}(V)$. This is functorial in the sense that*

$$(\phi \circ \psi)_\star = \psi_\star \circ \phi_\star$$

*Proof.* This follows immediately from (4.3.7) with $W = X$ and $T = Y$. ∎

**Proposition 4.3.10**
*Let $X$ be a local space with functions and $Z, W$ be irreducible subsets such that $Z \subset \overline{W}$. Then there is a canonical homomorphism $\mathcal{O}_{X,Z} \to \mathcal{O}_{X,W}$ making the following diagram commute*

$$\begin{array}{ccc} \mathcal{O}_X(U) & & \\ \downarrow & \searrow & \\ \mathcal{O}_{X,Z} & \dashrightarrow & \mathcal{O}_{X,W} \end{array}$$

*for all open subsets $U$ of $X$ meeting $Z$.*

*This holds in particular when $X$ is irreducible we may take $W = X$ and $\mathcal{O}_{X,W} = k(X)$.*

*Proof.* This follows immediately from (4.3.7) by considering the case $\phi = 1_X$. ∎

**Proposition 4.3.11 (Stalk Maps under Specialisation)**
*Let $X, Y$ be irreducible local spaces with functions, $\phi : X \to Y$ a dominant regular morphism and $x \in X$ such that $y \in \overline{\{\phi(x)\}}$. Then for all neighbourhoods $V$ of $\phi(x)$ we have a commutative diagram*

$$\begin{array}{ccc} \mathcal{O}_Y(V) & \xrightarrow{\phi_V^\sharp} & \mathcal{O}_X(\phi^{-1}(V)) \\ \downarrow & & \downarrow \\ \mathcal{O}_{Y,y} & \xrightarrow{\phi_x} & \mathcal{O}_{X,x} \\ \downarrow & & \downarrow \\ k(Y) & \xrightarrow{\phi_\star} & k(X) \end{array}$$

**Lemma 4.3.12 (Evaluation of Stalks)**
*Let $X$ be a local space of functions and $x \in X$. Then there is a unique function*

$$\mathrm{ev}_x : k\big(\mathfrak{m}_{X,x}\big) \hookrightarrow \overline{k}$$

*making the following diagram commute for all open neighbourhoods $U$ of $x$*

$$\begin{array}{ccccc} \mathcal{O}_X(U) & \longrightarrow & \mathcal{O}_{X,x} & \longrightarrow & k\big(\mathfrak{m}_{X,x}\big) \\ & \searrow^{\mathrm{ev}_x} & & & \downarrow{\scriptstyle \mathrm{ev}_x} \\ & & & & \overline{k} \end{array}$$

*This is functorial in the sense that for a regular morphism $\phi : X \to Y$ there is a commutative diagram*

$$\begin{array}{ccc} k\big(\mathfrak{m}_{Y,\phi(x)}\big) & \xrightarrow{\phi_x} & k\big(\mathfrak{m}_{X,x}\big) \\ \downarrow{\scriptstyle \mathrm{ev}_{\phi(x)}} & & \downarrow{\scriptstyle \mathrm{ev}_x} \\ \overline{k} & = & \overline{k} \end{array}$$

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### 4.3.2 Open Immersions

**Definition 4.3.13 (Open Immersion)**
*Let $X, Y$ be spaces with functions. Then we say a regular map $\phi : X \to Y$ is an **open immersion** if*

*a) $\phi$ is an open embedding, and*
*b) $\phi$ induces an isomorphism of sheaves*

$$\phi_\star\big(\mathcal{O}_Y|_{\phi(X)}\big) \xrightarrow{\sim} \mathcal{O}_X$$

**Proposition 4.3.14 (Open Subspace)**
*Let $X$ be a (local) space of functions and $U \subset X$ an open subset. Then $(U, \mathcal{O}_X|_U)$ is a (local) space of functions which we call an **open subspace**. We may write $\mathcal{O}_U \coloneqq \mathcal{O}_X|_U$.*

*The inclusion $i : U \hookrightarrow X$ is an open immersion and an open immersion $\phi : X \to Y$ induces an isomorphism between $X$ and the open subvariety $\phi(X)$.*

**Proposition 4.3.15**
*Let $X$ be a space of functions and $V \subset U$ open subsets. Then $(V, \mathcal{O}_U|_V) = (V, \mathcal{O}_X|_V)$.*

**Proposition 4.3.16**
*Let $\phi : X \to Y$ be an open embedding and $Z \subset Y$ closed. Then $Z$ is irreducible iff $\phi^{-1}(Z)$ is irreducible.*

*In particular if $Y$ is irreducible then so is $X$.*

*Proof.* We may reduce to the case $X = U$ is an open subset of $Y$. Then it follows from (4.1.73) and (4.1.70). ∎

**Proposition 4.3.17**
*Let $\phi : X \to Y$ be a regular open embedding. Then the following are equivalent*

*a) $\phi$ is an open immersion*
*b) for every irreducible subset $W \subset X$ the canonical map*

$$\mathcal{O}_{Y,Z} \to \mathcal{O}_{X,W}$$

*is an isomorphism for all $Z$ such that $\overline{Z} = \overline{\phi(W)}$*
*c) b) holds for some $Z$ such that $\overline{Z} = \overline{\phi(W)}$*

*In particular for all open subsets $V \subset Y$ and $U \subset \phi^{-1}(V)$ such that $U \cap W \neq \emptyset$ we have a commutative diagram*

$$\begin{array}{ccc} \mathcal{O}_Y(V) & \xrightarrow{\phi^\sharp} & \mathcal{O}_X(U) \\ \downarrow & & \downarrow \\ \mathcal{O}_{Y,\phi(W)} & \xrightarrow{\sim} & \mathcal{O}_{X,W} \end{array}$$

*Proof.* *b)* $\implies$ *a)* By definition we may factor into a regular homeomorphism $X \xrightarrow{\sim} V \subset Y$. It is an isomorphism of spaces by (4.2.11).

*a)* $\implies$ *b)* We may reduce to the case that $X \subset Y$ is an open subset. Because an open subset of $X$ intersects $W$ if and only if it intersects $Z$ we find that the map is bijective.

*c)* $\iff$ *b)* we simply observe that if $\overline{Z} = \overline{Z'}$ then the canonical maps are the same. ∎

**Proposition 4.3.18**
*Let $\phi : X \to Y$ be a dominant open immersion. If $Y$ is irreducible, then so is $X$ and the map*

$$\phi_\star : k(Y) \hookrightarrow k(X)$$

*is an isomorphism.*

*In particular if $X$ is irreducible and $U \subset X$ is a non-empty open subset, then there is a canonical isomorphism*

$$k(X) \xrightarrow{\sim} k(U)$$

*Proof.* By (4.3.16) $X$ is irreducible, so it is a special case of (4.3.17). ∎

**Proposition 4.3.19 (Stalk Maps under Open Immersions I)**
*Let $X$ be a local space with functions, $U \subset X$ an open subset and $W \subset X$ an irreducible subset such that $U \cap W \neq \emptyset$.*

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*Then $U \cap W$ is an irreducible subset of $U$ and we have canonical isomorphisms of stalks making the following diagram commute*

$$\begin{array}{ccc} \mathcal{O}_X(V) & \longrightarrow & \mathcal{O}_U(U \cap V) \\ \downarrow & & \downarrow \\ \mathcal{O}_{X,W} & \xrightarrow{\sim} & \mathcal{O}_{U,W\cap U} \end{array}$$

*for all open $V \subset X$ for which $V \cap W \neq \emptyset$.*

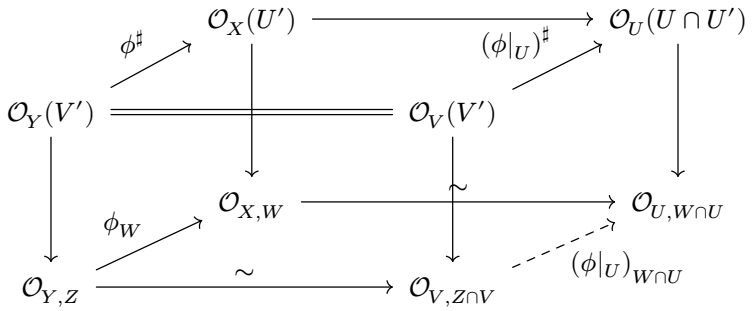
*Proof.* The inclusion $i : U \hookrightarrow X$ is an open immersion. Moreover $U \cap W$ is a non-empty open subset of the irreducible space $W$, and so is irreducible by (4.1.70). This is then a special case of (4.3.17). ∎

**Lemma 4.3.20**
*Let $U, Z$ be subsets of a topological space $X$, with $U$ open. Then*

$$\overline{Z} \cap U \subset \overline{Z \cap U}$$

*Proof.* Suppose $x \in \overline{Z} \cap U$ and $V$ is an open subset containing $x$. Then $U \cap V$ is an open set containing $x$ and therefore by (4.1.19) $Z \cap U \cap V \neq \emptyset$. As $V$ was an arbitrary neighbourhood of $x$ we conclude that $x \in \overline{Z \cap U}$. ∎

**Proposition 4.3.21 (Restricting Stalk Maps to Open Subspaces)**
*Let $\phi : X \to Y$ be a regular morphism of local space with functions. Suppose $U \subset X$ and $V \subset Y$ are open subsets such that $U \subset \phi^{-1}(V)$, and $W \subset X$, $Z \subset Y$ are irreducible subsets such that $Z \subset \overline{\phi(W)}$ and $U \cap W \neq \emptyset$. Then $\phi(W \cap U) \subset \overline{Z \cap V}$ and there is a commutative cube for every open set $V' \subset V$ such that $V' \cap Z \neq \emptyset$ and $U' \subset \phi^{-1}(V')$ such that $U' \cap W \neq \emptyset$*




*where the stalk isomorphisms are defined in (4.3.19) and $(\phi|_U)_{W\cap U}$ is the unique (local) homomorphism making the bottom square (or right side) commute*

*Proof.* Observe by (4.3.20) that

$$\phi(W \cap U) \subset \phi(W) \cap V \subset \overline{Z} \cap V \subset \overline{Z \cap V}$$

which means the stalk map of $\phi|_U$ is well-defined. The sides of the cube are well-defined and commute by (4.3.19) and (4.3.7), and the top by definition. As the horizontal arrows are isomorphisms then we see that the bottom square commutes, and $(\phi|_U)_{W\cap U}$ is the unique homomorphism for which this is the case. ∎

## 4.3.3 Closed Immersions

**Definition 4.3.22 (Closed Immersion)**
*Let $\phi : X \to Y$ be a regular morphism of spaces with functions. We say that $\phi$ is a **closed immersion** if*

*a) $\phi$ is a homeomorphism onto a closed subset $\phi(X)$ of $Y$*
*b) For all $x \in X$ we have the induced map on stalks*

$$\mathcal{O}_{Y,\phi(x)} \to \mathcal{O}_X$$

*is surjective.*

**Proposition 4.3.23 (Closed Subset Structure Sheaf)**
*Let $(X, \mathcal{O}_X)$ be a space of functions and $Z \subset X$ a closed subset. Then there exists a unique sheaf of functions $\mathcal{O}_Z$ such that the inclusion $j : Z \to X$ is a closed immersion.*

*More precisely for $U \subset Z$ an open subset*

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$$\mathcal{O}_Z(U) \coloneqq \left\{ f : U \to \overline{k} \;\middle|\; f(y) = g_i(y)\ \forall y \in Z \cap U_i \text{ for some } g_i \in \mathcal{O}_X(U_i) \text{ where } Z \cap \bigcup_{i\in I} U_i = U \right\}$$

*Proof.* Evidently $j$ is a homeomorphism onto $Z$. We first show that $j$ is regular. Suppose $g \in \mathcal{O}_X(U')$ then by definition we have $g \circ j \in \mathcal{O}_Z(j^{-1}(U')) = \mathcal{O}_Z(U' \cap Z)$.

Given $(V, f) \in \mathcal{O}_{Z,z}$ by taking a smaller open subset we may suppose that $f(y) = g(y)$ for all $y \in V$ for some $g \in \mathcal{O}_X(U)$ and $U \cap Z = V$. Then the image of $(U, g)$ under the stalk map is $(U \cap Z, g \circ j)$ which is precisely $(V, f)$. Therefore the stalk map $j_z$ is surjective for all $z \in Z$.

Let $\mathcal{F}$ be a sheaf of functions for which $j$ is a (regular) closed immersion and suppose $f \in \mathcal{O}_Z(U)$. Then $f|_{U_i\cap Z}(y) = g_i(y)$ for all $y \in Z \cap U_i$ and $g_i \in \mathcal{O}_X(U_i)$. By assumption $f|_{U_i\cap Z} = g_i \circ j \in \mathcal{F}(U_i \cap Z)$. By the sheaf condition we have $f \in \mathcal{F}(U)$ which shows $\mathcal{O}_Z$ is a subsheaf of $\mathcal{F}$.

We need to show that the inclusion of sheaves $\mathcal{O}_Z \hookrightarrow \mathcal{F}$ is an isomorphism. By (4.2.28) c) $\implies$ d) and (4.2.11) it is enough to show that it is surjective on stalks. However this follows by assumption by observing the composition $\mathcal{O}_{X,j(z)} \to \mathcal{O}_{Z,z} \to \mathcal{F}_z$ is surjective. ∎

**Remark 4.3.24**
*This can be defined formally as $j^{-1}(\mathcal{O}_X/\mathcal{I})$ where $\mathcal{I}$ is the presheaf of regular functions which vanish on $Z$. We show in the Algebraic Geometry chapter's treatment of closed subvarieties that this is consistent with the concept of affine subvariety.*

**Proposition 4.3.25**
*Let $\phi : X \to Y$ be a regular morphism of spaces with functions. Then the following are equivalent*

*a) $\phi$ is a closed immersion*
*b) $\phi(X)$ is closed and $\phi$ determines an isomorphism $X \xrightarrow{\sim} \phi(X)$ of spaces with functions using the structure defined in (4.3.23)*

*Proof.* *a)* $\implies$ *b)* By definition $\phi(X)$ is closed and $\phi$ determines a homeomorphism onto $\phi(X)$. Let $Z \coloneqq \phi(X)$ and $\mathcal{O}_Z$ denote the canonical structure sheaf. There is a well-defined map

$$\mathcal{O}_Z(U) \to \mathcal{O}_X(\phi^{-1}(U))$$

which is injective, and therefore injective on stalks (4.2.28) c) $\implies$ d). By (4.2.11) we need only show that the map $\mathcal{O}_{Z,\phi(x)} \to \mathcal{O}_{X,x}$ is surjective. However by hypothesis the composite map

$$\mathcal{O}_{Y,\phi(x)} \to \mathcal{O}_{Z,\phi(x)} \to \mathcal{O}_x$$

is surjective from which the result follows.

*b)* $\implies$ *a)* Evidently $\phi$ is a homeomorphism onto $\phi(X)$. Further we have a sequence of stalk maps

$$\mathcal{O}_{Y,\phi(x)} \xrightarrow{\sim} \mathcal{O}_{\phi(X),\phi(x)} \to \mathcal{O}_{X,x}$$

where the first is surjective by (4.3.23) and the second is an isomorphism by assumption. Therefore the composite is surjective and $\phi$ is a closed immersion. ∎

## 4.3.4 Locally Closed Subspace

**Lemma 4.3.26 (Locally Closed Structure Sheaf)**
*Let $(X, \mathcal{O}_X)$ be a space with functions, $(Z, \mathcal{O}_Z)$ a closed subset with canonical structure defined in (4.3.23) and $(U, \mathcal{O}_U)$ an open subset with canonical structure (4.3.14). Then*

$$\mathcal{O}_Z|_{Z\cap U} = \mathcal{O}_U|_{Z\cap U}$$

## 4.3.5 Glueing

**Proposition 4.3.27 (Patching Topologies)**
*Let $X = \bigcup_{i=1}^n U_i$ be a set with topologies given for each $U_i$. Suppose that for all $i, j = 1...n$ the following properties hold*

- *$U_i \cap U_j$ is open in $U_i$, and*
- *the identity map $1 : U_i \cap U_j \to U_i \cap U_j$ is continuous with respect to the subspace topologies on $U_i$ and $U_j$ respectively.*

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*Then there is a unique topology on $X$ such that*

- *the $U_i$ are open in $X$,*
- *the topology on $U_i$ coincides with the subspace topology from $X$*

*More precisely it is given by*

$$\{U \subset X \mid U \cap U_i \text{ open in } U_i \text{ for all } i = 1...n\}$$

**Proposition 4.3.28 (Glueing Topologies)**
*Let $X = \bigcup_{i=1}^n U_i$ be a set together with bijections $\theta_i : U_i \xrightarrow{\sim} W_i$ where $W_i$ are topological spaces. Suppose that for all $i, j = 1...n$ the following properties hold*

- *$\theta_i\big(U_i \cap U_j\big)$ is open in $W_i$,*
- *the map $\theta_{ij} := \theta_j \circ \theta_i^{-1} : \theta_i\big(U_i \cap U_j\big) \to \theta_j\big(U_i \cap U_j\big)$ is continuous.*

*Then there is a unique topology on $X$ such that*

- *$U_i$ are open in $X$,*
- *$\theta_i$ is a homeomorphism with respect to the subspace topology on $U_i$ for all $i = 1...n$.*

*More precisely the topology is given by*

$$\{U \subset X \mid \theta_i(U \cap U_i) \text{ is open in } W_i \text{ for all } i = 1...n\}$$

**Proposition 4.3.29 (Glueing Regular Maps)**
*Let $X, Y$ be spaces with functions and $X = \bigcup_{i=1}^n U_i$ an open cover. Suppose there exists regular maps*

$$\phi_i : \big(U_i, \mathcal{O}_X|_{U_i}\big) \to (Y, \mathcal{O}_Y)$$

*which satisfy $\phi_i|_{U_i \cap U_j} = \phi_j|_{U_i \cap U_j}$ for all $i, j = 1...n$. Then there exists a unique map $\phi : X \to Y$ such that $\phi|_{U_i} = \phi_i$, which is regular.*

*Proof.* As a function $\phi$ is clearly well-defined and unique. It is continuous by (4.1.29). Suppose $g \in \mathcal{O}_Y(V)$ then by definition $(g \circ \phi)|_{\phi^{-1}(V) \cap U_i} = g \circ \phi_i \in \mathcal{O}_{U_i}(\phi^{-1}(V) \cap U_i) = \mathcal{O}_X(\phi^{-1}(V) \cap U_i)$. As $\mathcal{O}_X$ is a sheaf this shows that $g \circ \phi$ is regular as required. ∎

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